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Integrated Likelihoods for Scalar Parameters of Interest in Count Distributions

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ABSTRACT

Integrated Likelihoods for Scalar Parameters of Interest in Count Distributions

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Acknowledgements

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2.1. Model Parameter Decomposition

It is often the case that we are not interested in estimating the full parameter $\boldsymbol{\theta} \in \boldsymbol{\theta} \subseteq \mathbb{R}^d$, but rather a different parameter ψ taking values in a set $\Psi \subseteq \mathbb{R}^p$, where $p < d$. In such an event, we refer to ψ as the *parameter of interest*. Crucially, as we will see, ψ can always be expressed as a function of $\boldsymbol{\theta}$.

Since ψ is of lower dimension than $\boldsymbol{\theta}$, it necessarily follows that there is another parameter λ, taking values in a set $\Lambda \subseteq \mathbb{R}^q$, where $p+q=d$, that is made up of whatever is "left over" from the full parameter $\boldsymbol{\theta}$. We refer to λ as the *nuisance parameter*, so named for its ability to complicate inference regarding the parameter of interest. Despite not being the object of study themselves, nuisance parameters are nevertheless capable of modifying the distributions of our observations and therefore must be accounted for when conducting inference or estimation regarding the parameter of interest.[1] The process by which this is accomplished is nontrivial and often represents a serious obstacle that must be overcome.

While not required, we will assume the parameter of interest ψ is always one-dimensional. That is, $\Psi \subseteq \mathbb{R}$ and consequently $\Lambda \subseteq \mathbb{R}^{d-1}$. This restriction reflects the common habit of researchers to focus on scalar-valued summaries of vector quantities. For example, suppose we observe data $Y = (y_1, ..., y_n)$, where each y_i is the outcome of some random variable $Y_i \sim N(\mu_i, \sigma_i^2)$, and we are interested in estimating the average of the population means, $\frac{1}{n}\sum_{i=1}^n \mu_i$. Rather than defining $\psi = (\mu_1, ..., \mu_n)$, we can instead define $\psi = \frac{1}{n}\sum_{i=1}^n \mu_i$ directly, bypassing the need to estimate each μ_i individually before taking their average. This does carry the trade-off of increasing the dimension of the nuisance parameter, which must be dealt with before conducting inference or estimation on ψ_0, the true value of ψ. We will examine some of the issues posed by high-dimensional nuisance parameters in greater detail in the next chapter.

[1]Nuisance parameters are not always uniquely defined. In fact, depending on the choice of parameter of interest, there may be multiple or even infinite ways to define one.

2.1.1. Explicit Parameters

Parameters of interest and nuisance parameters can be broadly classified into two categories, explicit or implicit. For a given statistical model, both types of parameter must occupy the same category - it is not possible for ψ to be explicit and λ to be implicit, or vice versa.

Let us first consider the case in which ψ and λ are *explicit* parameters. This means that ψ is a sub-vector of $\boldsymbol{\theta}$, so that all the components of ψ are also components of $\boldsymbol{\theta}$. Then there exists a set $I = \{I_1, ..., I_p\} \subsetneq \{1, ..., d\}$ such that

$$\psi = (\theta_{I_1}, ..., \theta_{I_p}). \tag{2.1.1}$$

It immediately follows that λ is the sub-vector of all components of $\boldsymbol{\theta}$ that are not part of ψ. More precisely, if we let $J = \{J_1, ..., J_q\} \subsetneq \{1, ..., d\}$ such that $I \cup J = \{1, ..., d\}$ and $I \cap J = \emptyset$, then

$$\lambda = (\theta_{J_1}, ..., \theta_{J_q}). \tag{2.1.2}$$

$\boldsymbol{\theta}$ can therefore be decomposed as $\boldsymbol{\theta} = (\psi, \lambda)$ when ψ and λ are explicit, provided we shuffle the indices appropriately.

2.1.2. Implicit Parameters

Now let us consider the case in which ψ and λ are *implicit* parameters. This means there exists some function $\varphi : \boldsymbol{\theta} \to \Psi$ for which the parameter of interest can be written as

$$\psi = \varphi(\boldsymbol{\theta}). \tag{2.1.3}$$

As before, Ψ is still assumed to be a subset of $\mathbb{R}^p$ where p is less than d. This reduction in dimension again implies the existence of a nuisance parameter $\lambda \in \Lambda \subseteq \mathbb{R}^{k-m}$. However, unlike in the explicit case, a closed form expression for λ in terms of the original components of $\boldsymbol{\theta}$ need not exist. For this reason, implicit nuisance parameters are in general more difficult to eliminate compared to their explicit counterparts.

When the parameter of interest and nuisance parameter are explicit, it is always possible to define a function φ such that

$$\varphi(\boldsymbol{\theta}) = (\theta_{I_1}, ..., \theta_{I_p}) = \psi, \tag{2.1.4}$$

where $\{I_1, ..., I_p\}$ is defined as above. Hence, the first case is really just a special example of this more general one in which $\psi = \varphi(\boldsymbol{\theta})$. With this understanding in mind, we will use the notation $\psi = \varphi(\boldsymbol{\theta})$ to refer to the parameter of interest in general, only making the distinction between implicitness and explicitness when the difference is relevant to the situation.

2.1.3. Nuisance Parameter Elimination

The natural solution to the hindrance nuisance parameters pose to making inferences on the parameter of interest is to find a method for eliminating them from the likelihood function altogether. The result of this elimination is what is known as a pseudolikelihood function.

In general, a *pseudolikelihood function* for ψ is a function of the data and ψ only, having properties resembling that of a genuine likelihood function. Suppose $\psi = \varphi(\boldsymbol{\theta})$ for some function φ and parameter $\boldsymbol{\theta} \in \Theta$. If we let $\Theta(\psi) = \{\boldsymbol{\theta} \in \Theta : \varphi(\boldsymbol{\theta}) = \psi\}$, then associated with each $\psi \in \Psi$ is the set of likelihoods $\mathcal{L}_\psi = \{L(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta(\psi)\}$.

Any summary of the values in $\mathcal{L}_\psi$ that does not depend on λ theoretically constitutes a pseudolikehood function for ψ. There exist a variety of methods to obtain this summary but among the most popular are profiling (maximization), conditioning, and integration, each with respect to the nuisance parameter. None of these summaries come without a cost though, meaning some information about ψ_0 is almost certainly sacrificed whenever a nuisance parameter is eliminated from a likelihood. One measure of a good pseudolikelihood, therefore, is how well it is able to retain information about ψ_0 without becoming overly complex in its computation.

In the previous chapter, we introduced the Bartlett identities as being a set of equations relating the derivatives of the log-likelihodd to one another. They can also be used to understand the difference between likelihood and pseudolikelihood functions. A genuine likelihood function of $\boldsymbol{\theta}$ can be characterized as any nonnegative random function of $\boldsymbol{\theta}$ for which all of the Bartlett identities hold. Similarly, we can think of a pseudolikelihood function of $\boldsymbol{\theta}$ as being any nonnegative random function of $\boldsymbol{\theta}$ for which at least one of the Bartlett identities does not hold. Hence, the identities act as a litmus test of sorts for determining the validity of a pseudolikelihood as an approximation to the genuine likelihood from which it originated - the more identities it does satisfy, the better the approximation. A pseudolikelihood that satisfies the first Bartlett identity is called *score-unbiased*; one that satisfies the second is called *information-unbiased*. Historically, more attention has been given to constructing pseudo-likelihoods that are score-unbiased, at least asymptotically (Cf. Kalbfleisch and Sprott, 1970; De Bin et al., 2015; Schumann et al., 2021, 2023).

2.2. The Marginal Likelihood

Suppose we observe some data $\mathbf{X} = \mathbf{x}$ such that the pair of statistics $(\mathbf{T}(\mathbf{X}), \mathbf{S}(\mathbf{X}))$ is sufficient for $\boldsymbol{\theta} = (\psi, \lambda)$. Then, abusing notation slightly, we can write

$$p_{\boldsymbol{\theta}}(\mathbf{X}) = p_{\boldsymbol{\theta}}(\mathbf{T}, \mathbf{S}). \tag{2.2.1}$$

Since the right-hand side is a joint density for $\mathbf{T}$ and $\mathbf{S}$, it can be factored into a product of a marginal and a conditional density as follows:

$$p_{\boldsymbol{\theta}}(\mathbf{T}, \mathbf{S}) = p_{\boldsymbol{\theta}}(\mathbf{T}|\mathbf{S}) p_{\boldsymbol{\theta}}(\mathbf{S}). \tag{2.2.2}$$

If the right-hand term in this product doesn't depend on λ, i.e., $p_{\boldsymbol{\theta}}(\mathbf{S}) = p_{\psi}(\mathbf{S})$, then we have

$$p_{\boldsymbol{\theta}}(\mathbf{T}, \mathbf{S}) = p_{\boldsymbol{\theta}}(\mathbf{T}|\mathbf{S}) p_{\psi}(\mathbf{S}). \tag{2.2.3}$$

In this case, a pseudolikelihood for ψ is simply

$$L_m(\psi) = p_{\psi}(\mathbf{S}). \tag{2.2.4}$$

$L_m(\psi)$ is called a *marginal likelihood* for ψ as it is based on the marginal distribution of $\mathbf{S} = \mathbf{S}(\mathbf{X})$. The conditional part of the density, $p_{\boldsymbol{\theta}}(\mathbf{T}|\mathbf{S})$, does still depend on ψ, however, and so by choosing to base our inferences regarding ψ_0 solely on $L_m(\psi)$, we have ignored some relevant information for ψ_0 that was contained in the data.

2.3. The Conditional Likelihood

If instead it is the left-hand term in Equation 2.2.2 that doesn't depend on λ, i.e., $p_{\boldsymbol{\theta}}(\mathbf{T}|\mathbf{S}) = p_{\psi}(\mathbf{T}|\mathbf{S})$, then we have

$$p_{\boldsymbol{\theta}}(\mathbf{T}, \mathbf{S}) = p_{\psi}(\mathbf{T}|\mathbf{S}) p_{\theta}(\mathbf{S}). \tag{2.3.1}$$

Here, a pseudolikelihood for ψ may be given by

$$L_C(\psi) = p_{\psi}(\mathbf{T}|\mathbf{S}). \tag{2.3.2}$$

$L_C(\psi)$ is called a *conditional likelihood* for ψ as it is based on the conditional distribution of $\mathbf{T}$ given $\mathbf{S}$. As with the marginal likelihood, some information has been disregarded through our omission of $p_{\theta}(\mathbf{S})$ in our inference for ψ_0.

Thus, the use of either a marginal or a conditional likelihood as a pseudolikelihood for a parameter of interest is theoretically only justified when the benefits of eliminating the nuisance parameter through marginalization or conditioning outweigh the corresponding loss in information. In practice, however, the real limiting constraint of using a marginal or conditional likelihood tends not to be the lack of clarity they confer to our inferences, but rather their lack of existence in the first place. The ability to factor a density as in Equation 2.2.3 or Equation 2.3.1 is a rather strong condition, and there are plenty of models for which it is impossible to construct a marginal or conditional likelihood for its parameter of interest. When they do exist, it is typically worthwhile to use them.

2.4. The Profile Likelihood

Profile likelihoods are one of the most straightforward methods for eliminating a nuisance parameter λ from a likelihood function. The idea is to summarize the set $\mathcal{L}_{\psi}$ by its maximum value. Formally, the profile likelihood is defined as

$$L_p(\psi) = \sup_{\boldsymbol{\theta} \in \Theta(\psi)} L(\boldsymbol{\theta}) = L(\hat{\boldsymbol{\theta}}_\psi),$$

where $\hat{\boldsymbol{\theta}}_\psi$ is the MLE of $\boldsymbol{\theta_0}$ for fixed ψ. If we define $\hat{\lambda}_\psi$ to be the conditional MLE for λ given ψ, meaning the value of λ that maximizes the likelihood for a particular value of ψ, then we must have $\hat{\boldsymbol{\theta}}_\psi = (\psi, \hat{\lambda}_\psi)$.

Much of the allure of a profile likelihood can be traced to its ease of computation. In the event of a regular model, finding the value of $\hat{\lambda}_\psi$ corresponding to a particular ψ is equivalent to solving a convex optimization problem. Either an analytical solution to this problem will exist or a numerical one can be obtained using modern computational tools. In both cases, $\hat{\lambda}_\psi$ can be found without much trouble, and from there it's just a matter of setting $\lambda = \hat{\lambda}_\psi$ inside $L(\psi, \lambda)$ to obtain $L_p(\psi)$.

The method is not without its drawbacks however. As noted by Kalbfleisch & Sprott (1970), a major disadvantage to profile likelihoods are their inability to take into account the dimensionality of the nuisance parameter. By effectively always assuming that $\lambda = \hat{\lambda}_\psi$, they fail to incorporate any uncertainty we might have in its value into the resulting pseudolikelihood, leading to overly confident estimates of ψ. This effect is especially pronounced when the dimension of Λ is large. Berger et al. (1999) also bring up the scenario in which the likelihood has a sharp ridge running in one direction as being one in which the profile likelihood will perform poorly. In such a situation, the profile of the likelihood along the ridge would not be representative of the shape of the likelihood elsewhere, and yet it is exactly that profile which will be obtained through maximization.

Nevertheless, a profile likelihood can still be a useful tool for conducting inference regarding ψ_0. At worst, it offers a baseline of sorts to assess the quality of other pseudolikelihoods. There is no point in using another pseudolikelihood that is harder to compute if it offers the same amount of information about ψ_0 - it must justify this increase in complexity with a corresponding increase in features that lend themselves to a greater degree of accuracy in our inference, such as higher level of peakedness around the value of ψ_0.

2.5. The Integrated Likelihood

The *integrated likelihood* for ψ seeks to summarize $\mathcal{L}_\psi$ by its average value with respect to some weight function π over $\Theta(\psi)$. From a theoretical standpoint, this is preferable to maximization as it incorporates (or at least is capable of incorporating) the uncertainty we have in the value of the nuisance parameter into the resulting pseudolikelihood in a very natural way. Formally, the integrated likelihood function is defined as

$$\bar{L}(\psi) = \int_\Lambda L(\psi, \lambda)\pi(\lambda|\psi)d\lambda, \tag{2.5.1}$$

where $\pi(\lambda|\psi)$ is a nonnegative function on Λ. $\pi(\lambda|\psi)$ is sometimes called a conditional prior density for λ given ψ, though it need not satisfy the requirements of a genuine density function.

Note the similarity in form between the integral in Equation 2.5.1 and the expression for the normalizing constant of a posterior distribution:

$$\int_\Theta L(\theta; X)\pi(\theta)d\theta.$$

This similarity lends weight to the idea that Bayesian techniques used to obtain empirical approximations to posterior distributions, such as Markov Chain Monte Carlo, could also be used to approximate an integrated likelihood function, with the result being useful for Bayesian and frequentist inference alike.

In general, the selection of the weight function plays an important role in the properties of the resulting integrated likelihood. The obvious first choice to make for it, and therefore the one that could be considered the “default”, is simply $\pi(\lambda|\psi) \propto 1$, i.e., a uniform density. This yields what is known as the uniform-integrated likelihood:

$$\bar{L}^U(\psi) = \int_\Lambda L(\psi, \lambda)d\lambda. \tag{2.5.2}$$

In the next chapter, we will discuss a re-parameterization of the nuisance parameter developed by Severini (2007) that makes the integrated likelihood relatively insensitive to the exact weight function chosen. Using this new parameterization, we have great flexibility in choosing our weight function; as long as it does not depend on the parameter of interest, the integrated likelihood that is produced will enjoy many desirable frequency properties.

CHAPTER 3

Integrated Likelihood Inference

3.1. Overview

The preceding chapter established the integrated likelihood as a principled approach to eliminating nuisance parameters from inference, provided an appropriate weight function can be identified. This chapter develops a general computational procedure for approximating the integrated likelihood in models where the nuisance parameter is high-dimensional and the integration cannot be carried out in closed form. The procedure is built on the zero-score-expectation (ZSE) parameterization of Severini (2007) and Severini (2018), which provides a canonical choice of weight function with favorable frequentist properties. Subsequent chapters specialize this general procedure to specific models and parameters of interest.

Throughout this chapter we work with a generic model parameterized by $\theta \in \Theta$ and a scalar parameter of interest $\psi = \varphi(\theta)$ for some function $\varphi : \Theta \to \mathbb{R}$. We denote the unrestricted MLE by $\hat{\theta}$, the induced MLE for ψ by $\hat{\psi} = \varphi(\hat{\theta})$, and the constrained parameter space at a fixed value of ψ by $\Theta(\psi) = \{\theta \in \Theta : \varphi(\theta) = \psi\}$.

3.2. The Zero-Score Expectation Parameterization

3.2.1. Explicit nuisance parameters

Severini (2007) considered the problem of selecting a weight function $\pi(\lambda \mid \psi)$ for the integrated likelihood

$$\bar{L}(\psi) = \int_{\Lambda} L(\psi, \lambda)\, \pi(\lambda \mid \psi)\, d\lambda$$

such that the resulting integrated likelihood possesses well-defined frequentist properties. He showed that a suitable weight function can be constructed by identifying a reparameterization $(\psi, \lambda) \mapsto (\psi, \phi)$ of the nuisance parameter with the property that $\hat{\phi}_\psi = \hat{\phi}$ — that is, the conditional MLE of ϕ given ψ equals its unrestricted MLE — and then choosing a prior $\pi(\phi)$ that does not depend on ψ. An integrated likelihood with the desired properties is then

$$\bar{L}(\psi) = \int_\Phi \tilde{L}(\psi, \phi)\, \pi(\phi)\, d\phi, \tag{3.2.1}$$

where $\tilde{L}(\psi, \phi)$ is the likelihood after reparameterization. The specific choice of $\pi(\phi)$ is secondary; the key requirement is that it not depend on ψ.

The reparameterization is achieved by defining ϕ as the solution to

$$\mathrm{E}_{(\psi_0, \lambda_0)}[\nabla_\lambda\, \ell(\psi, \lambda;\, \mathbf{X}_n)]\Big|_{(\psi_0,\, \lambda_0) \,=\, (\hat{\psi},\, \phi)} = \mathbf{0}, \tag{3.2.2}$$

where the expectation is taken with respect to the data distribution evaluated at the true parameter values $(\psi_0, \lambda_0) = (\hat{\psi}, \phi)$. The quantity ϕ is called the *zero-score-expectation parameter* because it is defined as the nuisance value at which the expected score for λ vanishes. Since ϕ satisfies Equation 3.2.2 for a given $(\psi, \lambda, \hat{\psi})$, it is implicitly a function of the data through $\hat{\psi}$. Once ϕ is determined, the corresponding nuisance parameter value is recovered as

$$\lambda(\psi, \phi) = \operatorname*{argmax}_{\lambda \in \Lambda} \mathrm{E}_{(\hat{\psi},\, \phi)}[\ell(\psi, \lambda;\, \mathbf{X}_n)]\,, \tag{3.2.3}$$

and Equation 3.2.1 can be written entirely in terms of the original parameterization:

$$\bar{L}(\psi) = \int_\Phi L(\psi,\, \lambda(\psi, \phi))\, \pi(\phi)\, d\phi. \tag{3.2.4}$$

3.2.2. Implicit nuisance parameters

Severini (2018) extended the ZSE construction to settings where ψ is an implicit function of a single model parameter θ, with no explicit separation into interest and nuisance components. In this setting, the role of the ZSE nuisance value ϕ is played by a point ω on the level set

$$\Omega_{\hat{\psi}} = \left\{\omega \in \Theta : \varphi(\omega) = \hat{\psi}\right\}, \tag{3.2.5}$$

and Equation 3.2.3 generalizes to

$$\theta^*(\psi, \omega) = \operatorname*{argmax}_{\varphi(\theta)=\psi} \mathrm{E}_\omega[\ell(\theta;\, \mathbf{X}_n)]\,. \tag{3.2.6}$$

The integrated likelihood for ψ is then

$$\bar{L}(\psi) = \int_{\Omega_{\hat{\psi}}} L(\theta^*(\psi, \omega))\, \pi(\omega)\, d\omega, \tag{3.2.7}$$

where $\pi(\omega)$ is a density on $\Omega_{\hat{\psi}}$ that does not depend on ψ. The expression in Equation 3.2.7 is the central object of interest in this paper.

3.3. Branches and the Profile Likelihood

3.3.1. Branch functions

The integrand in Equation 3.2.7 assigns to each $\omega \in \Omega_{\hat{\psi}}$ a likelihood value $L(\theta^*(\psi, \omega))$, obtained by evaluating the likelihood at the parameter value $\theta^*(\psi, \omega)$ that maximizes the expected log-likelihood subject to $\varphi(\theta) = \psi$. We refer to the function

$$b(\psi;\, \omega) = L(\theta^*(\psi, \omega))\,, \quad \psi \in \Psi, \tag{3.3.1}$$

as the *branch function* associated with ω. Each $\omega \in \Omega_{\hat{\psi}}$ induces a distinct branch function, and the integrated likelihood is the average of these branch functions over the level set:

$$\bar{L}(\psi) = \int_{\Omega_{\hat{\psi}}} b(\psi;\, \omega)\, \pi(\omega)\, d\omega. \tag{3.3.2}$$

The profile likelihood is the special case in which $\omega = \hat{\omega}_{\mathrm{MLE}}$ — the point on $\Omega_{\hat{\psi}}$ induced by the unrestricted MLE — and the expected log-likelihood reduces to the observed log-likelihood. In this case $\theta^*(\psi, \hat{\omega}_{\mathrm{MLE}}) = \hat{\theta}(\psi)$, the constrained MLE at ψ, and $b(\psi;\, \hat{\omega}_{\mathrm{MLE}}) = L_P(\psi)$ is the profile likelihood. The integrated likelihood can therefore be interpreted as a weighted average of profile-like likelihood curves, each anchored at a different point on the level set.

3.3.2. The branch mode

For a given ω, the branch function $b(\psi;\, \omega)$ has a maximum over $\psi \in \Psi$, which we call the *branch mode*:

$$\psi^*(\omega) = \underset{\psi \in \Psi}{\operatorname{argmax}}\, b(\psi;\, \omega). \tag{3.3.3}$$

The branch mode is the value of ψ at which the likelihood is highest along the constrained path parametrized by ω. For the profile likelihood branch, $\psi^*(\hat{\omega}_{\mathrm{MLE}}) = \hat{\psi}$ by construction. For other branches the mode may differ from $\hat{\psi}$, reflecting the fact that different nuisance configurations are compatible with different values of the parameter of interest.

3.4. Monte Carlo Approximation

3.4.1. The approximation problem

Direct evaluation of Equation 3.3.2 requires integration over $\Omega_{\hat{\psi}}$, which is a manifold of dimension $\dim(\Theta) - 1$ in general. In high-dimensional models this integration is intractable, and we instead approximate it by a Monte Carlo average over a finite collection of ω values sampled from $\Omega_{\hat{\psi}}$:

$$\bar{L}(\psi) \approx \frac{1}{R} \sum_{r=1}^{R} b(\psi;\, \omega^{(r)}), \tag{3.4.1}$$

where $\omega^{(1)}, \ldots, \omega^{(R)}$ are draws from a distribution supported on $\Omega_{\hat{\psi}}$. The quality of the approximation depends on (i) how densely and representatively the draws cover the level set and (ii) whether the branch functions associated with the drawn ω values are well-behaved.

3.4.2. Sampling omega-hat values

Sampling from $\Omega_{\hat{\psi}}$ proceeds in two stages. First, an initial candidate in the ambient parameter space Θ is generated by perturbing the MLE $\hat{\theta}$ in directions tangent to the level set. Second, the candidate is projected onto $\Omega_{\hat{\psi}}$ by solving a constrained feasibility problem with respect to ψ. The projection does not optimize the likelihood; it simply finds a feasible point consistent with the observed $\hat{\psi}$. The design of the initial generator is problem-specific and is discussed in detail for each model in the chapters that follow.

3.4.3. Evaluating branch functions

For each sampled $\omega^{(r)}$, the branch function $b(\psi; \omega^{(r)})$ is evaluated over a grid of ψ values. At each grid point ψ_k, the branch value is computed by solving Equation 3.2.6 — maximizing the expected log-likelihood subject to the constraint $\varphi(\theta) = \psi_k$ — and evaluating the likelihood at the resulting $\theta^*(\psi_k, \omega^{(r)})$. Solutions are chained across the grid: the optimizer at ψ_k is initialized from the solution at ψ_{k-1}, exploiting the continuity of $\theta^*(\psi; \omega)$ in ψ.

Before tracing a branch across the full grid, the algorithm first locates the branch mode $\psi^*(\omega^{(r)})$ via a one-dimensional search. The branch is then traced outward in both directions from the mode until the branch value falls below a threshold determined by the confidence levels of interest.

3.4.4. Branch screening

Not all sampled ω values yield well-behaved branch functions. A branch is considered valid if it has a unique interior mode, decreases monotonically on each side of the mode, and does not exhibit discontinuities or anomalous curvature. Draws that fail these criteria are discarded before the integrated likelihood is assembled. This screening step can be interpreted as a regularization of the integration measure: by restricting attention to well-behaved nuisance configurations, the approximation is made more stable at the cost of a slight bias in the weight function. The specific screening criteria are discussed in the context of each model in the chapters that follow.

3.4.5. Assembling the integrated likelihood

Valid branch functions are evaluated on a common ψ grid and averaged pointwise to form the integrated likelihood approximation Equation 3.4.1. The profile likelihood is constructed from the single branch corresponding to $\hat{\omega}_{\text{MLE}}$ and provides a baseline for comparison. Confidence intervals for ψ are obtained by inverting the likelihood ratio statistic against both the integrated and profile likelihoods, using the standard χ^2_1 reference distribution.

3.5. The Two-Space Problem

A structural subtlety arises in regression models where the level set $\Omega_{\hat{\psi}}$ lives in a different space from the model parameter θ. In the no-effects model, $\theta = \boldsymbol{\theta} \in \Delta^{J-1}$ and $\Omega_{\hat{\psi}} \subset \Delta^{J-1}$, so ω and θ inhabit the same space and the branch construction is straightforward. In regression models, θ is a matrix of regression coefficients while $\Omega_{\hat{\psi}}$ is a manifold of probability vectors at a reference covariate value $\mathbf{x}_0$.

Bridging this gap requires a map from $\Omega_{\hat{\psi}} \subset \Delta^{J-1}$ to Θ. The construction used throughout this paper is a rank-one adjustment of the MLE that matches the linear predictor at $\mathbf{x}_0$ to the logit vector implied by ω, while leaving the regression coefficients unchanged in all directions orthogonal to $\mathbf{x}_0$. The specific form of this adjustment and its consequences for the alignment between the surrogate and true log-likelihoods are discussed in the context of each model.

CHAPTER 4

Multinomial Dyads

4.1. Introduction

This chapter concerns parameters of interest derived from a multinomial distribution under a variety of model assumptions. The parameters of interest considered are Shannon entropy, Simpson's diversity index, and a shared effect coefficient in a multinomial logistic regression. We begin with a brief review of the multinomial distribution and its relevant properties.

Let $\mathbf{Y}$ denote the outcome of a single categorical trial with parameter $\boldsymbol{\theta} = (\theta_1, ..., \theta_J) \in \Delta^{J-1}$, where

$$\Delta^{J-1} = \left\{ \boldsymbol{\theta} \in \mathbb{R}^J : \theta_j > 0,\ \sum_{j=1}^{J} \theta_j = 1 \right\} \tag{4.1.1}$$

is the open $(J-1)$-dimensional probability simplex. The strict positivity constraint $\theta_j > 0$ reflects the modeling assumption that every included category is theoretically observable; a category believed to have zero probability would simply be omitted from the model.

We may represent $\mathbf{Y}$ as a random vector taking values in the canonical basis of $\mathbb{R}^J$, i.e.,

$$\mathbf{Y} \in \{\mathbf{e}_1, ..., \mathbf{e}_J\},$$

where $\mathbf{e}_j$ denotes the jth standard basis vector. The distribution of $\mathbf{Y}$ is specified by

$$\mathbb{P}(\mathbf{Y} = \mathbf{e}_j) = \theta_j, \quad j = 1, \dots, J, \tag{4.1.2}$$

and we write $\mathbf{Y} \sim \text{Categorical}(\boldsymbol{\theta})$.

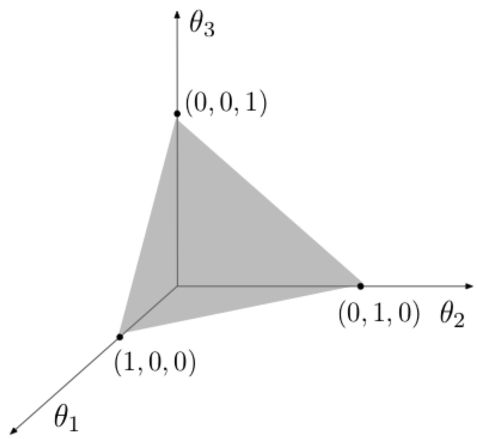

Figure 4.1. THE PROBABILITY SIMPLEX FOR A MULTINOMIAL MODEL WITH THREE CATEGORIES. The simplex Δ^2 is embedded in $\mathbb{R}^3$ with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. Each point on the surface corresponds to a probability vector $\boldsymbol{\theta}$ satisfying $\sum_j \theta_j = 1$.

A multinomial random vector arises as the sum of n i.i.d. categorical random vectors:

$$\sum_{i=1}^{n} \mathbf{Y}_i \sim \text{Multinomial}(n, \boldsymbol{\theta}), \quad \mathbf{Y}_i \overset{\text{i.i.d.}}{\sim} \text{Categorical}(\boldsymbol{\theta}), \quad i = 1, ..., n. \tag{4.1.3}$$

In particular, the categorical distribution is a special case of the multinomial distribution with $n = 1$, mirroring the relationship between the Bernoulli and binomial distributions.[1]

Let $(N_1, ..., N_J) \sim \text{Multinomial}(n, \boldsymbol{\theta})$, where N_j denotes the number of observations falling in category j and $n = \sum_{j=1}^{J} N_j$. Each marginal count satisfies $N_j \sim \text{Binomial}(n, \theta_j)$, with $\mathbb{E}_{\boldsymbol{\theta}}(N_j) = n\theta_j$. Since $\boldsymbol{\theta}$ represents category probabilities, the parameter space is $\Theta = \Delta^{J-1} \subset \mathbb{R}^J$. The likelihood and log-likelihood functions are given by

$$L(\boldsymbol{\theta}) = \prod_{j=1}^{J} \theta_j^{N_j} \tag{4.1.4}$$

and

$$\ell(\boldsymbol{\theta}) = \sum_{j=1}^{J} N_j \log \theta_j, \tag{4.1.5}$$

respectively. Maximization of Equation 4.1.5 subject to $\boldsymbol{\theta} \in \Delta^{J-1}$ yields the familiar maximum likelihood estimator

$$\hat{\theta}_j = \frac{N_j}{n}, \quad j = 1, ..., J. \tag{4.1.6}$$

[1]The categorical distribution is also known as the *multinoulli* distribution.

The multinomial likelihood is invariant under relabeling of the category indices: permuting the components of $\boldsymbol{\theta}$ leaves the likelihood unchanged. This symmetry implies that, absent a sample size large enough to detect fractional differences between cell probabilities, multiple configurations of the probability vector are observationally equivalent.[2] This equivalence is most pronounced near the boundary of the probability simplex. When one or more categories are rare or unobserved, the likelihood provides little information about which specific components of $\boldsymbol{\theta}$ are effectively zero, resulting in flat or weakly identified directions associated with different boundary faces.

4.2. Models

4.2.1. Baseline-category logit model with no effects

A natural starting point for entropy inference under a multinomial model is to treat the category probabilities as free parameters estimated directly from count data. This was the framework considered by Severini (2022), who found that an integrated likelihood based on the ZSE nuisance parameterization yields point and interval estimators for the entropy of a multinomial distribution that have improved frequency properties (e.g. reduced bias, lower root mean square error, closer-to-nominal empirical coverage rates, etc.) relative to their profile likelihood counterparts, provided that the sample size is small relative to the number of categories. We adopt this setting here for use as a benchmark but with a parameterization that is better suited to both numerical optimization and extension to more sophisticated models.

Suppose we observe $\mathbf{Y}_i \overset{\text{i.i.d.}}{\sim} \text{Categorical}(\boldsymbol{\theta})$ for $i = 1, \dots, n$, where the model parameter $\boldsymbol{\theta} = (\theta_1, \dots, \theta_J)^\top \in \Delta^{J-1}$ is a probability vector with $J - 1$ free components, with the remaining degree of freedom having been absorbed by the constraint $\sum_j \theta_j = 1$. Fixing category 1 as a reference, we define the logit parameters

$$\eta_j = \log\left(\frac{\theta_j}{\theta_1}\right), \quad j = 2, \dots, J, \tag{4.2.1}$$

[2]This invariance does not imply non-identifiability of the multinomial parameter, but rather reflects weak identification in directions that are immaterial for the likelihood yet consequential for certain parameters of interest.

so that each η_j represents the log-odds of category j relative to the reference category 1. Such a transformation maps the interior of Δ^{J-1} bijectively onto $\mathbb{R}^{J-1}$, allowing optimization of an objective function with respect to the model parameter to be carried out in unconstrained Euclidean space. The simplex constraints therefore no longer need to be enforced explicitly, and we avoid any boundary-related degeneracies that arise when one or more category probabilities are close to zero. In particular, rare or unobserved categories may be represented by negative logit values that, while very large, are ultimately still finite. The likelihood surfaces expressed in the logit parameterization tend to be smoother as a result, with no sharp corners to interfere with differentiability near the boundaries of the parameter space. This leads to improved numerical stability when approximating both the integrated and profile likelihoods for ψ, especially in settings where the interest function depends on logarithms of the category probabilities.

To recover the category probabilities from the logits, we invert Equation 4.2.1. Letting $\boldsymbol{\eta} = (\eta_2, \dots, \eta_J)^\top \in \mathbb{R}^{J-1}$, we have

$$\theta_1(\boldsymbol{\eta}) = \frac{1}{1 + \sum_{k=2}^{J} \exp(\eta_k)}, \qquad \theta_j(\boldsymbol{\eta}) = \frac{\exp(\eta_j)}{1 + \sum_{k=2}^{J} \exp(\eta_k)}, \quad j = 2, \dots, J. \tag{4.2.2}$$

This normalized exponential mapping is closely related to the softmax function, with the use of a reference category in place to ensure the model remains identifiable. Since $\boldsymbol{\eta}$ and $\boldsymbol{\theta}$ are related bijectively via Equation 4.2.1 and Equation 4.2.2, any statement about one is equivalent to a statement about the other; we work with $\boldsymbol{\eta} \in \mathbb{R}^{J-1}$ throughout for the computational reasons described above. We refer to the multinomial model equipped with this parameterization as the baseline-category logit model with no effects, reflecting the absence of covariate effects or additional regression structure.

The log-likelihood for this model is given by

$$\ell(\boldsymbol{\eta}) = \sum_{j=1}^{J} N_j \log \theta_j(\boldsymbol{\eta}), \tag{4.2.3}$$

where $N_j = \sum_{i=1}^{n} \mathbb{1}(y_i = j)$ is the observed count for category j and $\theta_j(\boldsymbol{\eta})$ is defined as in Equation 4.2.2. Since the log-likelihood depends on the data only through the category counts $N_1, \dots, N_J$, these form a sufficient statistic for $\boldsymbol{\theta}$. The MLE $\hat{\boldsymbol{\theta}}$ has the closed form $\hat{\theta}_j = N_j/n$, or equivalently $\hat{\eta}_j = \log(N_j/N_1)$ for $j = 2, \dots, J$, provided all categories are observed.

4.2.2. Baseline-category logit model with fixed effects

The baseline-category logit model with fixed effects extends the no-effects model by allowing the category probabilities to depend on observed covariates through a set of fixed regression coefficients. Inference is based on independent observations $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the covariate vector associated with observation i and $\mathbf{y}_i \in \{\mathbf{e}_1, \dots, \mathbf{e}_J\}$ denotes the corresponding categorical response. Fixing category 1 as a reference, the log-odds of category j relative to category 1 for observation i are modeled as

$$\log \frac{P(Y_i = j \mid \mathbf{x}_i)}{P(Y_i = 1 \mid \mathbf{x}_i)} = \alpha_j + \mathbf{x}_i^\top \boldsymbol{\beta}_j, \quad j = 2, \dots, J, \tag{4.2.4}$$

where $\alpha_j \in \mathbb{R}$ is a category-specific intercept and $\boldsymbol{\beta}_j \in \mathbb{R}^p$ is a category-specific vector of regression coefficients. The full model parameter is $(\mathbf{A}, \mathbf{B})$, where $\mathbf{A} = (\alpha_2, \dots, \alpha_J)^\top \in \mathbb{R}^{J-1}$ collects the intercepts and $\mathbf{B} = [\boldsymbol{\beta}_2 \mid \cdots \mid \boldsymbol{\beta}_J] \in \mathbb{R}^{p \times (J-1)}$ collects the regression coefficients, giving $(p+1)(J-1)$ free parameters in total — one intercept and p slopes for each of the $J-1$ non-baseline categories. The conditional category probabilities are obtained from Equation 4.2.4 via the softmax mappings

$$\theta_1(\mathbf{x}_i; \mathbf{A}, \mathbf{B}) = \frac{1}{1 + \sum_{k=2}^J \exp(\alpha_k + \mathbf{x}_i^\top \boldsymbol{\beta}_k)} \tag{4.2.5}$$

$$\theta_j(\mathbf{x}_i; \mathbf{A}, \mathbf{B}) = \frac{\exp(\alpha_j + \mathbf{x}_i^\top \boldsymbol{\beta}_j)}{1 + \sum_{k=2}^J \exp(\alpha_k + \mathbf{x}_i^\top \boldsymbol{\beta}_k)}, \quad j = 2, \dots, J, \tag{4.2.6}$$

which generalize Equation 4.2.2 to incorporate covariate effects. Since observations are independent, the log-likelihood decomposes as a sum over observations,

$$\ell(\mathbf{A}, \mathbf{B}) = \sum_{i=1}^n \sum_{j=1}^J y_{ij} \log \theta_j(\mathbf{x}_i; \mathbf{A}, \mathbf{B}), \tag{4.2.7}$$

where $y_{ij} = \mathbb{1}(Y_i = j)$.

4.2.3. Baseline-category logit model with random effects

The baseline-category logit model with random effects extends the fixed effects model to settings where observations are clustered, with multiple categorical responses observed per cluster. Clustering induces within-cluster correlation that violates the independence assumption of the fixed effects model: responses from the same cluster tend to resemble one another more than responses from different clusters, even after conditioning on observed covariates. We adopt the multinomial logit random effects framework of Hartzel et al. (2001), who present a general approach for logit random effects modeling of clustered nominal and ordinal responses within a unified multivariate generalized linear mixed model formulation.

The data consist of n independent clusters, with cluster i contributing m_i observations. Let $\mathbf{x}_{it} \in \mathbb{R}^p$ denote the covariate vector and $\mathbf{y}_{it} \in \{\mathbf{e}_1, \dots, \mathbf{e}_J\}$ the categorical response for observation t within cluster i, for $i = 1, \dots, n$ and $t = 1, \dots, m_i$. Fixing category 1 as the reference, the conditional log-odds of category j relative to category 1 for observation (i, t) are modeled as

$$\log \frac{P(Y_{it} = j \mid \mathbf{x}_{it}, \mathbf{u}_i)}{P(Y_{it} = 1 \mid \mathbf{x}_{it}, \mathbf{u}_i)} = \alpha_j + \mathbf{x}_{it}^\top \boldsymbol{\beta}_j + u_{ij}, \quad j = 2, \dots, J, \tag{4.2.8}$$

where $\alpha_j \in \mathbb{R}$ is a category-specific intercept, $\boldsymbol{\beta}_j \in \mathbb{R}^p$ is a category-specific vector of regression coefficients, and u_{ij} is a cluster- and category-specific random intercept. Comparing Equation 4.2.8 with Equation 4.2.4, the random effects model augments each fixed log-odds by a cluster-specific shift u_{ij}, inducing within-cluster correlation among responses that share the same $\mathbf{u}_i$.[3]

Both the fixed effects and the random effects are indexed by j because the baseline category is arbitrary, and in general there is no reason to expect effects to be similar for different categories. The random effects for cluster i are collected into the vector $\mathbf{u}_i = (u_{i2}, \dots, u_{iJ})^\top \in \mathbb{R}^{J-1}$, which is assumed to follow a multivariate normal distribution

$$\mathbf{u}_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{4.2.9}$$

[3]A more general version of the model would allow the random effects to enter the linear predictor through $\mathbf{w}_{it}^\top \mathbf{u}_i$ for a cluster- and observation-specific coefficient vector $\mathbf{w}_{it}$. We restrict attention to the random intercept case $\mathbf{w}_{it} = \mathbf{1}$, so that the random effect for category j in cluster i contributes a common shift u_{ij} to the log-odds of every observation in that cluster.

independently across clusters. Following Hartzel et al. (2001), we leave the covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{(J-1)\times(J-1)}$ unspecified, allowing different variances for random effects associated with different categories and arbitrary correlations among them, and ensuring that the model is structurally invariant to the choice of baseline category.

The full model parameter is $(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma})$, where $\mathbf{A}$ and $\mathbf{B}$ carry the same meaning as in the fixed effects model, giving an additional $\frac{1}{2}(J-1)J$ free parameters from the unique elements of the symmetric matrix $\boldsymbol{\Sigma}$ and $(p+1)(J-1) + \frac{1}{2}(J-1)J$ free parameters in total. The conditional category probabilities given the random effects are obtained from Equation 4.2.8 via the softmax mapping:

$$\theta_1(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i) = \frac{1}{1 + \sum_{k=2}^{J} \exp(\alpha_k + \mathbf{x}_{it}^\top \boldsymbol{\beta}_k + u_{ik})}, \tag{4.2.10}$$

$$\theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i) = \frac{\exp(\alpha_j + \mathbf{x}_{it}^\top \boldsymbol{\beta}_j + u_{ij})}{1 + \sum_{k=2}^{J} \exp(\alpha_k + \mathbf{x}_{it}^\top \boldsymbol{\beta}_k + u_{ik})}, \quad j = 2, \dots, J. \tag{4.2.11}$$

Comparing Equation 4.2.11 with Equation 4.2.6, the random effects enter the softmax in the same position as the fixed intercepts, shifting all conditional probabilities for cluster i away from the population mean in a category-specific direction.

4.2.3.1. Likelihood functions. Because the random effects $\mathbf{u}_i$ are unobserved, there are two natural ways to write the likelihood for this model, each treating the random effects differently. Understanding the distinction between them is important, as the two objects play different roles in estimation and in the construction of the integrated likelihood.

The first, and more fundamental, is the *complete-data likelihood* — the joint probability of the observed responses and the random effects, treating the latter as if they were observed:

$$L_{\mathrm{c}}(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma} \mid \mathbf{u}_1, \dots, \mathbf{u}_n) = \prod_{i=1}^{n} \left[\prod_{t=1}^{m_i} \prod_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i)^{y_{itj}} \right] \phi(\mathbf{u}_i; \mathbf{0}, \boldsymbol{\Sigma}). \tag{4.2.12}$$

This is the natural probability model for the data once the random effects are treated as known quantities. Since the clusters are independent and the random effects enter the model only through their own cluster, the complete-data likelihood factorizes across clusters.

The second is the *marginal likelihood* — the probability of the observed responses alone, obtained by integrating the complete-data likelihood over the distribution of the random effects:

$$L(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma}) = \prod_{i=1}^{n} \int_{\mathbb{R}^{J-1}} \left[\prod_{t=1}^{m_i} \prod_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i)^{y_{itj}} \right] \phi(\mathbf{u}_i; \mathbf{0}, \boldsymbol{\Sigma})\, d\mathbf{u}_i, \tag{4.2.13}$$

where $\phi(\cdot\,; \mathbf{0}, \boldsymbol{\Sigma})$ denotes the multivariate normal density with mean $\mathbf{0}$ and covariance $\boldsymbol{\Sigma}$, and $y_{itj} = \mathbb{1}(Y_{it} = j)$. Since the clusters are independent and the random effects are drawn independently from the same distribution, the marginal likelihood is equal to the product of cluster-level marginal likelihoods. The marginal likelihood is the appropriate object for inference about $(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma})$ when the random effects are a nuisance — it reflects the probability of the observed data under the model for a randomly drawn cluster, averaging over the distribution of $\mathbf{u}_i$.

4.2.3.2. Log-likelihoods. Taking logarithms of each in turn, the complete-data log-likelihood is

$$\ell_{\mathrm{c}}(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma} \mid \mathbf{u}_1, \dots, \mathbf{u}_n) = \sum_{i=1}^{n} \sum_{t=1}^{m_i} \sum_{j=1}^{J} y_{itj} \log \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i) + \sum_{i=1}^{n} \log \phi(\mathbf{u}_i; \mathbf{0}, \boldsymbol{\Sigma}), \tag{4.2.14}$$

and the marginal log-likelihood is

$$\ell(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma}) = \sum_{i=1}^{n} \log \int_{\mathbb{R}^{J-1}} \left[\prod_{t=1}^{m_i} \prod_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i)^{y_{itj}} \right] \phi(\mathbf{u}_i; \mathbf{0}, \boldsymbol{\Sigma})\, d\mathbf{u}_i. \tag{4.2.15}$$

The structural difference between these two expressions is fundamental. In Equation 4.2.14 the logarithm operates directly on the conditional probabilities θ_j and the normal density ϕ, making ℓ_{c} a linear function of the sufficient statistics y_{itj} that is well-defined only when $\mathbf{u}_i$ is treated as known. In Equation 4.2.15 the logarithm sits outside the integral: one first computes the marginalised probability of cluster i's responses by integrating over $\mathbf{u}_i$, and only then takes the log. The two quantities are related by Jensen's inequality — since the logarithm is concave, ℓ_{c} evaluated at any fixed $\mathbf{u}_i$ provides an upper bound for ℓ at the same parameter values — but they are not equal in general, and neither can be recovered from the other without knowledge of $\mathbf{u}_i$.

Estimation of $(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma})$ proceeds by maximizing ℓ, since $\mathbf{u}_i$ is never observed and the marginal log-likelihood is the correct likelihood for the observed data. The Laplace approximation to ℓ and the role of ℓ_{c} in the construction of the integrated likelihood are discussed in Section 4.4.

4.3. Parameters of interest

In the no-effects model, each parameter of interest is a scalar function of the probability vector $\boldsymbol{\theta} \in \Delta^{J-1}$, and its definition requires no further qualification. In the regression models described above, however, the probability vector $\boldsymbol{\theta}(\mathbf{x}; \mathbf{B})$ depends on the observed covariate value $\mathbf{x}$, so any scalar summary of the distribution is itself a function of $\mathbf{x}$ rather than a single number. Defining a scalar parameter of interest therefore requires first eliminating this dependence on $\mathbf{x}$. This is analogous to the problem of nuisance parameter elimination discussed in Chapter 2. In both cases an unwanted dependence must be removed before a scalar quantity of interest can be defined. Parallel to the options discussed there, two strategies present themselves here as well, namely marginalization and conditioning.

Marginalization in this context involves averaging the conditional probabilities over the joint covariate distribution $F_{\mathbf{X}}$ to obtain the marginal probability vector

$$\bar{\theta}_j(\mathbf{B}) = \mathbb{E}\left[\theta_j(\mathbf{X}; \mathbf{B})\right] = \int_{\mathbb{R}^p} \theta_j(\mathbf{x}; \mathbf{B})\, dF_{\mathbf{X}}(\mathbf{x}), \quad j = 1, \dots, J, \tag{4.3.1}$$

and defining the estimand as a function of $\bar{\boldsymbol{\theta}}(\mathbf{B})$.[4] Conditioning means fixing a reference covariate value $\mathbf{x}_0 \in \mathbb{R}^p$, evaluating the conditional probability vector $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B})$ at that point, and defining the estimand as a function of the resulting length-J probability vector. Rather than summarizing the distribution over the entire covariate space, this approach asks what the category probabilities look like at a single representative covariate value.

The choice between these strategies is dictated by whether the integrated likelihood under the resulting estimand can be approximated in a manner that is computationally tractable. The marginal

[4] Since $\sum_{j=1}^{J} \theta_j(\mathbf{x}; \mathbf{B}) = 1$ for every $\mathbf{x}$, linearity of the integral guarantees that $\sum_{j=1}^{J} \bar{\theta}_j(\mathbf{B}) = 1$, so $\bar{\boldsymbol{\theta}}(\mathbf{B}) \in \Delta^{J-1}$ is well-defined.

probability vector $\bar{\boldsymbol{\theta}}(\mathbf{B})$ depends on $\mathbf{B}$ through a nonlinear average over the covariate distribution. Because the softmax is nonlinear, this average has no closed form. Consequently, for any parameter of interest ψ, the level set $\{\mathbf{B} : \varphi(\bar{\boldsymbol{\theta}}(\mathbf{B})) = \hat{\psi}\}$ has no exploitable geometric structure in $\mathbf{B}$-space: sampling from it requires constrained optimization in $\mathbb{R}^{p(J-1)}$ with no warm-starting or analytic guidance, and maximizing the expected log-likelihood with respect to each sampled vector from $\Omega_{\hat{\psi}}$ requires numerical integration over $F_{\mathbf{X}}$ inside every function and gradient call of the inner optimization. The approximation procedure for an integrated likelihood function in this domain is therefore computationally prohibitive under a marginalization strategy.

The conditional probability vector $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B})$, by contrast, depends on $\mathbf{B}$ only through the single linear combination $\mathbf{B}^\top \mathbf{x}_0 \in \mathbb{R}^{J-1}$, and so the constraint $\varphi(\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B})) = \hat{\psi}$ is determined entirely by the corresponding probability vector $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B}) \in \Delta^{J-1}$. This suggests a reparameterization: rather than sampling from the level set of φ in $\mathbf{B}$-space directly, we sample a probability vector $\boldsymbol{\omega}$ from the level set of φ within Δ^{J-1} and then find a value of $\mathbf{B}$ consistent with $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B}) = \boldsymbol{\omega}$. The level set of φ within Δ^{J-1} is defined purely by the constraint on the probability vector, with no reference to $\mathbf{B}$ or $\mathbf{x}_0$, and so its geometry is identical to that of the corresponding level set in the no-effects model. Whatever sampling strategy is developed for the no-effects model therefore carries over to the fixed effects setting without modification. We also obviate the need to average the conditional class probabilities $\theta_j(\mathbf{x}_i; \mathbf{B})$ over the observed covariate values $\mathbf{x}_1, \dots, \mathbf{x}_n$ when evaluating the branch objective at each point along the ψ grid. Since this average would otherwise be recomputed at every function and gradient call inside the inner optimization, and the inner optimization is itself repeated across hundreds of branches and many ψ grid points, the computational savings are substantial.

Conditioning is not without theoretical cost. The branch objective, when computing the integrated likelihood at a given ψ, enforces the constraint $\varphi(\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B})) = \psi$, which is defined conditionally at $\mathbf{x}_0$. The true parameter value ψ_0, however, is defined as $\varphi(\bar{\boldsymbol{\theta}}(\mathbf{B}_0))$, a function of the marginal probability vector at the data-generating parameter $\mathbf{B}_0$. Since in general $\varphi(\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{B}_0)) \neq \varphi(\bar{\boldsymbol{\theta}}(\mathbf{B}_0))$, the integrated likelihood is evaluated on a scale that does not coincide with the scale on which ψ_0 is defined. The resulting estimator therefore targets a quantity that differs from the true parameter of interest, introducing a form of misspecification bias. Two observations temper this concern. First, the bias affects the profile and integrated likelihoods equally, since both evaluate the likelihood at the same covariate value $\mathbf{x}_0$; any systematic discrepancy between conditional and marginal quantities does not favor one method over the other. Second, the magnitude of the bias depends on how much the conditional probabilities at $\mathbf{x}_0$ deviate from their marginal averages, which in turn depends on the strength of the covariate effects and the variability of $\mathbf{X}$. When effects are modest or the covariate distribution is concentrated, the two quantities will be close. Ultimately, frequency properties — coverage rates, bias, mean squared error — are the appropriate criteria for evaluating estimators, and the simulation evidence presented in Section 4.5 provides the empirical verdict. A method that is theoretically misspecified but delivers superior frequency properties in finite samples is preferable to an unbiased but computationally intractable alternative.

In the random effects model, the probability vector $\boldsymbol{\theta}(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i)$ carries a further dependence on the unobserved cluster-specific random effects $\mathbf{u}_i$, in addition to the covariate dependence already present in the fixed effects model. Eliminating both dependencies is necessary before a scalar parameter of interest can be defined. The same two strategies apply. Marginalizing over the joint distribution of $\mathbf{X}$ and $\mathbf{u}$ yields a population-average probability vector that inherits the intractability of the marginal estimand in the fixed effects case, now compounded by the additional integration over $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$: the level set of any interest function of the resulting average has no exploitable geometric structure in $(\mathbf{A}, \mathbf{B}, \boldsymbol{\Sigma})$-space. Conditioning on $\mathbf{u}_i = \mathbf{0}$ — evaluating at the median cluster — eliminates the random effects dependence while preserving the sphere geometry of the level set, since the resulting probability vector $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{A}, \mathbf{B}, \mathbf{0})$ depends on $(\mathbf{A}, \mathbf{B})$ only through the single linear combination $\mathbf{A} + \mathbf{B}^\top \mathbf{x}_0 \in \mathbb{R}^{J-1}$, by the same argument as in the fixed effects case. The same sampling machinery therefore applies without modification.

For these reasons, all parameters of interest in the fixed effects and random effects settings are defined with respect to the conditional probability vector at the observed covariate mean $\mathbf{x}_0 = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$ and, in the random effects setting, at $\mathbf{u}_i = \mathbf{0}$. The definitions given below are stated in terms of a generic probability vector $\boldsymbol{\theta}$; in no-effects settings this is taken directly as the model parameter, in fixed effects settings it is understood to mean $\boldsymbol{\theta}(\bar{\mathbf{x}}; \mathbf{B})$, and in random effects settings it is understood to mean $\boldsymbol{\theta}(\bar{\mathbf{x}}; \mathbf{A}, \mathbf{B}, \mathbf{0})$.

4.3.1. Shannon entropy

The Shannon entropy of a discrete random variable X with support $\mathcal{X}$ and probability mass function p is given by

$$\mathrm{H}(X) = -\sum_{x\in\mathcal{X}} p(x)\log p(x), \tag{4.3.2}$$

with the convention that $0\log 0 = 0$. In general, $\mathrm{H}(X) \geq 0$, with equality if and only if $p(x^*) = 1$ for some $x^* \in \mathcal{X}$ and $p(x) = 0$ for all $x \neq x^*$. If $\mathcal{X}$ is finite with cardinality $J = |\mathcal{X}|$, then $\mathrm{H}(X) \leq J$, with equality if and only if $p(x) = 1/J$ for all $x \in \mathcal{X}$. Thus, the distribution's entropy is minimized when all its probability mass is concentrated on a single outcome and maximized when it is distributed uniformly across all possible outcomes. This property makes entropy a natural measure of the concentration or dispersion of probability mass over a finite set of categories. In applied settings, entropy is commonly used as an index of diversity, for example to quantify species heterogeneity in ecological communities. In the present setting, we are interested in the entropy associated with a multinomial model, or more precisely, the entropy of a single categorical trial underlying the multinomial construction in Equation 4.1.3.

Let $\mathbf{Y} \sim \mathrm{Categorical}(\boldsymbol{\theta})$ denote the outcome of one trial, with its distribution being as in Equation 4.1.2. Then the entropy of $\mathbf{Y}$ is given by

$$\begin{aligned}\mathrm{H}(\mathbf{Y}) &= -\sum_{\mathbf{y}\in\{e_1,\ldots,e_J\}} \mathbb{P}(\mathbf{Y}=\mathbf{y})\,\log\Big[\mathbb{P}(\mathbf{Y}=\mathbf{y})\Big] \\ &= -\sum_{j=1}^{J}\mathbb{P}(\mathbf{Y}=\mathbf{e}_j)\,\log\Big[\mathbb{P}(\mathbf{Y}=\mathbf{e}_j)\Big] \\ &= -\sum_{j=1}^{J}\theta_j\log\theta_j.\end{aligned}\tag{4.3.3}$$

It follows that our interest function for the entropy of a multinomial distribution with parameter $\boldsymbol{\theta}$ should be

$$\varphi(\boldsymbol{\theta}) = -\sum_{j=1}^{J}\theta_j\log\theta_j, \tag{4.3.4}$$

and so we may take our parameter of interest to be $\psi = \varphi(\boldsymbol{\theta})$. Note that while we may view the data as entering the model through the multinomial likelihood for $\boldsymbol{\theta}$, it is more appropriate to conceive of the entropy as being a property of the corresponding categorical distribution.

4.3.2. Simpson's diversity index

The index of diversity introduced by Simpson (1949) provides an alternative measure of the concentration of mass in a probability vector. For a vector $\boldsymbol{\theta} \in \Delta^{J-1}$, Simpson's index is defined as

$$D(\boldsymbol{\theta}) = \sum_{j=1}^{J} \theta_j^2 = \|\boldsymbol{\theta}\|^2. \tag{4.3.5}$$

Note that $1/J \leq D(\boldsymbol{\theta}) < 1$, with larger values indicating greater concentration of probability mass and smaller values reflecting greater diversity. The lower bound is attained at the uniform distribution $\theta_j = 1/J$ for all j, which is an interior point of Δ^{J-1}. The upper bound of 1 is never attained because it would require all mass to concentrate on a single category, forcing $\boldsymbol{\theta}$ to a vertex of the simplex, which is excluded by the open simplex assumption.

Unlike Shannon entropy, which depends logarithmically on the cell probabilities, Simpson's index is a quadratic function of $\boldsymbol{\theta}$. As a result, it places greater relative weight on large components of the probability vector and is less sensitive to the presence of categories with very small probability of observation. This property makes Simpson's index more robust to settings where rare categories are difficult to estimate reliably.

The statistical interpretation of Simpson's index is relatively straightforward: if two observations are drawn independently from a categorical distribution with probability vector $\boldsymbol{\theta}$, then $D(\boldsymbol{\theta})$ is the probability that the two observations fall in the same category. This makes $D(\boldsymbol{\theta})$ a natural scalar summary of the distribution, and thus

$$\psi = D(\boldsymbol{\theta}) \tag{4.3.6}$$

constitutes a sensible quantity for us to take as a parameter of interest. Note as well that the complementary quantity $1 - D(\boldsymbol{\theta})$ represents the probability of categorical disagreement between the two observations.

$D(\boldsymbol{\theta})$ is also invariant under permutations of the category labels. That is, if S_J denotes the symmetric group of permutations of the set $\{1, ..., J\}$, then for any $\pi \in S_J$,

$$D(\theta_{\pi(1)}, \dots, \theta_{\pi(J)}) = \sum_{i=1}^{J} \theta_{\pi(i)}^2 = \sum_{i=1}^{J} \theta_i^2 = D(\theta_1, \dots, \theta_J).$$

Hence, like Shannon entropy, Simpson's index measures the concentration of probability mass without regard to which specific categories carry that mass. The same symmetry carries over to the level set $\Omega_{\hat{\psi}}$: if $\boldsymbol{\omega} \in \Omega_{\hat{\psi}}$, then so is every permutation of its coordinates. This leads to a rich geometric interpretation and has direct implications for how elements of $\Omega_{\hat{\psi}}$ should be sampled. We summarize the key results here — see Appendix C for the full derivation.

Since $D(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|^2$, $\Omega_{\hat{\psi}}$ is the intersection of the open simplex Δ^{J-1} with a hypersphere of radius $\sqrt{\hat{\psi}}$ centered at the origin. Every point in this intersection lies at a common distance

$$r(\hat{\psi}, J) = \sqrt{\hat{\psi} - \tfrac{1}{J}} \tag{4.3.7}$$

from the centroid $\mathbf{c} = (1/J)\mathbf{1}$, so $\Omega_{\hat{\psi}}$ is a $(J-2)$-sphere of radius $r(\hat{\psi}, J)$ centered at $\mathbf{c}$ within the affine hyperplane $\Pi = \{\mathbf{v} \in \mathbb{R}^J : \sum_j v_j = 1\}$. Its shape depends on whether $r(\hat{\psi}, J)$ exceeds the inradius of Δ^{J-1}, which equals $1/\sqrt{J(J-1)}$. When $\hat{\psi} < 1/(J-1)$, the sphere lies entirely within Δ^{J-1} and $\Omega_{\hat{\psi}}$ is connected. When $\hat{\psi} \geq 1/(J-1)$, the sphere clips through all J facets of the simplex simultaneously, and $\Omega_{\hat{\psi}}$ breaks into J disjoint spherical caps, one near each vertex $\mathbf{e}_j$. We refer to these as the connected and disconnected regimes, respectively. The permutation symmetry of D under S_J ensures that all J caps are geometrically identical, related to one another by permutation of category labels.

When $\hat{\psi} < 1/(J-1)$, sampling from $\Omega_{\hat{\psi}}$ reduces to sampling uniformly from the full $(J-2)$-sphere of radius $r(\hat{\psi}, J)$ centered at $\mathbf{c}$ within Π. This is achieved by drawing $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$, projecting onto $\mathbf{1}^{\perp}$

by subtracting the mean,

$$\mathbf{z}' = \mathbf{z} - \bar{z}\mathbf{1},$$

and setting $\mathbf{v} = \mathbf{c} + r \cdot \mathbf{z}'/\|\mathbf{z}'\|$. The resulting draw is exactly uniform on $\Omega_{\hat{\psi}}$.

When $\hat{\psi} \geq 1/(J-1)$, each cap $\mathcal{C}_j$ is parametrized by a height coordinate $h \in [\cos\alpha, 1]$ along its axis and a directional component $\mathbf{u}$ within the subspace of $\mathbf{1}^\perp$ orthogonal to the cap axis $\mathbf{n}_j$, where $\alpha = \alpha(\hat{\psi}, J)$ is the angular radius of each cap (see Equation C.1.13 in Appendix C). A uniform draw from $\mathcal{C}_j$ is obtained by drawing h via truncated Beta inversion and $\mathbf{u}$ via Gaussian projection, then forming $\mathbf{v} = \mathbf{c} + r(h\,\mathbf{n}_j + \sqrt{1-h^2}\,\mathbf{u})$.

Rather than restricting draws to the dominant cap, we select cap j on each draw with probability proportional to the empirical category frequency $\hat{p}_j = N_j/n$, restricting to categories with above-average frequency $\hat{p}_j > 1/J$. This concentrates sampling on the caps whose dominant coordinates are most consistent with the observed data while preserving coverage of secondary caps and recovering uniform selection when the counts are approximately balanced.

4.3.3. Category-invariant effect

4.4. Profile and integrated likelihoods

This section specializes the general framework of Chapter 3 to each combination of model and estimand considered in this chapter. For each combination, we describe the level set geometry, the branch objective, and the approximation procedure used to construct the profile and integrated likelihood curves. The profile likelihood is treated throughout as the special case of the branch framework in which $\omega = \hat{\omega}_{\text{MLE}}$ — the point on $\Omega_{\hat{\psi}}$ induced by the unrestricted MLE — and serves as the primary comparator for the integrated likelihood in the simulation study of Section 4.5.

4.4.1. No-effects model

In the no-effects model ω and θ inhabit the same space Δ^{J-1}, so the level set geometry established in Section 4.3 applies directly. The branch objective reduces to the observed log-likelihood: since there are no regression coefficients to adjust, the expected log-likelihood under ω is maximized at $\theta = \omega$ itself, and no rank-one construction is needed. The profile likelihood is the special case $\omega = \hat{\theta}$, recovering the constrained MLE at each ψ. The integrated likelihood is the Monte Carlo average of branch values over draws from $\Omega_{\hat{\psi}}$ using the sampling procedure described in Section 4.3.

4.4.2. Fixed effects model

The fixed effects model introduces the two-space problem described in Section 3.5 of Chapter 3: ω is a probability vector in Δ^{J-1} while the model parameter $(\mathbf{A}, \mathbf{B})$ lives in $\mathbb{R}^{(p+1)(J-1)}$. For each $\omega \in \Omega_{\hat{\psi}}$, a unique coefficient pair $(\hat{\mathbf{A}}_{\omega}, \hat{\mathbf{B}}_{\omega})$ is recovered via a rank-one adjustment to the MLE that matches the linear predictor at $\mathbf{x}_0$ to the logit vector implied by ω while leaving the regression structure unchanged in all directions orthogonal to $\mathbf{x}_0$. The branch objective is the expected log-likelihood $E_{\omega}[\ell(\mathbf{A}, \mathbf{B})]$, which in the fixed effects case has a closed form — the sum of cross-entropies between the fitted probabilities under $(\hat{\mathbf{A}}_{\omega}, \hat{\mathbf{B}}_{\omega})$ and the log probabilities under $(\mathbf{A}, \mathbf{B})$ evaluated at each observation — and is maximized subject to $\varphi(\mathbf{A}, \mathbf{B}) = \psi$ by a constrained optimizer initialized from the MLE. The branch value is the observed log-likelihood $\ell(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega))$ evaluated at the branch mode. The level set geometry and sampling procedure are identical to those of the no-effects model.

4.4.3. Random effects model

The random effects model inherits the two-space structure of the fixed effects model and adds a further layer of complexity through the unobserved random effects $\mathbf{u}_i$. The four subsubsections below establish the estimand geometry, the branch objective, and the two Laplace approximations needed to evaluate it.

4.4.3.1. Estimand and level set geometry. The estimand is defined at the reference covariate value $\mathbf{x}_0 = \bar{\mathbf{x}}$ and at $\mathbf{u}_i = \mathbf{0}$, so the relevant probability vector is $\boldsymbol{\theta}(\mathbf{x}_0; \mathbf{A}, \mathbf{B}, \mathbf{0})$, which depends on $(\mathbf{A}, \mathbf{B})$ only through the linear combination $\mathbf{A} + \mathbf{B}^{\top}\mathbf{x}_0$. The level set $\Omega_{\hat{\psi}}$ is therefore a subset of Δ^{J-1} with the same geometry as in the no-effects and fixed effects models, and the same sampling procedure applies. The rank-one adjustment maps each $\omega \in \Omega_{\hat{\psi}}$ to a unique $(\hat{\mathbf{A}}_{\omega}, \hat{\mathbf{B}}_{\omega})$ satisfying $\boldsymbol{\theta}(\mathbf{x}_0; \hat{\mathbf{A}}_{\omega}, \hat{\mathbf{B}}_{\omega}, \mathbf{0}) = \omega$ exactly.

4.4.3.2. Branch objective. As discussed in Section 4.3, the branch objective in the random effects model is the expected complete-data log-likelihood $E_\omega[\ell_{\mathrm{c}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})]$ rather than the marginal log-likelihood ℓ. The reason is algebraic: ℓ_{c} is linear in y_{itj}, which allows soft responses to substitute cleanly for observed responses and gives rise to the zero-score-expectation property the integrated likelihood requires. The marginal log-likelihood does not share this linearity since the log sits outside the integral over $\mathbf{u}_i$.

The covariance matrix $\mathbf{\Sigma}$ may either be held fixed at its estimated value $\hat{\mathbf{\Sigma}}$ — treating it as a hyperparameter rather than a nuisance parameter in the IL sense — or jointly optimized alongside $(\mathbf{A}, \mathbf{B})$ subject to the same constraint. The former approach is computationally cheaper and treats $\mathbf{\Sigma}$ as profiled out by substitution; the latter treats it as a genuine nuisance parameter profiled out by optimization at each ω, consistent with the general framework of Equation 3.2.6.

Once the branch mode $(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega))$ has been located by maximizing $E_\omega[\ell_{\mathrm{c}}^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})]$ subject to the constraint $\varphi(\mathbf{A}, \mathbf{B}) = \psi$, the Laplace log-likelihood ℓ^{LA} is evaluated at $(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega), \mathbf{\Sigma})$ to form the branch value that enters the integrated likelihood.

4.4.3.3. Laplace approximation of the marginal log-likelihood. Closed-form evaluation of the integrals in Equation 4.2.15 is not available due to the nonlinearity of the softmax in $\mathbf{u}_i$. We therefore approximate each cluster's contribution separately using Laplace's method (Appendix A.2.1), and define the resulting approximation as the *Laplace log-likelihood* $\ell^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma}) = \sum_{i=1}^n \log L_i^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})$, where each cluster contribution is obtained as follows.

Writing the ith factor of Equation 4.2.13 as

$$L_i(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma}) = \int_{\mathbb{R}^{J-1}} \left[\prod_{t=1}^{m_i} \prod_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}_i)^{y_{itj}} \right] \phi(\mathbf{u}_i; \mathbf{0}, \mathbf{\Sigma})\, d\mathbf{u}_i, \tag{4.4.1}$$

the Laplace approximation to $\log L_i$ is

$$\log L_i^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma}) = g_i(\hat{\mathbf{u}}_i) - \tfrac{1}{2} \log |\mathbf{\Sigma}| - \tfrac{1}{2} \log \left| \textstyle\sum_t \mathbf{W}_t(\hat{\mathbf{u}}_i) + \mathbf{\Sigma}^{-1} \right|, \tag{4.4.2}$$

where

$$g_i(\mathbf{u}) = \sum_{t=1}^{m_i} \sum_{j=1}^{J} y_{itj} \log \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}) - \tfrac{1}{2}\mathbf{u}^\top \mathbf{\Sigma}^{-1} \mathbf{u}$$

is the log-integrand of Equation 4.4.1 — equivalently, the cluster i contribution to ℓ_c minus the log-normalizing constant of ϕ — $\hat{\mathbf{u}}_i = \operatorname{argmax}_{\mathbf{u}}\, g_i(\mathbf{u})$ is the posterior mode of $\mathbf{u}_i$ given cluster i's observations, $\mathbf{A}$, and $\mathbf{B}$, and

$$\mathbf{W}_t = \operatorname{diag}(\boldsymbol{\theta}_t) - \boldsymbol{\theta}_t \boldsymbol{\theta}_t^\top, \quad \boldsymbol{\theta}_t = \big(\theta_2(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \hat{\mathbf{u}}_i), \dots, \theta_J(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \hat{\mathbf{u}}_i)\big)^\top$$

is the multinomial covariance matrix of the non-baseline fitted probabilities at the posterior mode. The Laplace approximation therefore approximates the log of the marginalised cluster likelihood — the log-outside-the-integral quantity from Equation 4.2.15 — by evaluating g_i, the log-inside-the-integral quantity, at its mode and correcting for curvature. The approximation is accurate when the posterior of $\mathbf{u}_i$ given $\mathbf{y}_i$, $\mathbf{A}$, and $\mathbf{B}$ is close to Gaussian. This condition holds when m_i is moderate but may fail for the small cluster sizes ($m = 3$, $m = 5$) in the simulation study.

4.4.3.4. Laplace approximation of the expected complete-data log-likelihood. The branch objective requires evaluating $E_\omega[\ell_c(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})]$, the expectation of the complete-data log-likelihood over the joint distribution of $(\mathbf{y}, \mathbf{u})$ induced by ω. Under this distribution the conditional expectation of y_{itj} given $\mathbf{u}_i$ is the soft response

$$q_{itj}(\hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \mathbf{u}_i) = \mathbb{E}_\omega[y_{itj} \mid \mathbf{u}_i] = \theta_j(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \mathbf{u}_i),$$

where $(\hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega)$ is the coefficient pair recovered via the rank-one adjustment. Replacing y_{itj} with q_{itj} in Equation 4.2.14 and integrating over the distribution of $\mathbf{u}$ gives

$$\mathbb{E}_\omega[\ell_c(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})] = \sum_{i=1}^{n} \int_{\mathbb{R}^{J-1}} \underbrace{\left[\sum_{t=1}^{m_i} \sum_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \mathbf{u}) \log \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}) \right]}_{= \mathbb{E}_\omega[\ell_{c,i} \mid \mathbf{u}]} \phi(\mathbf{u}; \mathbf{0}, \mathbf{\Sigma})\, d\mathbf{u}. \tag{4.4.3}$$

No closed form exists for the integrals in Equation 4.4.3 because $\theta_j(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \mathbf{u})$ and $\log \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u})$ both depend on $\mathbf{u}$ through the softmax nonlinearity in different directions simultaneously.

Writing $\phi(\mathbf{u}; \mathbf{0}, \mathbf{\Sigma}) \propto \exp\{-\frac{1}{2}\mathbf{u}^\top \mathbf{\Sigma}^{-1}\mathbf{u}\}$ and absorbing the normalizing constant, the ith integral in Equation 4.4.3 has the form $\int \exp\{g_i^*(\mathbf{u})\}\, d\mathbf{u}$ with the cluster objective

$$g_i^*(\mathbf{u}) = \sum_{t=1}^{m_i} \sum_{j=1}^{J} \theta_j(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \mathbf{u}) \log \theta_j(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \mathbf{u}) - \tfrac{1}{2}\mathbf{u}^\top \mathbf{\Sigma}^{-1}\mathbf{u}. \tag{4.4.4}$$

Let $\hat{\mathbf{u}}_i^* = \operatorname{argmax}_{\mathbf{u}}\, g_i^*(\mathbf{u})$ denote the mode of the integrand. Applying the Laplace approximation (Appendix A.2) with $d = J-1$ and canceling the $(2\pi)^{(J-1)/2}$ factors from the normalizing constant of ϕ gives the Laplace approximation to the ith cluster contribution,

$$\int_{\mathbb{R}^{J-1}} \mathbb{E}_\omega[\ell_{\mathrm{c},i} \mid \mathbf{u}]\, \phi(\mathbf{u}; \mathbf{0}, \mathbf{\Sigma})\, d\mathbf{u} \approx g_i^*(\hat{\mathbf{u}}_i^*) - \tfrac{1}{2}\log|\mathbf{\Sigma}| - \tfrac{1}{2}\log\left|-\nabla^2 g_i^*(\hat{\mathbf{u}}_i^*)\right|. \tag{4.4.5}$$

Summing over clusters, the full Laplace-approximated expected complete-data log-likelihood is

$$E_\omega[\ell_{\mathrm{c}}^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})] = \sum_{i=1}^{n} \left[g_i^*(\hat{\mathbf{u}}_i^*) - \tfrac{1}{2}\log|\mathbf{\Sigma}| - \tfrac{1}{2}\log\left|-\nabla^2 g_i^*(\hat{\mathbf{u}}_i^*)\right| \right]. \tag{4.4.6}$$

Applying the Laplace gradient approximation (Appendix A.2.1) with $d\hat{\mathbf{u}}_i^*/d\mathbf{B} = \mathbf{0}$, the gradient of Equation 4.4.6 with respect to $\mathbf{B}$ reduces to

$$\nabla_{\mathbf{B}}\, E_\omega[\ell_{\mathrm{c}}^{\mathrm{LA}}(\mathbf{A}, \mathbf{B}, \mathbf{\Sigma})] \approx \sum_{i=1}^{n} \sum_{t=1}^{m_i} \tilde{\mathbf{x}}_{it} \big(\mathbf{q}_{it}(\hat{\mathbf{u}}_i^*) - \mathbf{p}_{it}(\hat{\mathbf{u}}_i^*)\big)^\top, \tag{4.4.7}$$

where

$$\mathbf{q}_{it} = (\theta_2(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \hat{\mathbf{u}}_i^*), \ldots, \theta_J(\mathbf{x}_{it}; \hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega, \hat{\mathbf{u}}_i^*))^\top$$

are soft responses and

$$\mathbf{p}_{it} = (\theta_2(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \hat{\mathbf{u}}_i^*), \ldots, \theta_J(\mathbf{x}_{it}; \mathbf{A}, \mathbf{B}, \hat{\mathbf{u}}_i^*))^\top$$

are fitted probabilities, both evaluated at $\hat{\mathbf{u}}_i^*$. The ZSE property is immediate from Equation 4.4.7: when $(\mathbf{A}, \mathbf{B}) = (\hat{\mathbf{A}}_\omega, \hat{\mathbf{B}}_\omega)$, we have $\mathbf{p}_{it} = \mathbf{q}_{it}$ at every observation and every $\mathbf{u}$, so the gradient vanishes identically. The branch mode $(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega))$ therefore satisfies

$$E_\omega\left[\nabla_{\mathbf{B}}\, \ell_c(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega), \boldsymbol{\Sigma})\right] = \mathbf{0}, \tag{4.4.8}$$

which is Severini's zero-score-expectation condition. Once the branch mode has been located by maximizing Equation 4.4.6 subject to the constraint, the Laplace log-likelihood ℓ^{LA} from Equation A.2.4 is evaluated at $(\mathbf{A}^*(\omega), \mathbf{B}^*(\omega), \boldsymbol{\Sigma})$ to form the branch value that enters the integrated likelihood.

4.5. Simulation evidence

The simulation evidence presented in this section evaluates the frequency properties of point and interval estimators derived from the profile and integrated likelihoods across seven studies, one for each combination of parameter of interest and model described earlier in this chapter. We first describe the elements common to all studies, then present the design and results for each combination in turn.

4.5.1. Simulation design

Each study employs a fully crossed factorial design, with factors chosen to span dimensions of the inferential problem along which the integrated and profile likelihoods are expected to differ. The specific factors and their levels vary across studies and are described in the corresponding subsections below. We simulate each design point 1,000 times, generating a fresh dataset from the true parameter value in each iteration, and approximate the integrated likelihood within each iteration using a Monte Carlo scheme with R replications, with R specified for each study below.

A consideration shared across all designs concerns categories that are modeled but absent from the generated data. In the baseline-category logit model with no effects, where the sufficient statistics are the category counts $(N_1, \dots, N_J)$, we correct for such absences by adding 1/2 to each count following Agresti (2002). In the baseline-category logit model with fixed effects, where the likelihood is expressed in terms of individual observations rather than aggregated counts, we instead add a single pseudo-observation to each absent category, setting the response to that category and all covariates to their observed column means for quantitative predictors and their observed modal values for categorical predictors. This places each pseudo-observation at the centroid of the observed covariate distribution, minimizing its leverage on the estimated regression coefficients while ensuring that all modeled categories are represented in the likelihood.

We also note the following concern regarding the quality of the Monte Carlo approximation to the integrated likelihood in settings where ψ has a natural boundary. In such settings, draws from $\Omega_{\hat{\psi}}$ near that boundary consistently failed to produce interior optima during the constrained optimization, instead converging to the boundary of the parameter space. Such boundary solutions indicate optimizer failure rather than genuine modes of the branch function, and including them in the Monte Carlo average would introduce approximation error not present in settings where sampling from $\Omega_{\hat{\psi}}$ is exact. To avoid this, we require that any draw from $\Omega_{\hat{\psi}}$ which results in a branch mode being located directly on a boundary of the ψ domain be excluded from the Monte Carlo approximation. We enforce this restriction at the time of the draw, using a quick optimizer to locate the position of the branch mode.

4.5.2. Shannon entropy

4.5.2.1. Baseline-category logit model with no effects.

4.5.2.2. Baseline-category logit model with fixed effects.

4.5.2.3. Baseline-category logit model with random effects.

4.5.3. Simpson's diversity index

4.5.3.1. Baseline-category logit model with no effects. The simulation study for Simpson's index employs the common factorial structure described above, with the true parameter value given by $\psi_0 = 1/J + f(1 - 1/J)$. The 48 resulting design points are listed in Table 4.1. The f factor covers low ($f = 0.10$), intermediate ($f = 0.30$ and 0.50), and high ($f = 0.70$) concentrations of the marginal probability distribution. As f approaches 1, nearly all probability mass concentrates in a single category, making zero-count cells increasingly likely. We cap f at 0.70 to avoid the most severe of these cases, noting that such extreme concentration is uncommon in practice.

4.5.3.2. Baseline-category logit model with fixed effects.

4.5.3.3. Baseline-category logit model with random effects.

4.5.4. Category-invariant effect

4.5.4.1. Baseline-category logit model with fixed effects.

4.5.5. Simulation results

4.5.5.1. Shannon entropy.

4.5.5.2. Simpson's diversity index.

4.5.5.3. Category-invariant effect.

4.6. Application

4.7. Summary and discussion

Table 4.1. SIMULATION DESIGN FOR SIMPSON'S INDEX UNDER BASELINE-CATEGORY LOGIT MODEL

			ψ_0			
J	n/J	n	$f = 0.10$	$f = 0.30$	$f = 0.50$	$f = 0.70$
	4	20				
5	6	30	0.28	0.44	0.6	0.76
	8	40				
	4	40				
10	6	60	0.19	0.37	0.55	0.73
	8	80				
	4	60				
15	6	90	0.16	0.347	0.533	0.72
	8	120				
	4	80				
20	6	120	0.145	0.335	0.525	0.715
	8	160				

Note: This table displays the design of the simulation study to investigate the frequency properties of point and interval estimators derived from the profile and the integrated likelihood functions for Simpson's index under the baseline-category logit model. The study employs a fully crossed $3 \times 4 \times 4$ factorial design over the sample size per category n/J, the number of categories J, and the fractional parameter value f, where $\psi_0 = 1/J + f(1 - 1/J)$. Each of the 48 design points is replicated 1,000 times.

CHAPTER 5

Poisson Dyads

5.1. Introduction

This chapter examines the utility of the integrated likelihood function as a tool for estimating parameters of interest in dyads with models of data generated from count distributions. In each dyad we investigate, we will compare and contrast the performances of the point and interval estimators produced by the integrated likelihood (or more commonly, our approximations to the integrated likelihood) for the dyad's parameter of interest to the analogous estimators produced by the profile likelihood function. Our use of the profile likelihood stems from its conceptual simplicity and ease of computation, making it a convenient choice for an estimation benchmark.

5.2. Models

To motivate our discussion, let us imagine a hypothetical scenario to which we will return at various points throughout this section for context. Suppose there has been an outbreak of disease among a group of hospitals, and we have been tasked with assessing its severity. A basic question to ask is, "What has been the overall rate of infection across all affected hospitals since the start of the outbreak?" A seemingly straightforward answer is the simple ratio of total infections to total patient-days reported across all the hospitals since the start of the outbreak.[1] We must tread with some caution here, however. The number of infections is clearly random, but what about the patient-days? The properties of our estimator hinge upon the answer so it would be prudent to give it some consideration.

To interpret the number of patient-days as a random variable is to declare that what really matters is not the rate of infection among hospitals in this outbreak in particular but rather the more abstract

[1]A patient-day is a single day spent by a patient in a hospital.

notion of the disease's "true" rate of infection in general. This would mean that both the numerator and denominator in our ratio are random variables, transforming it from an "estimator which happens to be a ratio" into an actual "ratio estimator". Conversely, to treat them as fixed parameters is to condition on their exact values reported by each hospital in the group, thereby restricting our attention to the rate of infection among these hospitals only. Neither choice is inherently right or wrong - it is up to us to decide where our interest lies - but once we have decided, we must honor the consequences of that choice in our analysis. Failure to do so would lead to bias in our estimates. We will begin with the assumption that the point of our analysis is to gauge the risk involved with this outbreak specifically, so until such time as it is explicitly noted otherwise, it is safe to regard the patient-days that each hospital reports as being fixed parameters.

The overall rate of infection among only the hospitals in our group is therefore what constitutes our parameter of interest. As usual we will call this value ψ and its estimator $\hat{\psi}$. In practice, especially for an example such as this, the line between ψ and $\hat{\psi}$ often gets blurred, with many treating the observed rate as though it is the same as the true underlying rate. From the perspective of assessing impact, this is understandable as the observed rate is by definition what the hospitals actually experience. However, for the sake of avoiding conceptual ambiguity in our language and intent, we will continue to treat them as two distinct entities. If nothing else, it serves as a friendly reminder of the importance of maintaining proper separation between estimand and estimator.

Assume there are J hospitals in the group, and let Y_j and t_j represent the number of infections and patient-days, respectively, reported by hospital j since the start of the outbreak. Then our estimator may be written as

$$
\begin{aligned}
\hat{\psi} &= \frac{\sum_{j=1}^{J} Y_j}{\sum_{j=1}^{J} t_j} \\
&= \frac{\sum_{j=1}^{J} t_j (Y_j/t_j)}{\sum_{j=1}^{J} t_j} \\
&= \frac{\sum_{j=1}^{J} t_j \hat{\theta}_j}{\sum_{j=1}^{J} t_j} \\
&= \sum_{j=1}^{J} t'_j \hat{\theta}_j,
\end{aligned}
\tag{5.2.1}
$$

where $t'_j = \frac{t_j}{\sum_{j=1}^{J} t_j}$ and $\hat{\theta}_j = Y_j/t_j$ is the observed rate of infection at hospital j. If θ_j represents its associated estimand, we can similarly rewrite our expression for ψ in a similar form as

$$
\psi = \sum_{j=1}^{J} t'_j \theta_j. \tag{5.2.2}
$$

Thus, the overall rate of infection can be decomposed as the weighted *average* of the individual hospitals' rates of infection. Note that the term "average" is proper here since $\sum_{j=1}^{J} t'_j = 1$. This follows directly from our requirement that each weight t'_j be proportional to the number of patient-days contributed by hospital j to the total number of patient-days across all the hospitals since the outbreak began. We made this restriction as a consequence of the question that motivated our analysis, but it was entirely arbitrary. There is nothing actually requiring the weights to sum to one, or indeed that they depend on the patient-days at all. We could just as easily have asked about an aggregated infection rate relative to some other hospital-specific metric (e.g. number of available beds, population of surrounding region, etc.) that would in turn inform the precise values of the weights we choose for our parameter of interest.

And of course the utility of a weighted sum as a parameter of interest is not unique to our example of evaluating the severity of a disease outbreak in a group of hospitals either. In general, we may consider any set of J independent count-generating processes with distinct rates (countable events per unit time) of $\theta_1, ..., \theta_J$ and corresponding weights $\alpha_1, ..., \alpha_J$ as inducing a parameter of interest of the form

$$\psi = \sum_{j=1}^{J} \alpha_j \theta_j. \tag{5.2.3}$$

It is not necessary to assume that the weights are normalized, though we will still require that $\alpha_j > 0$ for all j in order to retain the interpretation of ψ as being a weighted *sum* of individual rates.

In the following sections, we will assess the ability of the integrated likelihood to produce accurate estimates for ψ under three separate model frameworks. The models we will consider are (i) a naive one in which we estimate the rate in each group using only the group's observed count and exposure time, (ii) a fixed effects regression model in which we assume there is an underlying covariate structure influencing the observed counts in each group, and (iii) a hierarchical mixed effects regression model in which we assume some or all of the intercepts and slopes from the previous model are random effects instead of fixed. We will also take into account the robustness of the pseudolikelihood-derived estimators to model misspecification when evaluating their performance.

5.2.1. Simple rates model

We begin with a model characterized by its sole use of a process's observed count of events over some recorded interval of time to estimate its associated rate. This is perhaps the simplest nontrivial model one can employ for our parameter of interest, so we will refer to it as the *naive rates model*.

Let Y_j denote the count of events that we record from the j-th process during an exposure time t_j (the patient-days in our outbreak example). We will proceed for the moment under the assumption that the mean of the random variable underlying each count-generating process is roughly equal to its variance, meaning there is no risk of overdispersion. It is therefore safe to assume that the counts follow a Poisson distribution, i.e.

$$Y_j \sim \text{Poisson}(t_j \theta_j).$$

The log-likelihood for θ then becomes

$$\ell(\theta; Y) = \sum_{j=1}^{J} (Y_j \log \theta_j - t_j \theta_j),$$

where we have discarded any additive terms not depending on θ. The unconstrained MLE for θ_j is given by $\hat{\theta}_j = Y_j/t_j$, with the corresponding MLE for ψ being obtained by plugging these estimates into Equation 5.2.3, yielding

$$\hat{\psi} = \sum_{j=1}^{J} \alpha_j \hat{\theta}_j. \tag{5.2.4}$$

Since ψ is a function of the full model parameter, we can employ the ZSE parameterization method for an implicit nuisance parameter introduced in Severini (2018) to approximate the integrated likelihood. Recall that elements of the subspace $\Omega_{\hat{\psi}} = \{\omega \in \Theta : \alpha^\top \omega = \hat{\psi}\}$ play the role of the ZSE parameter in models for which the nuisance parameter is implicit. Let $Q(\theta; \omega)$ represent the expected value of $\ell(\theta; Y)$ under the probability distribution indexed by an element $\omega \in \Omega_{\hat{\psi}}$. That is,

$$\begin{aligned}
Q(\theta; \omega) &= \mathrm{E}_\omega[\ell(\theta; Y)] \\
&= \mathrm{E}_\omega\left[\sum_{g=1}^{G} (Y_g \log \theta_g - t_g \theta_g)\right] \\
&= \sum_{g=1}^{G} (\mathrm{E}_{\omega_g}[Y_g] \log \theta_g - t_g \theta_g) \\
&= \sum_{g=1}^{G} (t_g \omega_g \log \theta_g - t_g \theta_g).
\end{aligned}$$

For a given value of ψ, the ZSE-constrained optimizer is

$$\tilde{\theta}(\psi; \omega) = \arg\max_{\theta} Q(\theta; \omega) \;\text{ s.t. }\; \alpha^\top \theta = \psi.$$

Evaluating $L(\theta; Y) = \exp(\ell(\theta; Y))$ at $\theta = \tilde{\theta}(\psi; \omega)$ gives us a mapping from ψ to its associated likelihood value under the distribution indexed by ω. If we integrate this mapping over $\Omega_{\hat{\psi}}$ with respect to some weight function $\pi(\omega)$ that doesn't depend on ψ, we can obtain the integrated likelihood function for ψ based on the ZSE parameterization, i.e.

$$\bar{L}(\psi) = \int_{\Omega_{\hat{\psi}}} L(\tilde{\theta}(\psi; \omega))\pi(\omega) d\omega.$$

Unsurprisingly, no closed-form solution exists for this integral. Instead, we can use Monte Carlo integration to approximate it with the following estimator:

$$\hat{\bar{L}}(\psi) = \frac{1}{R} \sum_{j=1}^{R} L(\tilde{\theta}(\omega^{(j)}; \psi); Y),$$

where the choice of $\pi(\omega)$ is now understood to be defined implicitly as the density function admitted by the probability measure $d\Pi(\omega)$ that governs how each $\omega^{(j)}$ is drawn from $\Omega_{\hat{\psi}}$. As long as the selection does not depend on the value of ψ, we are essentially free to choose any (non-degenerate) probability distribution we wish, and the form of $\pi(\omega)$ will be adjusted accordingly.

Hence, under a simple MC approximation scheme, the functions $L(\tilde{\theta}(\omega^{(j)}; \psi); Y)$ for $j = 1, ..., R$ are individual "branches" of the integrated likelihood that we average to obtain an estimate of its true underlying value at a particular value of ψ. Let

$$b_{\omega}(\psi) = \log L(\tilde{\theta}(\psi; \omega^{(j)}); Y)$$

denote the branch curve generated by the value $\omega \in \Omega_{\hat{\psi}}$, and let its mode be given by

$$\psi^*_{\omega} = \arg\max_{\psi} b_{\omega}(\psi).$$

For an arbitrary dyad, two distinct values $\omega^{(1)}, \omega^{(2)} \in \Omega_{\hat{\psi}}$ will in general produce branch curves with modes $\psi^*_{\omega^{(1)}}$ and $\psi^*_{\omega^{(2)}}$, respectively, that are also distinct. In other words, there is no reason to expect that branch curves corresponding to different values drawn from $\Omega_{\hat{\psi}}$ will be maximized at the same location. However, if our dyad is such that the model belongs to the exponential family and our parameter of interest ψ is a linear function of the full model parameter, then this is no longer case. Indeed, it can be shown that under these two conditions, the branch curves of the integrated likelihood function for ψ will all have the same mode regardless of the values of ω that generated them, and furthermore that mode will simply be the unconstrained MLE for ψ, given by $\hat{\psi} = \alpha^{\top}\hat{\theta}$ (see appendix:A for a proof). Therefore our MC approximation to the integrated likelihood for ψ, being nothing more a simple average of these branches at each value of ψ, must also be maximized at $\hat{\psi}$.

Of course, by definition $\hat{\psi}$ is also the maximizer of the profile likelihood, meaning that the integrated likelihood and profile likelihood will produce identical maximum likelihood estimates for ψ in any dyad satisfying these criteria. We can further use a quadratic expansion of an arbitrary branch curve to show that the branch's curvature around its peak at $\hat{\psi}$ will be equal to that of the profile likelihood up to a second-order approximation (again, see appendix:A for a proof). It follows that the behavior of our MC approximation to the integrated likelihood for ψ, itself being nothing more than a simple average of these branches, will be virtually indistinguishable from that of the profile likelihood near $\hat{\psi}$ since each branch contributes a quadratic with the same peak and very similar curvature. Thus it is clear that for any dyad satisfying these criteria, it does not matter whether we use the integrated or the profile likelihood to conduct inference regarding its parameter of interest - our conclusions will be identical. However, the profile likelihood is in general both conceptually and computationally easier to implement, making it the better choice in such cases.

5.2.2. Poisson regression model with fixed effects

In the previous section, we relied solely on the observed counts and exposure times to estimate ψ. In practice, however, additional external factors often influence the counts we observe from each process. For instance, in the disease-outbreak example, variations in patient age distributions or other comorbidities across hospitals could affect the number of reported infections. To account for such influences, we turn to the standard framework of linear regression, which formally relates covariates to systematic variation in the observed counts.

Let Y_{ij} denote the i-th count from process j observed over an exposure time t_{ij}. Unlike the naive rates model in which only the total count per process was relevant, the presence of covariates that vary by observation requires us to retain each individual Y_{ij} and t_{ij}. If we were to ignore the covariates and sum over the index i, we would recover the structure of the naive model. Suppose that for each count we also observe a vector of covariates X_{ij} that may affect the rate at which process j generates events. We model these effects using a *fixed effects regression* framework, in which the coefficients associated with each covariate are treated as constant but unknown parameters to be estimated from the data. Some covariates may have *process-specific* effects (their impact varies across j), while others may have *process-agnostic* effects (affecting Y_{ij} uniformly across all processes $j = 1, ..., J$). We partition the covariate vector as

$$X_{ij} = \begin{pmatrix} X_{ij}^{(s)} \\ X_{ij}^{(a)} \end{pmatrix} \tag{5.2.5}$$

where $X_{ij}^{(s)} \in \mathbb{R}^{p\times 1}$ and $X_{ij}^{(a)} \in \mathbb{R}^{q\times 1}$ collect the process-specific and process-agnostic covariates, respectively.

Because covariates can influence the number of events recorded in each observation, the underlying rate

$$\theta_{ij} = \theta_j(X_{ij}) \tag{5.2.6}$$

may vary with i. Each observed count Y_{ij} then has a conditional mean

$$\mu_{ij} = \mathrm{E}(Y_{ij}|X_{ij}) = t_{ij}\theta_{ij}. \tag{5.2.7}$$

We define the *marginal rate* of process j as the expected rate averaged over the distribution of covariates, i.e.,

$$\bar{\theta}_j = \mathrm{E}_{X_j}[\theta_j(X_j)] = \int \theta_j(X_j)dF_{X_j}(X_j), \tag{5.2.8}$$

where X_j represents a generic covariate vector associated with process j, drawn according to the joint distribution F_{X_j}. These marginal rates generalize the naive rates from the previous model by averaging over covariates rather than relying on raw totals. Our parameter of interest for this model may then be written as

$$\psi = \sum_{j=1}^{J} \alpha_j \bar{\theta}_j, \tag{5.2.9}$$

for some positive weights $\alpha_1, ..., \alpha_J$.

Up to this point, we have refrained from placing any distributional assumption on the observed counts. In this chapter, we focus on two commonly used distributions for count data: the Poisson and the negative binomial. Since both distributions belong to the exponential family, our model fits naturally within the framework of generalized linear models (GLMs), which provide a formal connection between the covariate-dependent rates $\theta_j(X_{ij})$ and the observed counts. In the following subsections, we examine each distribution in turn and assess how well an integrated likelihood function can produce estimators of weighted sums of the marginal rates $\bar{\theta}_j$ under these standard count models.

Suppose the conditional distribution of Y_{ij} given X_{ij} is $Y_{ij}|X_{ij} \sim \text{Poisson}(\mu_{ij})$. Then the conditional probability mass function of Y_{ij} can be expressed in canonical exponential-family form as

$$\begin{aligned}
\mathrm{P}(Y_{ij} = y|X_{ij}) &= \frac{e^{-\mu_{ij}}\mu_{ij}^y}{y!} \\
&= \exp\left[y\log(\mu_{ij}) - \mu_{ij} - \log(y!)\right] \\
&= \exp\left[y\log(t_{ij}\theta_{ij}) - t_{ij}\theta_{ij} - \log(y!)\right] \\
&= \exp\left[y\log(\theta_{ij}) - t_{ij}\theta_{ij} + y\log(t_{ij}) - \log(y!)\right] \\
&= \exp\left[y\eta_{ij} - t_{ij}\exp(\eta_{ij}) + y\log(t_{ij}) - \log(y!)\right],
\end{aligned}$$

where $\eta_{ij} = \log(\theta_{ij})$ is the natural parameter of the model.

As a transformation of the rate, η_{ij} governs the mean of the distribution and thus captures how the expected count depends on the covariates. Following from the bedrock regression principle that systematic variation in the mean response should be explained linearly in terms of the predictors, we formalize this dependence by defining the *linear predictor* for observation i from process j as $X_{ij}^\top\beta_j$, a linear combination of the covariates with a vector of unknown coefficients β_j.

In a GLM, the systematic component represented by $X_{ij}^\top\beta_j$ must be connected to its random component μ_{ij}, the conditional mean of the response. This is done via a *link function* $g(\cdot)$, which specifies the relationship

$$g(\mu_{ij}) = X_{ij}^\top\beta_j.$$

The *canonical link function* is the special choice of $g(\cdot)$ that sets the natural parameter equal to the linear predictor, i.e.,

$$\eta_{ij} = X_{ij}^\top\beta_j. \tag{5.2.10}$$

For the Poisson distribution, we have already shown that the natural parameter is related to the rate through $\eta_{ij} = \log(\theta_{ij})$. Because the conditional mean of each Poisson process includes the exposure time, $\mu_{ij} = t_{ij}\theta_{ij}$, the model can be equivalently written as

$$\begin{aligned}
\log(\mu_{ij}) &= \log(t_{ij}\theta_{ij}) \\
&= \log(t_{ij}) + \log(\theta_{ij}) \\
&= \log(t_{ij}) + \eta_{ij} \\
&= \log(t_{ij}) + X_{ij}^{\top}\beta_j, \quad \text{(using the canonical link)}
\end{aligned} \tag{5.2.11}$$

where $\log(t_{ij})$ acts as a known offset. It follows that the canonical link for a Poisson GLM is the *log link* $g(\mu_{ij}) = \log(\mu_{ij}/t_{ij})$, under which additive effects of X_{ij} correspond to multiplicative effects on μ_{ij}.

To simplify notation, we include a leading 1 in the process-specific covariate vector $X_{ij}^{(s)}$ that corresponds to the intercept term for process j. We then partition the coefficient vector β_j as

$$\beta_j = \begin{pmatrix} \beta_j^{(s)} \\ \beta^{(a)} \end{pmatrix} \in \mathbb{R}^{(p+q)\times 1}, \tag{5.2.12}$$

where $\beta_j^{(s)} \in \mathbb{R}^{p\times 1}$ contains all the coefficients specific to process j (including the intercept) and $\beta^{(a)} \in \mathbb{R}^{q\times 1}$ contains the shared, process-agnostic slopes. The linear predictor for observation i from process j can then be written as

$$\eta_{ij} = \left(X_{ij}^{(s)}\right)^{\top}\beta_j^{(s)} + \left(X_{ij}^{(a)}\right)^{\top}\beta^{(a)}. \tag{5.2.13}$$

Stacking the observations for process j, we can express the model in matrix form as

$$\eta_j = X_j^{(s)}\beta_j^{(s)} + X_j^{(a)}\beta^{(a)}, \tag{5.2.14}$$

where $X_j^{(s)} \in \mathbb{R}^{n_j\times p}$ and $X_j^{(a)} \in \mathbb{R}^{n_j\times q}$ are the design matrices of process-specific and process-agnostic covariates, respectively, and $\eta_j = (\eta_{1j}, \dots, \eta_{n_j j})^{\top}$ is the vector of linear predictors for process j. If we concatenate the design matrices so that $X_j = [X_j^{(s)}\ X_j^{(a)}] \in \mathbb{R}^{n_j\times(p+q)}$, we can further simplify Equation 6.2.5

as

$$\eta_j = X_j\beta_j. \tag{5.2.15}$$

Collecting all coefficients across all processes gives the global parameter vector

$$\beta = \begin{pmatrix} \beta_1^{(s)} \\ \vdots \\ \beta_J^{(s)} \\ \beta^{(a)} \end{pmatrix} \in \mathbb{R}^{(pJ+q)\times 1},$$

and we view $\beta_j = (\beta_j^{(s)\top}, \beta^{(a)\top})^\top$ as the subvector of β relevant to process j.

The log-likelihood for an individual observation Y_{ij} follows directly from the exponential-family representation of the Poisson distribution:

$$\ell_{ij}(\beta_j; Y_{ij}) = Y_{ij}X_{ij}^\top\beta_j - t_{ij}\exp(X_{ij}^\top\beta_j)), \tag{5.2.16}$$

where we have discarded any additive terms not depending on the parameters. Summing over all observations within process j gives the process-level log-likelihood,

$$\begin{aligned} \ell_j(\beta_j; Y_j) &= \sum_{i=1}^{n_j} \ell_{ij}(\beta_j; Y_{ij}) \\ &= \sum_{i=1}^{n_j} \left[Y_{ij}X_{ij}^\top\beta_j - t_{ij}\exp(X_{ij}^\top\beta_j)) \right] \\ &= Y_j^\top X_j\beta_j - t_j^\top \exp(X_j\beta_j), \end{aligned} \tag{5.2.17}$$

where $Y_j \in \mathbb{R}^{n_j\times 1}$ and $t_j \in \mathbb{R}^{n_j\times 1}$ collect the observed counts and exposure times, respectively, for process j, and $\exp(X_j\beta_j)$ is understood to mean the exponential function applied component-wise to the vector $X_j\beta_j$.

Finally, aggregating across all processes yields the full log-likelihood,

$$\begin{aligned}\ell(\beta; Y) &= \sum_{j=1}^{J} \ell_j(\beta_j; Y_j) \\ &= \sum_{j=1}^{J} \left[Y_j^\top X_j \beta_j - t_j^\top \exp(X_j \beta_j) \right] \\ &= Y^\top X\beta - t^\top \exp(X\beta),\end{aligned} \tag{5.2.18}$$

where $n = \sum_{j=1}^{J} n_j$, $Y = (Y_1^\top, ..., Y_J^\top)^\top \in \mathbb{R}^{n\times 1}$ is the stacked response vector, $t = (t_1^\top, ..., t_J^\top)^\top \in \mathbb{R}^{n\times 1}$ is the stacked vector of exposure times, and $X = \text{blockdiag}(X_1, ..., X_j) \in \mathbb{R}^{n\times(pJ+q)}$ is the block-diagonal design matrix with process-specific blocks, augmented with shared covariates $X_j^{(a)}$ repeated across processes as necessary.

Let $\hat{\beta}_j$ denote the MLE of the coefficient vector β_j. Because the exponential term in the log-likelihood in Equation 6.2.9 makes it a nonlinear function of each β_j, the score equations for this model cannot be solved algebraically, and thus no closed-form expressions for $\hat{\beta}_j$ exists. Instead, $\hat{\beta}_j$ can be obtained numerically using the `glm()` function from the `stats` package in R, which implements the standard iteratively reweighted least squares (IRLS) algorithm for approximating the MLEs in the GLM framework.

From Equation 6.2.1, the corresponding estimated linear predictors for process j are given by

$$\hat{\eta}_{ij} = X_{ij}^\top \hat{\beta}_j. \tag{5.2.19}$$

Since $\eta_{ij} = \log(\theta_{ij})$, the estimated observation-level rate is

$$\hat{\theta}_{ij} = \exp(X_{ij}^\top \hat{\beta}_j). \tag{5.2.20}$$

An estimate of the marginal rate $\bar{\theta}_j$ defined in Equation 5.2.8 can then be obtained by averaging these observation-level estimates over the index i:

$$\hat{\theta}_{\bullet j} = \frac{1}{n_j} \sum_{i=1}^{n_j} \exp(X_{ij}^\top \hat{\beta}_j) = \frac{1}{n_j} \mathbf{1}_{n_j}^\top \exp(X_j \hat{\beta}_j). \tag{5.2.21}$$

Finally, substituting these into Equation 5.2.9 yields the MLE for ψ:

$$\hat{\psi} = \sum_{j=1}^{J} \alpha_j \hat{\theta}_{\bullet j} = \alpha^\top \hat{\theta}_{\bullet}, \tag{5.2.22}$$

where

$$\alpha = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_J \end{pmatrix} \quad \text{and} \quad \hat{\theta}_{\bullet} = \begin{pmatrix} \hat{\theta}_{\bullet 1} \\ \vdots \\ \hat{\theta}_{\bullet J} \end{pmatrix}.$$

If we let

$$\gamma = \begin{pmatrix} \frac{\alpha_1}{n_1} \mathbf{1}_{n_1} \\ \vdots \\ \frac{\alpha_J}{n_J} \mathbf{1}_{n_J} \end{pmatrix} \in \mathbb{R}^{n \times 1}, \tag{5.2.23}$$

we can rewrite our expression for $\hat{\psi}$ directly in terms of the MLE for β:

$$\hat{\psi} = \gamma^\top \exp(X\hat{\beta}). \tag{5.2.24}$$

This allows us to write our manifold of parameter vectors that produce the same estimate of ψ with the compact notation

$$\Omega_{\hat{\psi}} = \Big\{ \omega \in \mathbb{R}^{(pJ+q) \times 1} : \gamma^\top \exp(X\omega) = \gamma^\top \exp(X\hat{\beta}) \Big\}. \tag{5.2.25}$$

To estimate the value of the ZSE-parameterized integrated likelihood at a specific value of ψ, say ψ_1, using the method in Chapter 3, we use the following procedure.

7.1.2.1 Integrated Likelihood Calculation Procedure

1. Draw a random variate $\hat{\omega} \in \Omega_{\hat{\psi}}$ according to a distribution not depending on ψ. The method we have chosen for our simulations is:
   (i) Draw a random vector $u = (u_1, \dots, u_m)$, where $m = pJ + q$ and each $u_k \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0, 1)$, $k = 1, \dots, m$.
   (ii) Pass u as the initial guess to an `auglag()` call from the `nloptr` R package that minimizes a dummy function $f(\hat{\omega}) = 0$ subject to the constraint

$$h(\hat{\omega}) = \gamma^\top \exp(X\hat{\omega}) - \hat{\psi} = 0.$$

2. Solve

$$\tilde{\beta} = \underset{\beta \in \mathbb{R}^m}{\arg\max} \; \left(t \odot \exp(X\hat{\omega})\right)^\top X\beta - t^\top \exp(X\beta) \quad \text{subject to } \gamma^\top \exp(X\hat{\omega}) = \psi.$$

3. Define the branch evaluated at ψ_1 as

$$\ell_b^{(1)}(\psi_1) = Y^\top X\tilde{\beta} - t^\top \exp(X\tilde{\beta}).$$

4. Repeat steps (1) - (3) $R - 1$ times to obtain $\ell_b^{(1)}(\psi_1), \dots, \ell_b^{(R)}(\psi_1)$, then calculate

$$\hat{\bar{L}}(\psi_1) = \frac{1}{R} \sum_{r=1}^{R} \exp\left(\ell_b^{(r)}(\psi_1)\right).$$

7.1.2.1 Profile Likelihood Calculation Procedure

1. Solve

$$\tilde{\beta} = \underset{\beta \in \mathbb{R}^m}{\arg\max} \; \left(t \odot \exp(X\hat{\beta})\right)^\top X\beta - t^\top \exp(X\beta) \quad \text{subject to } \gamma^\top \exp(X\hat{\beta}) = \psi.$$

2. Obtain the profile likelihood evaluated at ψ_1 by calculating

$$L_p(\psi_1) = \exp\left(Y^\top X\tilde{\beta} - t^\top \exp(X\tilde{\beta})\right).$$

5.2.3. Poisson regression model with mixed effects

5.2.3.1. Fixed intercepts and random slopes.

5.2.3.2. Random intercepts and fixed slopes.

5.2.4. Poisson regression model with random effects

5.3. Estimating weighted sums of Poisson rates

5.3.1. Definition

5.3.2. Simulation design

5.3.3. Simulation results

5.3.4. Discussion

5.4. Detecting overdispersion

5.4.1. Definition

5.4.2. Simulation design

5.4.3. Simulation results

5.4.4. Discussion

5.5. Detecting zero-inflation

5.5.1. Definition

5.5.2. Simulation design

5.5.3. Simulation results

5.5.4. Discussion

5.6. Summary and discussion

CHAPTER 6

Negative Binomial Dyads

6.1. Introduction

6.2. Models

6.2.1. Simple rates model

6.2.2. Negative binomial regression model with fixed effects

Suppose the conditional distribution of Y_{ij} given X_{ij} is $Y_{ij}|X_{ij} \sim \mathrm{NB}(\mu_{ij}, r_j)$. We may continue to interpret μ_{ij} as the expected value of Y_{ij} conditional on X_{ij}, while r_j is best interpreted as a dispersion parameter that measures the excess variation across counts within process j that a Poisson random variable would fail to capture. Then the conditional probability mass function of Y_{ij} can be expressed in canonical exponential-family form as

$$\mathrm{P}(Y_{ij} = y|X_{ij}) = \frac{\Gamma(y + r_j)}{\Gamma(r_j)y!}\left(\frac{r_j}{r_j + \mu_{ij}}\right)^{r_j}\left(\frac{\mu_{ij}}{r_j + \mu_{ij}}\right)^{y}, \ \ y = 0, 1, 2, ...$$

where $\eta_{ij} = \log(\theta_{ij})$ is the natural parameter of the model.

As a transformation of the rate, η_{ij} governs the mean of the distribution and thus captures how the expected count depends on the covariates. Following from the bedrock regression principle that systematic variation in the mean response should be explained linearly in terms of the predictors, we formalize this dependence by defining the *linear predictor* for observation i from process j as $X_{ij}^\top\beta_j$, a linear combination of the covariates with a vector of unknown coefficients β_j.

In a GLM, the systematic component represented by $X_{ij}^\top\beta_j$ must be connected to its random component μ_{ij}, the conditional mean of the response. This is done via a *link function* $g(\cdot)$, which specifies the relationship

$$g(\mu_{ij}) = X_{ij}^{\top}\beta_j.$$

The *canonical link function* is the special choice of $g(\cdot)$ that sets the natural parameter equal to the linear predictor, i.e.,

$$\eta_{ij} = X_{ij}^{\top}\beta_j. \tag{6.2.1}$$

For the Poisson distribution, we have already shown that the natural parameter is related to the rate through $\eta_{ij} = \log(\theta_{ij})$. Because the conditional mean of each Poisson process includes the exposure time, $\mu_{ij} = t_{ij}\theta_{ij}$, the model can be equivalently written as

$$\begin{aligned}
\log(\mu_{ij}) &= \log(t_{ij}\theta_{ij}) \\
&= \log(t_{ij}) + \log(\theta_{ij}) \\
&= \log(t_{ij}) + \eta_{ij} \\
&= \log(t_{ij}) + X_{ij}^{\top}\beta_j, \qquad \text{(using the canonical link)}
\end{aligned} \tag{6.2.2}$$

where $\log(t_{ij})$ acts as a known offset. It follows that the canonical link for a Poisson GLM is the *log link* $g(\mu_{ij}) = \log(\mu_{ij}/t_{ij})$, under which additive effects of X_{ij} correspond to multiplicative effects on μ_{ij}.

To simplify notation, we include a leading 1 in the process-specific covariate vector $X_{ij}^{(s)}$ that corresponds to the intercept term for process j. We then partition the coefficient vector β_j as

$$\beta_j = \begin{pmatrix} \beta_j^{(s)} \\ \beta^{(a)} \end{pmatrix} \in \mathbb{R}^{(p+q)\times 1}, \tag{6.2.3}$$

where $\beta_j^{(s)} \in \mathbb{R}^{p\times 1}$ contains all the coefficients specific to process j (including the intercept) and $\beta^{(a)} \in \mathbb{R}^{q\times 1}$ contains the shared, process-agnostic slopes. The linear predictor for observation i from process j can then be written as

$$\eta_{ij} = \left(X_{ij}^{(s)}\right)^{\top}\beta_j^{(s)} + \left(X_{ij}^{(a)}\right)^{\top}\beta^{(a)}. \tag{6.2.4}$$

Stacking the observations for process j, we can express the model in matrix form as

$$\eta_j = X_j^{(s)}\beta_j^{(s)} + X_j^{(a)}\beta^{(a)}, \tag{6.2.5}$$

where $X_j^{(s)} \in \mathbb{R}^{n_j \times p}$ and $X_j^{(a)} \in \mathbb{R}^{n_j \times q}$ are the design matrices of process-specific and process-agnostic covariates, respectively, and $\eta_j = (\eta_{1j}, \dots, \eta_{n_j j})^\top$ is the vector of linear predictors for process j. If we concatenate the design matrices so that $X_j = [X_j^{(s)}\, X_j^{(a)}] \in \mathbb{R}^{n_j \times (p+q)}$, we can further simplify Equation 6.2.5 as

$$\eta_j = X_j\beta_j. \tag{6.2.6}$$

Collecting all coefficients across all processes gives the global parameter vector

$$\beta = \begin{pmatrix} \beta_1^{(s)} \\ \vdots \\ \beta_J^{(s)} \\ \beta^{(a)} \end{pmatrix} \in \mathbb{R}^{(pJ+q)\times 1},$$

and we view $\beta_j = (\beta_j^{(s)\top}, \beta^{(a)\top})^\top$ as the subvector of β relevant to process j.

The log-likelihood for an individual observation Y_{ij} follows directly from the exponential-family representation of the Poisson distribution:

$$\ell_{ij}(\beta_j; Y_{ij}) = Y_{ij}X_{ij}^\top\beta_j - t_{ij}\exp(X_{ij}^\top\beta_j)), \tag{6.2.7}$$

where we have discarded any additive terms not depending on the parameters. Summing over all observations within process j gives the process-level log-likelihood,

$$\begin{aligned} \ell_j(\beta_j; Y_j) &= \sum_{i=1}^{n_j} \ell_{ij}(\beta_j; Y_{ij}) \\ &= \sum_{i=1}^{n_j} \Big[Y_{ij}X_{ij}^\top\beta_j - t_{ij}\exp(X_{ij}^\top\beta_j))\Big] \\ &= Y_j^\top X_j\beta_j - t_j^\top \exp(X_j\beta_j), \end{aligned} \tag{6.2.8}$$

where $Y_j \in \mathbb{R}^{n_j \times 1}$ and $t_j \in \mathbb{R}^{n_j \times 1}$ collect the observed counts and exposure times, respectively, for process j, and $\exp(X_j\beta_j)$ is understood to mean the exponential function applied component-wise to the vector $X_j\beta_j$.

Finally, aggregating across all processes yields the full log-likelihood,

$$\begin{aligned}\ell(\beta; Y) &= \sum_{j=1}^{J} \ell_j(\beta_j; Y_j) \\ &= \sum_{j=1}^{J} \left[Y_j^\top X_j\beta_j - t_j^\top \exp(X_j\beta_j)\right] \\ &= Y^\top X\beta - t^\top \exp(X\beta),\end{aligned} \tag{6.2.9}$$

where $n = \sum_{j=1}^{J} n_j$, $Y = (Y_1^\top, ..., Y_J^\top)^\top \in \mathbb{R}^{n\times 1}$ is the stacked response vector, $t = (t_1^\top, ..., t_J^\top)^\top \in \mathbb{R}^{n\times 1}$ is the stacked vector of exposure times, and $X = \text{blockdiag}(X_1, ..., X_j) \in \mathbb{R}^{n\times(pJ+q)}$ is the block-diagonal design matrix with process-specific blocks, augmented with shared covariates $X_j^{(a)}$ repeated across processes as necessary.

Let $\hat{\beta}_j$ denote the MLE of the coefficient vector β_j. Because the exponential term in the log-likelihood in Equation 6.2.9 makes it a nonlinear function of each β_j, the score equations for this model cannot be solved algebraically, and thus no closed-form expressions for $\hat{\beta}_j$ exists. Instead, $\hat{\beta}_j$ can be obtained numerically using the `glm()` function from the `stats` package in R, which implements the standard iteratively reweighted least squares (IRLS) algorithm for approximating the MLEs in the GLM framework.

From Equation 6.2.1, the corresponding estimated linear predictors for process j are given by

$$\hat{\eta}_{ij} = X_{ij}^\top \hat{\beta}_j. \tag{6.2.10}$$

Since $\eta_{ij} = \log(\theta_{ij})$, the estimated observation-level rate is

$$\hat{\theta}_{ij} = \exp(X_{ij}^\top \hat{\beta}_j). \tag{6.2.11}$$

An estimate of the marginal rate $\bar{\theta}_j$ defined in Equation 5.2.8 can then be obtained by averaging these observation-level estimates over the index i:

$$\hat{\theta}_{\bullet j} = \frac{1}{n_j}\sum_{i=1}^{n_j}\exp(X_{ij}^\top\hat{\beta}_j) = \frac{1}{n_j}\mathbf{1}_{n_j}^\top\exp(X_j\hat{\beta}_j). \tag{6.2.12}$$

Finally, substituting these into Equation 5.2.9 yields the MLE for ψ:

$$\hat{\psi} = \sum_{j=1}^{J}\alpha_j\hat{\theta}_{\bullet j} = \alpha^\top\hat{\theta}_\bullet, \tag{6.2.13}$$

where

$$\alpha = \begin{pmatrix}\alpha_1\\ \vdots\\ \alpha_J\end{pmatrix} \quad \text{and} \quad \hat{\theta}_\bullet = \begin{pmatrix}\hat{\theta}_{\bullet 1}\\ \vdots\\ \hat{\theta}_{\bullet J}\end{pmatrix}.$$

If we let

$$\gamma = \begin{pmatrix}\frac{\alpha_1}{n_1}\mathbf{1}_{n_1}\\ \vdots\\ \frac{\alpha_J}{n_J}\mathbf{1}_{n_J}\end{pmatrix} \in \mathbb{R}^{n\times 1}, \tag{6.2.14}$$

we can rewrite our expression for $\hat{\psi}$ directly in terms of the MLE for β:

$$\hat{\psi} = \gamma^\top\exp(X\hat{\beta}). \tag{6.2.15}$$

This allows us to write our manifold of parameter vectors that produce the same estimate of ψ with the compact notation

$$\Omega_{\hat{\psi}} = \Big\{\omega \in \mathbb{R}^{(pJ+q)\times 1} : \gamma^\top\exp(X\omega) = \gamma^\top\exp(X\hat{\beta})\Big\}. \tag{6.2.16}$$

To estimate the value of the ZSE-parameterized integrated likelihood at a specific value of ψ, say ψ_1, using the method in Chapter 5, we use the following procedure.

7.1.2.1 Integrated Likelihood Calculation Procedure

1. Draw a random variate $\hat{\omega} \in \Omega_{\hat{\psi}}$ according to a distribution not depending on ψ. The method we have chosen for our simulations is:
   (i) Draw a random vector $u = (u_1, \dots, u_m)$, where $m = pJ + q$ and each $u_k \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0, 1)$, $k = 1, \dots, m$.
   (ii) Pass u as the initial guess to an `auglag()` call from the `nloptr` R package that minimizes a dummy function $f(\hat{\omega}) = 0$ subject to the constraint
   $$h(\hat{\omega}) = \gamma^\top \exp(X\hat{\omega}) - \hat{\psi} = 0.$$
2. Solve
$$\tilde{\beta} = \underset{\beta \in \mathbb{R}^m}{\arg\max} \; \left(t \odot \exp(X\hat{\omega})\right)^\top X\beta - t^\top \exp(X\beta) \quad \text{subject to } \gamma^\top \exp(X\hat{\omega}) = \psi.$$
3. Define the branch evaluated at ψ_1 as
$$\ell_b^{(1)}(\psi_1) = Y^\top X\tilde{\beta} - t^\top \exp(X\tilde{\beta}).$$
4. Repeat steps (1) - (3) $R - 1$ times to obtain $\ell_b^{(1)}(\psi_1), \dots, \ell_b^{(R)}(\psi_1)$, then calculate
$$\hat{\bar{L}}(\psi_1) = \frac{1}{R} \sum_{r=1}^{R} \exp\left(\ell_b^{(r)}(\psi_1)\right).$$

7.1.2.1 Profile Likelihood Calculation Procedure

1. Solve
$$\tilde{\beta} = \underset{\beta \in \mathbb{R}^m}{\arg\max} \; \left(t \odot \exp(X\hat{\beta})\right)^\top X\beta - t^\top \exp(X\beta) \quad \text{subject to } \gamma^\top \exp(X\hat{\beta}) = \psi.$$
2. Obtain the profile likelihood evaluated at ψ_1 by calculating
$$L_p(\psi_1) = \exp\left(Y^\top X\tilde{\beta} - t^\top \exp(X\tilde{\beta})\right).$$

6.2.3. Negative binomial regression model with mixed effects

6.2.3.1. Fixed intercepts and random slopes.

6.2.3.2. Random intercepts and fixed slopes.

6.2.4. Negative binomial regression model with random effects

6.3. Estimating weighted sums of negative binomial rates

6.3.1. Definition

6.3.2. Simulation design

6.3.3. Simulation results

6.3.4. Discussion

6.4. Estimating dispersion in a negative binomial model

6.4.1. Definition

6.4.2. Simulation design

6.4.3. Simulation results

6.4.4. Discussion

6.5. Detecting zero-inflation in a negative binomial model

6.5.1. Definition

6.5.2. Simulation design

6.5.3. Simulation results

6.5.4. Discussion

6.6. Summary and discussion

References

Agresti, A. (2002). Categorical data analysis. In *Wiley Series in Probability and Statistics* (2nd ed.). John Wiley & Sons. https://doi.org/10.1002/0471249688

Berger, J. O., Liseo, B., & Wolpert, R. L. (1999). Integrated likelihood methods for eliminating nuisance parameters. *Statistical Science*, *14*(1), 1–22. http://www.jstor.org/stable/2676641

Breslow, N. E., & Clayton, D. G. (1993). Approximate inference in generalized linear mixed models. *Journal of the American Statistical Association*, *88*(421), 9–25. https://doi.org/10.1080/01621459.1993.10594284

De Bin, R., Sartori, N., & Severini, T. A. (2015). Integrated likelihoods in models with stratum nuisance parameters. *Electronic Journal of Statistics*, *9*(1), 1474–1491. https://doi.org/10.1214/15-EJS1045

Hartzel, J., Agresti, A., & Caffo, B. (2001). Multinomial logit random effects models. *Statistical Modelling*, *1*(2), 81–102. https://doi.org/10.1177/1471082X0100100201

Kalbfleisch, J. D., & Sprott, D. A. (1970). Application of likelihood methods to models involving large numbers of parameters. *Journal of the Royal Statistical Society. Series B (Methodological)*, *32*(2), 175–208. http://www.jstor.org/stable/2984524

Sartori, N. (2003). Modified profile likelihoods in models with stratum nuisance parameters. *Biometrika*, *90*(3), 533–549. http://www.jstor.org/stable/30042064

Severini, T. A. (2007). Integrated likelihood functions for non-bayesian inference. *Biometrika*, *94*(3), 529–542. http://www.jstor.org/stable/20441394

Severini, T. A. (2018). Integrated likelihoods for functions of a parameter. *Stat*, *7*(1), e212. https://doi.org/10.1002/sta4.212

Severini, T. A. (2022). Integrated likelihood inference in multinomial distributions. *Metron*. https://doi.org/10.1007/s40300-022-00236-x

Simpson, E. H. (1949). Measurement of diversity. *Nature*, *163*(4148), 688–688. https://doi.org/10.1038/163688a0

Tierney, L., & Kadane, J. B. (1986). Accurate approximations for posterior moments and marginal densities. *Journal of the American Statistical Association*, *81*(393), 82–86. http://www.jstor.org/stable/2287970

APPENDIX A

Chapter 2

A.1. The Likelihood Function

A.1.1. Definition

Upon choosing a statistical model that we think best characterizes our population of interest, the obvious next step is to identify the true distribution in $\mathscr{P}$ or at the very least, the one that best approximates the truth. This is equivalent to making inferences about $\boldsymbol{\theta}_0$ in the case where the model is parametric and identifiable. That is, given the particular form(s) we have chosen for the distributions in $\mathscr{P}$, the only unknown remaining is the value of $\boldsymbol{\theta}_0$ itself. Since this value is ultimately what controls the mechanism generating any sample of data $\mathbf{x}_n = (x_1, ..., x_n)$ that we might observe from the population, it stands to reason that information regarding $\boldsymbol{\theta}_0$ can be inferred from the specific values of $x_1, ..., x_n$ that we obtain. To make this notion more rigorous, we require some method of analyzing the joint probability of our sample as a function of our parameter $\boldsymbol{\theta}$.

Given some observed data $\mathbf{x}_n$, the *likelihood function* for $\boldsymbol{\theta}$ is defined as

$$L(\boldsymbol{\theta}) = L(\boldsymbol{\theta}; \mathbf{x}_n) = p(\mathbf{x}_n; \boldsymbol{\theta}), \;\; \boldsymbol{\theta} \in \Theta. \tag{A.1.1}$$

In other words, the value of the likelihood function evaluated at a particular $\boldsymbol{\theta} \in \Theta$ is simply equal to the output of the model's density function evaluated at the same inputs. However, while $p(\mathbf{x}_n; \boldsymbol{\theta})$ is viewed primarily as a function of $\mathbf{x}_n$ for fixed $\boldsymbol{\theta}$, the reverse is actually true for $L(\boldsymbol{\theta}; \mathbf{x}_n)$. Indeed, we regard the likelihood as being a function of the parameter $\boldsymbol{\theta}$ for fixed $\mathbf{x}_n$. The reversal of the order of the arguments $\boldsymbol{\theta}$ and x is a reflection of this difference in perspectives.

When X is discrete, we may interpret $L(\boldsymbol{\theta}; x)$ as the probability that $X = x$ given that $\boldsymbol{\theta}$ is the true parameter value.[1] Crucially, this is *not* equivalent to the inverse probability that $\boldsymbol{\theta}$ is the true parameter value given $X = x$. The likelihood does not directly tell us anything about the probability that $\boldsymbol{\theta}$ assumes any particular value at all. Though intuitively appealing, this interpretation constitutes a fundamental misunderstanding of what a likelihood function is, and great care must be taken to avoid it.

When X is continuous, the likelihood for $\boldsymbol{\theta}$ may still be defined as it is in Equation A.1.1. However, we must forfeit our previous interpretation of $L(\boldsymbol{\theta})$ as a probability since the probability that X takes on any particular value is now 0. We may however still think of the likelihood as being proportional to the probability that X takes on a value "close" to x, meaning that that X is within a tiny ball centered at x. Specifically, for two different observations x_1 and x_2, if $L(\boldsymbol{\theta}; x_1) = c \cdot L(\boldsymbol{\theta}; x_2)$, where $c > 1$, then under this model we may conclude X is c times more likely to assume a value closer to x_1 than x_2 given that $\boldsymbol{\theta}$ is the true value of the parameter.

As in the discrete case, we must also be careful when X is continuous to avoid using $L(\boldsymbol{\theta}; \mathbf{x}_n)$ to make probabilistic assertions regarding $\boldsymbol{\theta}$. Despite our use of probability in its definition, the likelihood itself is *not* a probability density function for the parameter $\boldsymbol{\theta}$ and is subject to neither the same rules nor interpretations as one.

A.1.2. Transformations

There are a few useful transformations of the likelihood function that we will define here for use in future sections. The first is the *log-likelihood function*, which is defined as the natural logarithm of the likelihood function:

$$\ell(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}; \mathbf{x}_n) = \log L(\boldsymbol{\theta}; \mathbf{x}_n). \tag{A.1.2}$$

In practice, we will typically eschew direct analysis of the likelihood in favor of the log-likelihood due to the nice mathematical properties logarithms possess. Chief among these properties is the ability to turn products into sums (i.e. $\log(ab) = \log(a) + \log(b)$ for $a, b > 0$). Sums tend to be easier to differentiate than

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[1] Whichever value of $\boldsymbol{\theta}$ we choose to plug into $L(\boldsymbol{\theta}; x)$ is the value that we are currently "pretending" is the true one, regardless of whether or not it actually equals $\boldsymbol{\theta}_0$ in reality.

products, making this is a particularly useful feature for likelihood functions, which are often expressed as the product of marginal density functions when the observations are independent.

The other key property of logarithms that makes the log-likelihood so useful is that they are strictly increasing functions of their arguments (i.e. $\log x > \log y$ for $x > y > 0$). This monotonicity ensures that the locations of a function's extrema are preserved when the function is passed to the argument of a logarithm. For example, for a positive function f with a global maximum, $\underset{x}{\operatorname{argmax}} f(x) = \underset{x}{\operatorname{argmax}} \log f(x)$.

In the general case in which $\boldsymbol{\theta}$ is a d-dimensional vector, where d is an integer greater than 1, it follows that the first derivative of the log-likelihood with respect to $\boldsymbol{\theta}$ will also be a d-dimensional vector, the second derivative will be a $d \times d$ matrix, the third derivative will be a $d \times d \times d$ array, and so forth. To emphasize the multidimensional nature of these results, we will use notation typically associated with partial derivatives involving functions of more than one variable (e.g. $\nabla, \mathbf{J}, \mathbf{H}$, etc.) along with subscripts that indicate the variable with respect to which the partial derivatives are being taken. See Appendix A for a review of this notation.

The gradient of ℓ with respect to $\boldsymbol{\theta}$ appears frequently enough in the analysis of likelihood functions that it has earned its own name - the *score function*, or just the *score*. Formally, it is defined as

$$\mathcal{S}(\boldsymbol{\theta}; \mathbf{x}_n) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}; \mathbf{x}_n) = \begin{pmatrix} \frac{\partial}{\partial \boldsymbol{\theta}_1} \ell(\boldsymbol{\theta}; \mathbf{x}_n) \\ \vdots \\ \frac{\partial}{\partial \boldsymbol{\theta}_d} \ell(\boldsymbol{\theta}; \mathbf{x}_n) \end{pmatrix} = \begin{pmatrix} \mathcal{S}_1(\boldsymbol{\theta}; \mathbf{x}_n) \\ \vdots \\ \mathcal{S}_d(\boldsymbol{\theta}; \mathbf{x}_n) \end{pmatrix}, \tag{A.1.3}$$

where we think of each component as being a function $S_j : \Theta \to \mathbb{R}$.

Similarly, the Hessian matrix of the log-likelihood function with respect to $\boldsymbol{\theta}$ (i.e. the transpose of the Jacobian matrix of the score) multiplied by -1 is called the *observed information*, or just the *information*, and is denoted by

$$\mathcal{J}_{\mathbf{X}_n}(\boldsymbol{\theta}) = -\mathbf{H}_{\boldsymbol{\theta}}\Big(\ell(\boldsymbol{\theta};\mathbf{x}_n)\Big) = -\mathbf{J}_{\boldsymbol{\theta}}\Big(\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n)\Big)^{\top} = -\begin{pmatrix} \frac{\partial^2 \ell}{\partial \theta_1^2} & \frac{\partial^2 \ell}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 \ell}{\partial \theta_1 \partial \theta_d} \\ \frac{\partial^2 \ell}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 \ell}{\partial \theta_2^2} & \cdots & \frac{\partial^2 \ell}{\partial \theta_2 \partial \theta_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \ell}{\partial \theta_d \partial \theta_1} & \frac{\partial^2 \ell}{\partial \theta_d \partial \theta_2} & \cdots & \frac{\partial^2 \ell}{\partial \theta_d^2} \end{pmatrix}. \tag{A.1.4}$$

The use of the term "information" here derives from the fact that the second partial derivatives of ℓ with respect to the components of $\boldsymbol{\theta}$ are all related to the curvature of ℓ near its maximum - the sharper the curve, the less uncertainty and therefore more information we have about $\boldsymbol{\theta}$.

Recall that $L(\boldsymbol{\theta};\mathbf{x}_n)$ is defined as a function of $\boldsymbol{\theta}$ for a fixed sample of observations $\mathbf{x}_n = (x_1, ..., x_n)$, where we think of each x_i as being a realization of a random variable X_i. We may therefore interpret $L(\boldsymbol{\theta};\mathbf{x}_n)$ as a random variable in the following sense: for a given $\boldsymbol{\theta}$, the value of $L(\boldsymbol{\theta};\mathbf{x}_n)$ depends entirely on the values of $X_1, ..., X_n$ that we happened to observe, and so $L(\boldsymbol{\theta};\mathbf{X}_n)$ is itself a random variable with respect to the joint probability distribution of $\mathbf{X}_n = (X_1, ..., X_n)$. The same is also true for any function or estimate based on the likelihood, as they ultimately will all depend on the data through it as well. Going forward, we will use capital letters inside these functions when we want to emphasize this interpretation. For example, $\mathcal{S}(\boldsymbol{\theta};\mathbf{X}_n)$ is a random variable for which we have observed the value $\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n)$.

The random nature of these likelihood-based quantities further implies that finding their expectations and variances with respect to $p_{\boldsymbol{\theta}}(\mathbf{x}_n)$ is a well-defined, nontrivial task. The variance of the score function will be of particular importance, as it also relates to the amount of information pertaining to $\boldsymbol{\theta}_0$ that is contained within the log-likelihood function of our model. Properly known as the *Fisher information* or the *expected information*, it is defined as

$$\mathscr{I}_{\mathbf{X}_n}(\boldsymbol{\theta}) = \mathrm{Var}_{\boldsymbol{\theta}}[\mathcal{S}(\boldsymbol{\theta};\mathbf{X}_n)]. \tag{A.1.5}$$

Since we are working in the more general framework in which $\mathcal{S}(\boldsymbol{\theta})$ is a $d \times 1$ random vector, it would be more accurate to speak of the *Fisher information matrix*, which is equal to the variance-covariance matrix of $\mathcal{S}(\boldsymbol{\theta})$. Hence, we have

$$
\begin{aligned}
\mathscr{I}_{\mathbf{X}_n}(\boldsymbol{\theta}) &= \mathrm{Var}_{\boldsymbol{\theta}}[\mathcal{S}(\boldsymbol{\theta};\mathbf{X}_n)] && \text{(by Eq. 2.2.4)} \\
&= \mathrm{Cov}_{\boldsymbol{\theta}}\left[\left(\frac{\partial \ell}{\partial \theta_1,}\cdots,\frac{\partial \ell}{\partial \theta_d}\right)^T\right] && \text{(by Eq. 2.2.2)} \\
&= \begin{pmatrix}
\mathrm{Var}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_1}\right) & \mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_1},\frac{\partial \ell}{\partial \theta_2}\right) & \cdots & \mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_1},\frac{\partial \ell}{\partial \theta_d}\right) \\
\mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_2},\frac{\partial \ell}{\partial \theta_1}\right) & \mathrm{Var}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_2}\right) & \cdots & \mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_2},\frac{\partial \ell}{\partial \theta_d}\right) \\
\vdots & \vdots & \ddots & \vdots \\
\mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_d},\frac{\partial \ell}{\partial \theta_1}\right) & \mathrm{Cov}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_d},\frac{\partial \ell}{\partial \theta_2}\right) & \cdots & \mathrm{Var}_{\boldsymbol{\theta}}\left(\frac{\partial \ell}{\partial \theta_d}\right)
\end{pmatrix}. && \text{(A.1.6)}
\end{aligned}
$$

If the observations are independent, the Fisher information of the whole sample is equal to the sum of the Fisher information values for each of the observations individually. That is,

$$\mathscr{I}_{\mathbf{X}_n}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \mathscr{I}_{X_i}(\boldsymbol{\theta}). \tag{A.1.7}$$

If the observations are also identically distributed according to the distribution of some random variable X, then $\mathscr{I}_{X_i}(\boldsymbol{\theta}) = \mathscr{I}_X(\boldsymbol{\theta})$ for all i, and so the Fisher information for the entire sample is simply equal to the Fisher information for a single observation of X multiplied by a factor of n:

$$\mathscr{I}_{\mathbf{X}_n}(\boldsymbol{\theta}) = n\mathscr{I}_X(\boldsymbol{\theta}). \tag{A.1.8}$$

The above two equations hold true for the observed information under the same conditions as well.

A.1.3. Maximum Likelihood Estimation

A.1.3.1. Motivation. Maximum likelihood estimation is one of the most powerful and widespread techniques for obtaining point estimates of model parameters. The original intuition behind the method derives from the observation that when faced with a choice between two possible values of a parameter, the sensible choice is the one which makes the data we actually did observe more probable to have been observed. We have already defined the likelihood function as a means of capturing this probability, which makes expressing this decision rule in terms of it very easy - we simply choose for our estimate the option that produces the higher value of the likelihood function. That is, if $L(\boldsymbol{\theta}_1; \mathbf{x}_n) > L(\boldsymbol{\theta}_2; \mathbf{x}_n)$, then under the preceding logic, $\boldsymbol{\theta}_1$ is the better estimate of the true parameter value.

This can be extended to include as many candidate parameter values as we would like. For n potential estimates of $\boldsymbol{\theta}_0$, the best is the one that corresponds to the highest value of the likelihood function. Following this line of reasoning to its natural conclusion, a sensible choice for an estimate of $\boldsymbol{\theta}_0$ is any value that maximizes the likelihood function based on an observed dataset $\mathbf{x}_n$. Hence, let $\hat{\boldsymbol{\theta}} \in \Theta$ be any parameter value that renders the likelihood at the observed $\mathbf{X}_n = \mathbf{x}_n$ as large as possible, i.e.,

$$L(\hat{\boldsymbol{\theta}}; \mathbf{x}_n) = \sup_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta}; \mathbf{x}_n). \tag{A.1.9}$$

We call such a value a *maximum likelihood estimate* of $\boldsymbol{\theta}_0$. This definition of $\hat{\boldsymbol{\theta}}$ as a maximizer of $L(\hat{\boldsymbol{\theta}}; \mathbf{x}_n)$ necessarily makes it a function of the observed data. When this function is measurable, then we can further define the *maximum likelihood estimator* (MLE) of $\boldsymbol{\theta}_0$ as the statistic $\hat{\boldsymbol{\theta}}(\mathbf{X}_n)$ for which we observe the value $\hat{\boldsymbol{\theta}}(\mathbf{x}_n)$.

A.1.3.2. Regularity Conditions. As a consequence of the random variable interpretation of likelihood-based quantities, a natural line of inquiry to investigate is how the behavior of these random variables changes as the sample size n increases. Of particular interest is the distribution to which the MLE converges, if any, as n tends toward infinity. To that end, it will be useful to establish some *regularity conditions* for our models. We can think of these conditions as being assumptions similar to those we discussed in the introduction to this paper that, when satisfied, endow our models with certain properties that enable us, among other things, to determine the aforementioned distribution.

For our purposes, we will call a model *regular* if it satisfies the following conditions:

RC1) Any observations $x_1, ..., x_n$ belonging to a sample that has been drawn from the model's sample space are independent and identically distributed (i.i.d.) realizations of a random variable X with density function $p_{\boldsymbol{\theta}}(x)$.

RC2) $P_{\boldsymbol{\theta}_1} = P_{\boldsymbol{\theta}_2} \implies \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$ for all $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \Theta$.

RC3) The distributions in $\mathcal{P}$ have a common support $\mathcal{X} = \{x : p_{\boldsymbol{\theta}}(x) > 0\} \subseteq \mathbb{R}$ not depending on $\boldsymbol{\theta}$.

RC4) There exists an open set $\Theta^* \subseteq \Theta$ of which $\boldsymbol{\theta}_0$ is an interior point.

RC5) $p(x;\boldsymbol{\theta})$ is twice continuously differentiable with respect to $\boldsymbol{\theta}$ for all $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_0$.

RC6) There exists a random function $M(x)$ (that does not depend on $\boldsymbol{\theta}$) satisfying $\mathrm{E}[M(X)] < \infty$ such that each third partial derivative of $\ell(\boldsymbol{\theta};x)$ is bounded in absolute value by $M(x)$ uniformly in some neighborhood of $\boldsymbol{\theta}_0$.

RC7) The integral $\int_{\mathcal{X}} p(x;\boldsymbol{\theta})dx$ can be differentiated twice under the integral sign with respect to the components of $\boldsymbol{\theta} \in \Theta^*$.

RC8) $\mathscr{I}_X(\boldsymbol{\theta})$ is positive definite for all $\boldsymbol{\theta} \in \Theta$.

RC9) Θ is a compact and convex subset of $\mathbb{R}^d$.

While not strictly necessary, **RC1** is often assumed as a matter of convenience since it tends to simplify calculations greatly. We include it here for that purpose and to frame our discussion in the context of a standard case in which likelihood theory holds. Of course, it is possible to construct models lacking i.i.d. observations yet still possessing real world applications for which the results discussed in this paper hold.

The implication in **RC2** is simply the identifiability property we mentioned in Chapter 2. We repeat it here as it is necessary to guarantee the consistency of the MLE, i.e., that it converges in probability to $\boldsymbol{\theta}_0$ as $n \to \infty$.

RC3 requires that the distributions in $\mathscr{P}$ be supported on a common subset of the real line, and the definition of this subset cannot depend on $\boldsymbol{\theta}$. This is to prevent situations in which, for example, the event $\{X_i \leq x_i\}$ occurs with positive probability when $\boldsymbol{\theta} = \boldsymbol{\theta}_1$ but not $\boldsymbol{\theta} = \boldsymbol{\theta}_2$.

RC4 guarantees the existence of an open subset Θ^* of Θ containing $\boldsymbol{\theta}_0$ as an interior point. The fact that $\boldsymbol{\theta}_0$ is an interior point of Θ^* further implies that it is possible to find a neighborhood of $\boldsymbol{\theta}_0$ that is contained in Θ^*. **RC5** then goes on to assert the existence and continuity of the first two partial derivatives with respect to the components of $\boldsymbol{\theta}$ of $p(x;\boldsymbol{\theta})$ in this neighborhood. This is a necessary requirement for defining a second-order Taylor series expansion of the score function around $\boldsymbol{\theta}_0$.

Another way of stating **RC6** is that for all $\boldsymbol{\theta}$ in a neighborhood $N_{\boldsymbol{\theta}_0}$, there exists a random function $M(x)$ with finite expectation such that

$$\sup_{\boldsymbol{\theta} \in N_{\boldsymbol{\theta}_0}} \left| \frac{\partial^3 \ell(\boldsymbol{\theta}; x)}{\partial \theta_i \partial \theta_j \partial \theta_k} \right| \leq M(x)$$

for all integers $1 \leq i, j, k \leq d$. Equivalently, we could say that the entries of the Hessian matrix for each component of the score function are all bounded by $M(x)$ as well. This ensures the remainder terms in a second-order Taylor series expansion of the score function around $\boldsymbol{\theta_0}$ become negligible as the sample size increases to infinity.

RC7 grants us the ability to freely interchange integration and second-order partial differentiation with respect to the components of $\boldsymbol{\theta}$, i.e.,

$$\frac{\partial^2}{\partial \theta_i \partial \theta_j} \int_{\mathcal{X}} p(x; \boldsymbol{\theta}) dx = \int_{\mathcal{X}} \frac{\partial^2}{\partial \theta_i \partial \theta_j} p(x; \boldsymbol{\theta}) dx$$

for all $\boldsymbol{\theta} \in \Theta^*$ and $i, j = 1, ..., d$. This implies first-order partial derivatives can be passed under the integral sign as well. This will prove useful in our discussion of the Bartlett identities in Section 3.5.

Finally, **RC8-9** play an important role in ensuring the existence and uniqueness of the maximum likelihood estimator of $\boldsymbol{\theta}_0$.

A.1.3.3. Properties. In general, there is no guarantee that an MLE for a model's parameter will exist, and even if it does, it will not necessarily be unique. However, since maximum likelihood estimation is critical to our discussion in Chapters 5 and 6, it would come as a great convenience if we possessed the ability to speak freely of *the* MLE of a model's parameter without having to clarify *which* MLE we mean or whether one even exists at all. Hence, some discussion of the conditions under which the MLE of a model's parameter exists and is unique is warranted.

The extreme value theorem (see Appendix A) implies that sufficient conditions for the existence of a model's MLE are that Θ is compact, and $\ell(\boldsymbol{\theta}; x)$ is continuous on Θ. The former is satisfied directly by **RC9** and the latter is implied through our assumption in **RC5** of the differentiability of $p(x; \boldsymbol{\theta})$ in $\boldsymbol{\theta}$. Therefore, at least one MLE will always exist for the true parameter value of a regular model. These are only sufficient conditions, however, and not necessary; MLEs may exist for parameters of non-regular models as wel.

Similarly, when the MLE does exist, a sufficient condition for its uniqueness is that Θ is convex, and $\ell(\boldsymbol{\theta}; x)$ is strictly concave on Θ, as this ensures that it has exactly one global maximum, which it attains at the $\hat{\boldsymbol{\theta}}$. **RC9** directly satisfies the compactness criterion. Our assumption in **RC8** that the Fisher information matrix is positive definite forces the Hessian matrix of $\ell(\boldsymbol{\theta}; x)$ to be negative definite. This in turn implies that $\ell(\boldsymbol{\theta}; x)$ is strictly concave, so the requirement is met. Hence, a regular model will always have a unique MLE for its parameter.

One last useful property of the maximum likelihood estimator is its functional invariance. If $\hat{\boldsymbol{\theta}}$ is an MLE of $\boldsymbol{\theta}_0$, then any function $h(\boldsymbol{\theta}_0)$ will have $h(\hat{\boldsymbol{\theta}})$ as its MLE. Hence, it is straightforward to find the MLE of a model that has undergone a reparameterization given that we know the MLE of the original parameter.

A.1.4. The Bartlett Identities

The Bartlett identities are a set of equations relating to the expectations of the derivatives of a log-likelihood function to one another. In general, there is no guarantee that an arbitrary function of a random variable X and its parameter $\boldsymbol{\theta}$ will satisfy the Bartlett identities. It is guaranteed, however, that the log-likelihood function associated with X and $\boldsymbol{\theta}$ will satisfy them, provided that the model is regular. This means we can think of any function that does satisfy the Bartlett identities (or at least some of them) as resembling that of a genuine log-likelihood.

Consider the case where a random variable X has density function $p_\theta(x)$, where θ is a scalar. For a single observation $X = x$, the expectation of $\frac{\partial}{\partial\theta}\ell(\theta; X)$ gives

$$
\begin{aligned}
E_\theta\left[\frac{\partial}{\partial\theta}\ell(\theta;X)\right] &= \int_{\mathbb{R}}\left[\frac{\partial}{\partial\theta}\log p(x;\theta)\right]p(x;\theta)dx \\
&= \int_{\mathbb{R}}\frac{\frac{\partial}{\partial\theta}p(x;\theta)}{p(x;\theta)}p(x;\theta)dx \\
&= \int_{\mathbb{R}}\frac{\partial}{\partial\theta}p(x;\theta)dx \\
&= \frac{d}{d\theta}\int_{\mathbb{R}}p(x;\theta)dx \\
&= \frac{d}{d\theta}1 \\
&= 0.
\end{aligned}
\tag{A.1.10}
$$

Equation A.1.10 is called the first Bartlett identity. In words, it states that the expectation of the first partial derivative of the log-likelihood function of a statistical model with respect to the parameter will always be 0. Since the score is defined as $\frac{\partial}{\partial\theta}\ell(\theta;x)$, any function that satisfies the first Bartlett identity is said to be *score-unbiased*.

For any model with a log-likelihood satisfying the first Bartlett identity, the expected information for its parameter θ may be rewritten as

$$
\begin{aligned}
\mathscr{I}_X(\theta) &= \mathrm{Var}_\theta[\mathcal{S}(\theta;X)] \\
&= \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\ell(\theta;X)\right] \\
&= \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\ell(\theta;X)\right] + \left(\mathrm{E}_\theta\left[\frac{\partial}{\partial\theta}\ell(\theta;X)\right]\right)^2 \quad \text{(by the first Bartlett identity)} \\
&= \mathrm{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\ell(\theta;X)\right)^2\right].
\end{aligned}
\tag{A.1.11}
$$

If we now consider the second partial derivative of $\ell(\theta; x)$ with respect to θ, we have

$$\begin{aligned}
\frac{\partial^2}{\partial\theta^2}\ell(\theta; x) &= \frac{\partial}{\partial\theta}\left[\frac{\partial}{\partial\theta}\ell(\theta; x)\right] \\
&= \frac{\partial}{\partial\theta}\left[\frac{\partial}{\partial\theta}\log p(x; \theta)\right] \\
&= \frac{\partial}{\partial\theta}\left[\frac{\frac{\partial}{\partial\theta}p(x; \theta)}{p(x; \theta)}\right] \\
&= \frac{\left[\frac{\partial^2}{\partial\theta^2}p(x; \theta)\right]p(x; \theta) - \left[\frac{\partial}{\partial\theta}p(x; \theta)\right]\left[\frac{\partial}{\partial\theta}p(x; \theta)\right]}{\left[p(x; \theta)\right]^2} \\
&= \frac{\frac{\partial^2}{\partial\theta^2}p(x; \theta)}{p(x; \theta)} - \left[\frac{\frac{\partial}{\partial\theta}p(x; \theta)}{p(x; \theta)}\right]^2 \\
&= \frac{\frac{\partial^2}{\partial\theta^2}p(x; \theta)}{p(x; \theta)} - \left[\frac{\partial}{\partial\theta}\log p(x; \theta)\right]^2 \\
&= \frac{\frac{\partial^2}{\partial\theta^2}p(x; \theta)}{p(x; \theta)} - \left[\frac{\partial}{\partial\theta}\ell(\theta; x)\right]^2 .
\end{aligned}$$

Rearranging terms and taking expectations yields

$$\begin{aligned}
\mathrm{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\ell(\theta; X)\right] + \mathrm{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\ell(\theta; X)\right)^2\right] &= \mathrm{E}_\theta\left[\frac{\frac{\partial^2}{\partial\theta^2}p(X; \theta)}{p(X; \theta)}\right] \\
&= \int_{\mathbb{R}}\left[\frac{\frac{\partial^2}{\partial\theta^2}p(x; \theta)}{p(x; \theta)}\right]p(x; \theta)dx \\
&= \int_{\mathbb{R}}\frac{\partial^2}{\partial\theta^2}p(x; \theta)dx \\
&= \frac{d^2}{d\theta^2}\int_{\mathbb{R}}p(x; \theta)dx \\
&= \frac{d^2}{d\theta^2}1 \\
&= 0.
\end{aligned}$$

Therefore,

$$\mathrm{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\ell(\theta; X)\right] + \mathrm{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\ell(\theta; X)\right)^2\right] = 0. \tag{A.1.12}$$

Equation A.1.11 is called the second Bartlett identity. Any function that satisfies it is said to be *information-unbiased*. Regular models as we have defined them will automatically satisfy both the first and second Bartlett identities. Hence, for any regular model, the statements in Equation A.1.5, Equation A.1.11, and Equation A.1.12 imply that the following definitions for its expected information regarding its parameter θ are all equivalent:

$$\mathscr{I}_X(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\ell(\theta;X)\right] = \mathrm{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\ell(\theta;X)\right)^2\right] = \mathrm{E}_\theta\left[-\frac{\partial^2}{\partial\theta^2}\ell(\theta;X)\right] = \mathrm{E}_\theta[\mathcal{J}_X(\theta)]. \tag{A.1.13}$$

It is possible to derive further Bartlett identities by continuing in this manner for an arbitrary number of partial θ-derivatives of the log-likelihood function, provided that they exist. However, the first two are sufficient for our purposes of evaluating the validity of approximations to genuine likelihoods so we will not go further here. While the above derivations were performed under the assumption that θ is a scalar, the Bartlett identities also hold in the case where $\boldsymbol{\theta}$ is a $d \times 1$ vector.

A.2. Laplace's Method

Laplace's method approximates integrals of the form

$$I = \int_{\mathbb{R}^d} \exp\{f(\mathbf{u})\}\, d\mathbf{u}$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is smooth and has a unique global maximum at $\hat{\mathbf{u}}$. Expanding f to second order around $\hat{\mathbf{u}}$, we have

$$\begin{aligned} f(\mathbf{u}) &\approx f(\hat{\mathbf{u}}) + (\mathbf{u}-\hat{\mathbf{u}})^\top \nabla f(\hat{\mathbf{u}}) + \tfrac{1}{2}(\mathbf{u}-\hat{\mathbf{u}})^\top \nabla^2 f(\hat{\mathbf{u}})(\mathbf{u}-\hat{\mathbf{u}}) \\ &= f(\hat{\mathbf{u}}) + \tfrac{1}{2}(\mathbf{u}-\hat{\mathbf{u}})^\top \nabla^2 f(\hat{\mathbf{u}})(\mathbf{u}-\hat{\mathbf{u}}), \end{aligned}$$

where the linear term vanishes because $\hat{\mathbf{u}}$ maximizes f, so $\nabla f(\hat{\mathbf{u}}) = \mathbf{0}$ by first-order optimality. Substituting and integrating the resulting Gaussian yields

$$I \approx (2\pi)^{d/2} |-\nabla^2 f(\hat{\mathbf{u}})|^{-1/2} \exp\{f(\hat{\mathbf{u}})\},$$

so that

$$\log I \approx f(\hat{\mathbf{u}}) + \tfrac{d}{2}\log(2\pi) - \tfrac{1}{2}\log\lvert -\nabla^2 f(\hat{\mathbf{u}})\rvert. \tag{A.2.1}$$

The approximation is exact when f is quadratic, and has relative error $O(n^{-1})$ in general, where n is the effective sample size driving the concentration of $\exp\{f\}$ around $\hat{\mathbf{u}}$. Tierney & Kadane (1986) showed that for integrals arising in Bayesian computation — specifically, ratios of the form

$$\frac{\int g(\mathbf{u}) \exp\{n\,\ell(\mathbf{u})\}\,d\mathbf{u}}{\int \exp\{n\,\ell(\mathbf{u})\}\,d\mathbf{u}}$$

— the relative error improves to $O(n^{-2})$ under mild regularity conditions, because certain odd-order error terms cancel in the ratio.

A.2.1. Application to the Random Effects Multinomial Model

For notational compactness, let $\mathbf{C} = [\mathbf{A} \mid \mathbf{B}] \in \mathbb{R}^{p\times(J-1)}$ denote the full coefficient matrix and let $\tilde{\mathbf{x}}_{it} = (1, \mathbf{x}_{it}^\top)^\top$ denote the design vector with intercept prepended, so that the linear predictor for observation (i,t) is $\mathbf{C}^\top\tilde{\mathbf{x}}_{it} + \mathbf{u}_i$. The cluster-level integral from Equation 4.4.1 has the form $\int \exp\{g_i(\mathbf{u})\}\,d\mathbf{u}$, where

$$g_i(\mathbf{u}) = \sum_{t=1}^{m_i}\sum_{j=1}^{J} y_{itj} \log\theta_j(\tilde{\mathbf{x}}_{it}; \mathbf{C}, \mathbf{u}) - \tfrac{1}{2}\mathbf{u}^\top\boldsymbol{\Sigma}^{-1}\mathbf{u}, \tag{A.2.2}$$

obtained by writing $\phi(\mathbf{u}_i; \mathbf{0}, \boldsymbol{\Sigma}) = (2\pi)^{-(J-1)/2}\lvert\boldsymbol{\Sigma}\rvert^{-1/2}\exp\{-\tfrac{1}{2}\mathbf{u}_i^\top\boldsymbol{\Sigma}^{-1}\mathbf{u}_i\}$ and absorbing the normalizing constants. Note that $g_i(\mathbf{u})$ is precisely the complete-data log-likelihood ℓ_{c} from Equation 4.2.14 evaluated at cluster i, minus the log-prior on $\mathbf{u}_i$. The Laplace approximation therefore approximates the log of the marginal likelihood — the log-outside-the-integral quantity from Equation 4.2.15 — by evaluating the log-inside-the-integral quantity g_i at its mode and correcting for curvature. The mode $\hat{\mathbf{u}}_i = \operatorname{argmax}_{\mathbf{u}}\, g_i(\mathbf{u})$ is the posterior mode of $\mathbf{u}_i$ given cluster i's observations and $\mathbf{C}$. Differentiating Equation A.2.2 twice with respect to $\mathbf{u}$ and evaluating at $\hat{\mathbf{u}}_i$ gives the negative Hessian

$$-\nabla^2 g_i(\hat{\mathbf{u}}_i) = \sum_{t=1}^{m_i}\mathbf{W}_t(\hat{\mathbf{u}}_i) + \boldsymbol{\Sigma}^{-1}, \tag{A.2.3}$$

where

$$\mathbf{W}_t = \operatorname{diag}(\boldsymbol{\pi}_t) - \boldsymbol{\pi}_t\boldsymbol{\pi}_t^\top, \quad \boldsymbol{\pi}_t = \left(\theta_2(\tilde{\mathbf{x}}_{it};\mathbf{C},\hat{\mathbf{u}}_i), \dots, \theta_J(\tilde{\mathbf{x}}_{it};\mathbf{C},\hat{\mathbf{u}}_i)\right)^\top$$

is the multinomial covariance matrix of the non-baseline fitted probabilities at the posterior mode. The term $\boldsymbol{\Sigma}^{-1}$ reflects the curvature of the random effects prior and $\sum_t \mathbf{W}_t$ reflects the curvature of the data likelihood; together they determine the sharpness of the posterior around $\hat{\mathbf{u}}_i$. Applying Equation A.2.1 with $d = J-1$ and canceling the $(2\pi)^{(J-1)/2}$ factors from the normalizing constant of ϕ yields

$$\log L_i(\mathbf{C},\boldsymbol{\Sigma}) \approx g_i(\hat{\mathbf{u}}_i) - \tfrac{1}{2}\log|\boldsymbol{\Sigma}| - \tfrac{1}{2}\log\left|\textstyle\sum_t \mathbf{W}_t(\hat{\mathbf{u}}_i) + \boldsymbol{\Sigma}^{-1}\right|. \tag{A.2.4}$$

When $\mathbf{C}$ is treated as variable rather than fixed, the gradient of $\log I$ with respect to $\mathbf{C}$ requires accounting for the dependence of $\hat{\mathbf{u}}_i(\mathbf{C})$ on $\mathbf{C}$ through the implicit equation $\nabla_{\mathbf{u}} g_i(\hat{\mathbf{u}}_i;\mathbf{C}) = \mathbf{0}$. By the implicit function theorem,

$$\frac{d\hat{\mathbf{u}}_i}{d\mathbf{C}} = -\left[\nabla^2_{\mathbf{u}} g_i(\hat{\mathbf{u}}_i)\right]^{-1} \nabla_{\mathbf{C}}\nabla_{\mathbf{u}} g_i(\hat{\mathbf{u}}_i).$$

The standard approximation from Breslow & Clayton (1993) sets $d\hat{\mathbf{u}}_i/d\mathbf{C} = 0$, treating the posterior mode as fixed when differentiating with respect to $\mathbf{C}$. Under this approximation the gradient of ℓ^{LA} with respect to $\mathbf{C}$ reduces to

$$\nabla_{\mathbf{C}}\ell^{\mathrm{LA}}(\mathbf{C},\boldsymbol{\Sigma}) \approx \sum_{i=1}^{n}\sum_{t=1}^{m_i} \tilde{\mathbf{x}}_{it}\left(\mathbf{Y}_{it} - \boldsymbol{\pi}_{it}\right)^\top, \tag{A.2.5}$$

where $\mathbf{Y}_{it} \in \{0,1\}^{J-1}$ is the response indicator vector with the baseline category dropped and $\boldsymbol{\pi}_{it} = \boldsymbol{\pi}_t(\hat{\mathbf{u}}_i)$ is the corresponding vector of non-baseline fitted probabilities at the posterior mode. Gradients with respect to $\mathbf{A}$ and $\mathbf{B}$ are the corresponding row submatrices of $\nabla_{\mathbf{C}}\ell^{\mathrm{LA}}$. The expression in Equation A.2.5 has the same form as the score of an ordinary multinomial logistic regression, with $\hat{\mathbf{u}}_i$ acting as a cluster-level offset in the linear predictor. The approximation error is $O(m_i^{-1})$ per cluster, matching the order of the Laplace approximation itself.

In the context of this paper, Equation A.2.5 is used in two places: in estimation of $\mathbf{C}$, where it drives the optimization of ℓ^{LA} with respect to the fixed effects, and in the computation of profile and integrated likelihoods, where it is supplied to the constrained optimizer at each point of the ψ grid. In the former case an inexact gradient affects the path to convergence but not the solution itself, since convergence is assessed by the log-likelihood value. In both cases the approximation is unlikely to be consequential for the designs considered in the simulation study.

A.3. Properties of the Integrated Likelihood

A.3.1. Asymptotic Theory

A.3.1.1. One-Index Asymptotics. The one-index asymptotics framework describes the behavior of likelihood-based statistics as the sample size (n) grows to infinity. The aim of this section is to present an overview of some of the theory's classic results. This provides us with a readily available baseline against which to compare the results of the following section discussing the two-index asymptotics framework, in which the full model parameter is partitioned into a parameter of interest and a nuisance parameter, and the dimension of the nuisance parameter is allowed to increase with the sample size.

Let $\boldsymbol{\theta}$ be the parameter of a regular model with true value $\boldsymbol{\theta}_0$. As a partial justification for our use of the MLE in estimating $\boldsymbol{\theta}_0$, we will start by showing that with probability tending to 1 as n tends toward infinity, the likelihood for $\boldsymbol{\theta}$ is strictly larger at $\boldsymbol{\theta}_0$ than for any other $\boldsymbol{\theta} \in \Theta$. **RC1** implies

$$L(\boldsymbol{\theta}; \mathbf{x}_n) = p(\mathbf{x}_n; \boldsymbol{\theta}) = \prod_{i=1}^{n} p(x_i; \boldsymbol{\theta}) \tag{A.3.1}$$

and

$$\ell(\boldsymbol{\theta}; \mathbf{x}_n) = \log p(\mathbf{x}_n; \boldsymbol{\theta}) = \sum_{i=1}^{n} \log p(x_i; \boldsymbol{\theta}). \tag{A.3.2}$$

It follows that

$$\begin{aligned} L(\boldsymbol{\theta}; \mathbf{x}_n) < L(\boldsymbol{\theta}_0; \mathbf{x}_n) &\iff \ell(\boldsymbol{\theta}; \mathbf{x}_n) < \ell(\boldsymbol{\theta}_0; \mathbf{x}_n) \\ &\iff \sum_{i=1}^{n} \log p(x_i; \boldsymbol{\theta}) - \sum_{i=1}^{n} \log p(x_i; \boldsymbol{\theta}_0) < 0 \\ &\iff \sum_{i=1}^{n} \big[\log p(x_i; \boldsymbol{\theta}) - \log p(x_i; \boldsymbol{\theta}_0)\big] < 0 \\ &\iff \sum_{i=1}^{n} \log \frac{p(x_i; \boldsymbol{\theta})}{p(x_i; \boldsymbol{\theta}_0)} < 0 \\ &\iff \frac{1}{n} \sum_{i=1}^{n} \log \frac{p(x_i; \boldsymbol{\theta})}{p(x_i; \boldsymbol{\theta}_0)} < 0. \end{aligned} \tag{A.3.3}$$

RC3 guarantees that the ratio $p(x;\boldsymbol{\theta})/p(x;\boldsymbol{\theta}_0)$ is well-defined and finite for all $x \in \mathcal{X}$, the region of common support. Then by the Weak Law of Large Numbers,

$$\frac{1}{n}\sum_{i=1}^{n}\log\frac{p(X_i;\boldsymbol{\theta})}{p(X_i;\boldsymbol{\theta}_0)} \to \mathrm{E}_{\boldsymbol{\theta}_0}\left[\log\frac{p(X;\boldsymbol{\theta})}{p(X;\boldsymbol{\theta}_0)}\right] \tag{A.3.4}$$

in probability as $n \to \infty$. Furthermore,

$$\mathrm{E}_{\boldsymbol{\theta}_0}\left[\frac{p(X;\boldsymbol{\theta})}{p(X;\boldsymbol{\theta}_0)}\right] = \int_{\mathcal{X}}\left[\frac{p(x;\boldsymbol{\theta})}{p(x;\boldsymbol{\theta}_0)}\right]p(x;\boldsymbol{\theta}_0)dx = \int_{\mathcal{X}}p(x;\boldsymbol{\theta})dx = 1. \tag{A.3.5}$$

Since $\log(x)$ is a strictly concave function, it follows from Jensen's inequality (see Appendix A) and Equation A.3.5

$$\mathrm{E}_{\boldsymbol{\theta}_0}\left[\log\frac{p(X;\boldsymbol{\theta})}{p(X;\boldsymbol{\theta}_0)}\right] < \log\mathrm{E}_{\boldsymbol{\theta}_0}\left[\frac{p(X;\boldsymbol{\theta})}{p(X;\boldsymbol{\theta}_0)}\right] = \log 1 = 0. \tag{A.3.6}$$

Hence, the quantity on the left-hand side of Equation A.3.4 is converging in probability to a constant that is less than 0 as n tends to infinity. From this and the equivalence we established in Equation A.3.3, it follows that

$$\lim_{n\to\infty}P_{\boldsymbol{\theta}_0}[L(\boldsymbol{\theta};\mathbf{x}_n) < L(\boldsymbol{\theta}_0;\mathbf{x}_n)] = \lim_{n\to\infty}P_{\boldsymbol{\theta}_0}\left[\frac{1}{n}\sum_{i=1}^{n}\log\frac{p(X_i;\boldsymbol{\theta})}{p(X_i;\boldsymbol{\theta}_0)} < 0\right] = 1, \tag{A.3.7}$$

which proves the claim.

As a global maximizer of the log-likelihood function, the MLE $\hat{\boldsymbol{\theta}}$ of a regular model must be a root of the log-likelihood function, i.e., it must satisfy the *likelihood equation*,

$$\nabla_{\boldsymbol{\theta}}\ell(\hat{\boldsymbol{\theta}}) = \mathbf{0}. \tag{A.3.8}$$

whenever it exists. For an arbitrary model, there may be other roots as well, even when the MLE doesn't exist. Assuming **RC2** and **RC5-8**, it can be shown that there will always be at least one sequence of roots $\hat{\boldsymbol{\theta}}_n$ of its log-likelihood such that $\hat{\boldsymbol{\theta}}_n$ tends to $\boldsymbol{\theta}_0$ in probability as $n \to \infty$ (Cf. Cramér 1945). The MLE will not necessarily be a part of this sequence though, even if it exists. Adding **RC9** is enough to ensure the MLE must be the unique solution to the likelihood equation, however, and therefore this sequence of roots will also be unique and for a given sample $\mathbf{x}_n$, the corresponding root $\hat{\boldsymbol{\theta}}_n = \hat{\boldsymbol{\theta}}(\mathbf{x}_n)$ will be the unique MLE of $\boldsymbol{\theta}_0$. It follows that the MLE is a consistent estimator of $\boldsymbol{\theta}_0$ for regular models.

We now turn our attention to the asymptotic distribution of the MLE. Similarly to the log-likelihood function, **RC1** implies the score function is equal to

$$\begin{aligned}
\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n) &= \nabla_{\boldsymbol{\theta}}\ell(\boldsymbol{\theta};\mathbf{x}_n) \\
&= \nabla_{\boldsymbol{\theta}}\sum_{i=1}^{n}\ell(\boldsymbol{\theta};x_i) \\
&= \sum_{i=1}^{n}\nabla_{\boldsymbol{\theta}}\ell(\boldsymbol{\theta};x_i) \\
&= \sum_{i=1}^{n}\mathcal{S}(\boldsymbol{\theta};x_i).
\end{aligned} \tag{A.3.9}$$

where the last equality is true by . In other words, the score function for the parameter $\boldsymbol{\theta}$ based on data $x_1, ..., x_n$ can be written as the sum of independent contributions $\mathcal{S}(\boldsymbol{\theta};x_i)$ $(i = 1, ..., n)$ where each $\mathcal{S}(\boldsymbol{\theta};x_i)$ can be thought of as the score function for $\boldsymbol{\theta}$ based only on observation x_i. This implies that a Taylor series expansion of $\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n)$ will be equal to the sum of the Taylor series expansions of its individual contributions, plus a remainder term that depends on n. Since the observations are identically distributed, it suffices to consider the expansion for an arbitrary contribution, $\mathcal{S}(\boldsymbol{\theta};x_i)$.

RC4-5 guarantee the existence of a neighborhood of $\boldsymbol{\theta}_0$ on which the first two partial derivatives of $\mathcal{S}(\boldsymbol{\theta};x_i)$ with respect to $\boldsymbol{\theta}$ exist and are continuous. Without loss of generality, we may assume this neighborhood, call it $N_{\boldsymbol{\theta}_0}$, is convex so that it contains all of the line segments connecting any two of its points. In particular, for any $\boldsymbol{\theta} \in N_{\boldsymbol{\theta}_0}$, $\mathrm{LS}(\boldsymbol{\theta}, \boldsymbol{\theta}_0) \subset N_{\boldsymbol{\theta}_0}$. Then by Taylor's theorem with the Lagrange form of the remainder, there exists $\tilde{\boldsymbol{\theta}}_j \in \mathrm{LS}(\boldsymbol{\theta}, \boldsymbol{\theta}_0)$ such that the j-th component of $\mathcal{S}(\boldsymbol{\theta};x_i)$ may be expanded as

$$\begin{aligned}\mathcal{S}_j(\boldsymbol{\theta}; x_i) &= \mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)\nabla_{\boldsymbol{\theta}}\mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\mathbf{H}_{\boldsymbol{\theta}}\Big(\mathcal{S}_j(\tilde{\boldsymbol{\theta}}_j; x_i)\Big)(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top \\ &= \mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)\Big[\nabla_{\boldsymbol{\theta}}\mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + \frac{1}{2}\mathbf{H}_{\boldsymbol{\theta}}\Big(\mathcal{S}_j(\tilde{\boldsymbol{\theta}}_j; x_i)\Big)(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top\Big] \\ &= \mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)\Big[\nabla_{\boldsymbol{\theta}}\mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + M(x_i)O(||\boldsymbol{\theta} - \boldsymbol{\theta}_0||)\Big] \qquad \text{(by } \mathbf{RC6}\text{)} \\ &= \mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + \Big[\nabla_{\boldsymbol{\theta}}^\top\mathcal{S}_j(\boldsymbol{\theta}_0; x_i) + M(x_i)O(||\boldsymbol{\theta} - \boldsymbol{\theta}_0||)\Big](\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top.\end{aligned}$$

When we stack each of these individual equations into a system of equations, we get

$$\begin{pmatrix}\mathcal{S}_1(\boldsymbol{\theta}; x_i)\\ \vdots \\ \mathcal{S}_d(\boldsymbol{\theta}; x_i)\end{pmatrix} = \begin{pmatrix}\mathcal{S}_1(\boldsymbol{\theta}_0; x_i)\\ \vdots \\ \mathcal{S}_d(\boldsymbol{\theta}_0; x_i)\end{pmatrix} + \left[\begin{pmatrix}\nabla^\top\mathcal{S}_1(\boldsymbol{\theta}_0; x_i)\\ \vdots \\ \nabla^\top\mathcal{S}_d(\boldsymbol{\theta}_0; x_i)\end{pmatrix} + M(x_i)O(||\boldsymbol{\theta} - \boldsymbol{\theta}_0||)\mathbf{1}_{d\times d}\right](\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top. \tag{A.3.10}$$

The matrix of gradient vectors in the second term on the right-hand side of the above equation is simply the Jacobian of the score function evaluated at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, i.e., $\mathbf{J}_{\boldsymbol{\theta}}\Big(\mathcal{S}(\boldsymbol{\theta}_0; x_i)\Big)$. However, by Equation A.1.4, this is just the negative transpose of the observed information matrix also evaluated at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, $\mathcal{I}(\boldsymbol{\theta}_0)$. Furthermore, since we have assumed ℓ is continuous in $\boldsymbol{\theta}$, we can freely swap the order of differentiation in all of its second partial derivatives with respect to $\boldsymbol{\theta}$. This implies the Jacobian of the score will be a symmetric matrix, and so we simply have $\mathbf{J}_{\boldsymbol{\theta}}\Big(\mathcal{S}(\boldsymbol{\theta}_0; x_i)\Big) = -\mathcal{I}_i(\boldsymbol{\theta}_0)$. Hence, using more compact notation, Equation A.3.10 becomes

$$\begin{aligned}\mathcal{S}(\boldsymbol{\theta}; x_i) &= \mathcal{S}(\boldsymbol{\theta}_0; x_i) + \Big[\mathbf{J}_{\boldsymbol{\theta}}\Big(\mathcal{S}(\boldsymbol{\theta}_0; x_i)\Big) + M(x_i)O(||\boldsymbol{\theta} - \boldsymbol{\theta}_0||)\mathbf{1}_{d\times d}\Big](\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top \\ &= \mathcal{S}(\boldsymbol{\theta}_0; x_i) - \Big[\mathcal{I}_i(\boldsymbol{\theta}_0) + M(x_i)O(||\boldsymbol{\theta} - \boldsymbol{\theta}_0||)\mathbf{1}_{d\times d}\Big](\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top\end{aligned} \tag{A.3.11}$$

The above represents the second-order Taylor expansion around $\boldsymbol{\theta}_0$ for an individual observation x_i's contribution to the score function. Summing over all of these contributions yields

$$\begin{aligned}
\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n) &= \sum_{i=1}^{n} \mathcal{S}(\boldsymbol{\theta};x_i) \\
&= \sum_{i=1}^{n} \left[\mathcal{S}(\boldsymbol{\theta}_0;x_i) - \left[\mathcal{I}_{X_i}(\boldsymbol{\theta}_0) + M(x_i)O(||\boldsymbol{\theta}-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] (\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\top} \right] \\
&= \mathcal{S}(\boldsymbol{\theta}_0;\mathbf{x}_n) - \left[\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + \left\{ \sum_{i=1}^{n} M(x_i) \right\} O(||\boldsymbol{\theta}-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] (\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\top} \\
&= \mathcal{S}(\boldsymbol{\theta}_0;\mathbf{x}_n) - \left[\frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + \left\{ \frac{1}{n}\sum_{i=1}^{n} M(x_i) \right\} O(||\boldsymbol{\theta}-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] n(\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\top}.
\end{aligned} \tag{A.3.12}$$

If we divide through by $\sqrt{n}$, we arrive at

$$\frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta};\mathbf{x}_n) = \frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta}_0;\mathbf{x}_n) - \left[\frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + \left\{ \frac{1}{n}\sum_{i=1}^{n} M(x_i) \right\} O(||\boldsymbol{\theta}-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] \sqrt{n}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)^{\top}. \tag{A.3.13}$$

We previously established that regular models will always have a sequence of MLEs $\hat{\boldsymbol{\theta}}_n$ that converge in probability to $\boldsymbol{\theta}_0$ as $n \to \infty$, and that each $\hat{\boldsymbol{\theta}}_n$ in this sequence will satisfy $\mathcal{S}(\hat{\boldsymbol{\theta}}_n;\mathbf{x}_n) = \mathbf{0}$. Plugging $\hat{\boldsymbol{\theta}}_n$ in for $\boldsymbol{\theta}$ in Equation A.3.13 gives

$$\frac{1}{\sqrt{n}}\mathcal{S}(\hat{\boldsymbol{\theta}}_n;\mathbf{x}_n) = \frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta}_0;\mathbf{x}_n) - \left[\frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + \left\{ \frac{1}{n}\sum_{i=1}^{n} M(x_i) \right\} O(||\hat{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] \sqrt{n}(\hat{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0)^{\top}$$

and therefore,

$$\frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta}_0;\mathbf{x}_n) = \left[\frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + \left\{ \frac{1}{n}\sum_{i=1}^{n} M(x_i) \right\} O_p(||\hat{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0||)\mathbf{1}_{d\times d} \right] \sqrt{n}(\hat{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0)^{\top}. \tag{A.3.14}$$

For the terms in the square brackets in the line above, we have the following observations:

1) By Equation 3.5.4, $\mathrm{E}_{\boldsymbol{\theta}_0}[\mathcal{I}_X(\boldsymbol{\theta}_0)] = \mathscr{I}_X(\boldsymbol{\theta}_0)$, and thus $\frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{I}_{X_i}(\boldsymbol{\theta}_0)$ is converging in probability to $\mathscr{I}_X(\boldsymbol{\theta}_0)$ as $n \to \infty$ by the Weak Law of Large Numbers, i.e., $\mathscr{I}_X(\boldsymbol{\theta}_0) = \frac{1}{n}\mathcal{I}_{\mathbf{X}_n}(\boldsymbol{\theta}_0) + o_p(1)$.

2) $M(x_i)$ has finite expectation and does not depend on $\boldsymbol{\theta}$ by **RC6**, which implies through Markov's inequality that it is bounded in probability as $n \to \infty$, i.e., $M(x_i) = O_p(1)$. It follows that $\frac{1}{n}\sum_{i=1}^{n} M(x_i) = O_p(1)$ as well.

3) The fact that $\hat{\boldsymbol{\theta}}_n$ is converging in probability to $\boldsymbol{\theta}_0$ as $n \to \infty$ implies that $||\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0||$ is $o_p(1)$.

From these three facts we can conclude that the entire term inside the square brackets is converging in probability to $\mathscr{I}_X(\boldsymbol{\theta}_0)$ as $n \to \infty$. This allows us to rewrite Equation A.3.14 as

$$\frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta}_0; \mathbf{x}_n) = \Big[\mathscr{I}_X(\boldsymbol{\theta}_0) + o_p(1)\Big]\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^\top. \tag{A.3.15}$$

This is useful because if we know the distribution to which the term on the left-hand side is converging as $n \to \infty$, we can deduce the asymptotic distribution of $\hat{\boldsymbol{\theta}}_n$ using Slutsky's theorem.

By definition, $\mathrm{Var}_{\boldsymbol{\theta}_0}[\mathcal{S}(\boldsymbol{\theta}_0; x_i)] = \mathscr{I}_X(\boldsymbol{\theta}_0)$. From Equation A.3.9, we have that $\frac{1}{n}\mathcal{S}(\boldsymbol{\theta}_0; \mathbf{x}_n) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{S}(\boldsymbol{\theta}_0; x_i)$. It follows from the Central Limit Theorem that

$$\sqrt{n}\left(\frac{1}{n}\mathcal{S}(\boldsymbol{\theta}_0; \mathbf{x}_n) - \mathrm{E}_{\boldsymbol{\theta}_0}\Big[\mathcal{S}(\boldsymbol{\theta}_0; x_i)\Big]\right) \xrightarrow{d} \mathrm{N}\Big(\mathbf{0}, \mathscr{I}_X(\boldsymbol{\theta}_0)\Big) \text{ as } n \to \infty. \tag{A.3.16}$$

But by the first Bartlett identity, $\mathrm{E}_{\boldsymbol{\theta}_0}\Big[\mathcal{S}(\boldsymbol{\theta}_0; x_i)\Big] = \mathbf{0}$, and therefore

$$\frac{1}{\sqrt{n}}\mathcal{S}(\boldsymbol{\theta}_0; \mathbf{x}_n) \xrightarrow{d} \mathrm{N}\Big(\mathbf{0}, \mathscr{I}_X(\boldsymbol{\theta}_0)\Big) \text{ as } n \to \infty. \tag{A.3.17}$$

Combining the results of Equation A.3.15 and Equation A.3.17, we see that

$$\Big[\mathscr{I}_X(\boldsymbol{\theta}_0) + o_p(1)\Big]\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^\top \xrightarrow{d} \mathrm{N}\Big(\mathbf{0}, \mathscr{I}_X(\boldsymbol{\theta}_0)\Big) \text{ as } n \to \infty. \tag{A.3.18}$$

Since the term in square brackets is converging in probability to $\mathscr{I}(\boldsymbol{\theta}_0)$, which by **RC8** is a positive definite matrix, a minor extension of Slutsky's Theorem (see Appendix A) allows us to deduce that the asymptotic distribution of the MLE for the true parameter value of a regular model is given by

$$\begin{aligned}\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)^\top &\xrightarrow{d} \mathscr{I}_X(\boldsymbol{\theta}_0)^{-1}\mathrm{N}\Big(\mathbf{0}, \mathscr{I}_X(\boldsymbol{\theta}_0)\Big) \\ &\stackrel{d}{=} \mathrm{N}\Big(\mathbf{0}, \mathscr{I}_X(\boldsymbol{\theta}_0)^{-1}\Big) \text{ as } n \to \infty.\end{aligned} \tag{A.3.19}$$

A.3.1.2. Two-Index Asymptotics. Earlier we discussed the one-index asymptotics setting, in which the sample size (n) of the model diverged to infinity while the dimension of the nuisance parameter (q) remained fixed. Now we turn our attention to the two-index asymptotics setting which describes the behavior of likelihood and pseudolikelihood functions as n and q both tend to infinity, with q growing at least as fast as n. Under such a framework, De Bin et al. (2015) showed that estimates for ψ based on a suitably constructed integrated likelihood function will outperform those coming from more traditional pseudolikelihoods, such as the profile likelihood. Such findings provide the motivation for our ensuing examination of two-index asymptotics theory, insofar as it relates to the performance of the integrated likelihood function as a method of inference regarding a parameter of interest.

To mirror the strategy used by Sartori (2003) and De Bin et al. (2015), we will frame our discussion in terms of a stratified sample of data in which each stratum contributes one component to the overall nuisance parameter. Consider a model with parameter $\boldsymbol{\theta} = (\psi, \lambda)$ where ψ is the parameter of interest and $\lambda = (\lambda_1, ..., \lambda_q)$ is a q-dimensional nuisance parameter. For the sake of reducing complexity in our notation, we will only consider the case in which ψ and the individual components of λ are scalar parameters, though the results of this section should hold in the case where they are all vectors as well. Suppose that we have divided the model's population into q strata and collected a sample of size m_i from each stratum such that observation j from stratum i may be modeled as

$$X_{ij} \sim p_{ij}(x_{ij}; \psi, \lambda_i), \tag{A.3.20}$$

where $i = 1, ..., q$ and $j = 1, ..., m_i$, making the total sample size $n = \sum_{i=1}^{q} m_i$.[2] Hence, there is a one-to-one correspondence between the strata and the components of λ; assume that each $\lambda_i \in \Lambda$, where the space Λ is the same for all i, and they all have the same interpretation within their respective strata.

Assume that all the regularity conditions set forth in Section 3.4 apply, except possibly for **RC1** - it is not necessary to assume that the observations are i.i.d. here, and in fact it is perfectly acceptable for the

[2] It will be convenient to work under the restriction that the stratum sample sizes are all identical, meaning there exists some positive integer m such that $m_i = m$ for all i. However, as both Sartori (2003) and De Bin et al. (2015) note, we could also assume a looser condition in which $m_i = K_i m$ where $0 < K_i < \infty$ without compromising our results.

p_{ij}'s in Equation A.3.20 to differ from one another. We will also allow for the possibility of dependence among observations within a stratum, though not between them.

Let $\mathbf{x}_i = (x_{i1}, ..., x_{im})$ denote the sample of observations from stratum i, so that their joint density may be written as $p_i(\mathbf{x}_i; \psi, \lambda_i)$. Therefore, the likelihood and log-likelihood for the ith stratum are

$$L^{(i)}(\psi, \lambda_i) = p_i(\mathbf{x}_i; \psi, \lambda_i), \tag{A.3.21}$$

and

$$\ell^{(i)}(\psi, \lambda_i) = \log L^{(i)}(\psi, \lambda_i), \tag{A.3.22}$$

respectively. For a particular choice of weight function $g(\lambda_i; \psi)$, the integrated likelihood for ψ in stratum i is given by

$$\bar{L}^{(i)}(\psi) = \int_\Lambda L^{(i)}(\psi, \lambda_i) g(\lambda_i; \psi) d\lambda_i. \tag{A.3.23}$$

From here, we proceed by using Laplace's method as described by Tierney & Kadane (1986) (see Appendix B for a brief review) to obtain an analytic approximation to $\bar{L}^{(i)}(\psi)$. Setting $h(\lambda_i) = -\frac{1}{m}\ell^{(i)}(\psi, \lambda_i)$ and $f(\lambda_i) = g(\lambda_i; \psi)$, we may rewrite the integral in Equation A.3.23 as

$$\bar{L}^{(i)}(\psi) = \int_\Lambda f(\lambda_i) \exp[-mh(\lambda_i)] d\lambda_i.$$

One consequence of the regularity conditions we have assumed is that $L^{(i)}(\psi, \lambda_i)$ is that an MLE for $\boldsymbol{\theta}_0$, $\hat{\boldsymbol{\theta}}$, exists and is unique. This further implies the existence and uniqueness of a conditional MLE for the true value of each stratum-specific nuisance parameter given ψ - denote this value for the ith stratum by $\hat{\lambda}_{i\psi}$. By definition this value maximizes $\ell^{(i)}(\psi, \lambda_i)$ as a function of λ_i, and so it also maximizes $-h(\lambda_i)$ since the two functions differ only by a multiplicative constant $\frac{1}{m}$.

The Laplace approximation to $\bar{L}^{(i)}(\psi)$ is then given by

$$\hat{\bar{L}}^{(i)}(\psi) = f(\hat{\lambda}_{i\psi}) \sqrt{\frac{2\pi}{m}} \sigma \exp[-mh(\hat{\lambda}_{i\psi})], \tag{A.3.24}$$

where

$$\begin{aligned}\sigma &= \left[\frac{\partial^2 h}{\partial \lambda_i^2}\bigg|_{\lambda_i=\hat{\lambda}_{i\psi}}\right]^{-1/2}\\ &= \left[-\frac{1}{m}\frac{\partial^2 \ell^{(i)}(\psi,\lambda_i)}{\partial \lambda_i^2}\bigg|_{\lambda_i=\hat{\lambda}_{i\psi}}\right]^{-1/2}\\ &= \left[\frac{1}{m}\mathcal{I}(\hat{\lambda}_{i\psi})\right]^{-1/2}.\end{aligned}$$

Here, $\mathcal{I}(\hat{\lambda}_{i\psi})$ denotes the observed information function for λ_i only (i.e. the negative second partial derivative of the log-likelihood with respect to λ_i) evaluated at $\hat{\lambda}_{i\psi}$. Plugging in the appropriate quantities for f, h, and σ into Equation A.3.24, we arrive at

$$\begin{aligned}\hat{\bar{L}}^{(i)}(\psi) &= f(\hat{\lambda}_{i\psi})\sqrt{\frac{2\pi}{m}}\sigma\exp[-mh(\hat{\lambda}_{i\psi})]\\ &= g(\hat{\lambda}_{i\psi};\psi)\sqrt{\frac{2\pi}{m}}\left[\frac{1}{m}\mathcal{I}(\hat{\lambda}_{i\psi})\right]^{-1/2}\exp\left\{-m\cdot-\frac{1}{m}\ell^{(i)}(\psi,\hat{\lambda}_{i\psi})\right\}\\ &= \frac{\sqrt{2\pi}}{m}L^{(i)}(\psi,\hat{\lambda}_{i\psi})g(\hat{\lambda}_{i\psi};\psi)[\mathcal{I}(\hat{\lambda}_{i\psi})]^{-1/2}\\ &= \frac{\sqrt{2\pi}}{m}L_P^{(i)}(\psi)g(\hat{\lambda}_{i\psi};\psi)[\mathcal{I}(\hat{\lambda}_{i\psi})]^{-1/2},\end{aligned} \tag{A.3.25}$$

where $L_P^{(i)}(\psi) = L^{(i)}(\psi,\hat{\lambda}_{i\psi})$ is the profile likelihood for ψ. The error in this approximation is

$$\bar{L}^{(i)}(\psi) = \hat{\bar{L}}^{(i)}(\psi)\left\{1 + O\left(\frac{1}{m}\right)\right\} \quad \text{as} \quad m \to \infty. \tag{A.3.26}$$

Let

$$\bar{\ell}^{(i)}(\psi) = \log \bar{L}^{(i)}(\psi) \tag{A.3.27}$$

denote the integrated log-likelihood for ψ. Putting the results in Equation A.3.25, Equation A.3.26, and Equation A.3.27 together, we have

$$\begin{aligned}
\bar{\ell}^{(i)}(\psi) &= \log \bar{L}^{(i)}(\psi) && \text{(by Equation 4.2.8)} \\
&= \log\left(\hat{\bar{L}}^{(i)}(\psi)\left\{1 + O\left(\frac{1}{m}\right)\right\}\right) && \text{(by Equation 4.2.7)} \\
&= \log \hat{\bar{L}}^{(i)}(\psi) + \log\left\{1 + O\left(\frac{1}{m}\right)\right\} && \text{(by Equation 4.2.6)} \\
&= \log\left\{\frac{\sqrt{2\pi}}{m} L_P^{(i)}(\psi) g(\hat{\lambda}_{i\psi}; \psi) \left[\mathcal{J}(\hat{\lambda}_{i\psi})\right]^{-1/2}\right\} + O\left(\frac{1}{m}\right) \\
&= \frac{1}{2}\log(2\pi) - \log(m) + \log L_P^{(i)}(\psi) + \log g(\hat{\lambda}_{i\psi}; \psi) - \frac{1}{2}\log \mathcal{J}(\hat{\lambda}_{i\psi}) + O\left(\frac{1}{m}\right) \\
&= \ell_P^{(i)}(\psi) + \log g(\hat{\lambda}_{i\psi}; \psi) - \frac{1}{2}\log \mathcal{J}(\hat{\lambda}_{i\psi}) + \frac{1}{2}\log(2\pi) - \log(m) + O\left(\frac{1}{m}\right) \quad \text{as} \quad m \to \infty,
\end{aligned}$$

where $\ell_P^{(i)}(\psi) = \ell^{(i)}(\psi, \hat{\lambda}_{i\psi})$ is the profile log-likelihood for ψ. Since log-likelihoods are equivalent up to additive constants, we can discard the $\frac{1}{2}\log(2\pi)$ and $\log(m)$ terms in the final line above to arrive at our final approximation for the integrated log-likelihood in stratum i:

$$\hat{\bar{\ell}}^{(i)}(\psi) = \ell_P^{(i)}(\psi) + \log g(\hat{\lambda}_{i\psi}; \psi) - \frac{1}{2}\log \mathcal{J}(\hat{\lambda}_{i\psi}). \tag{A.3.28}$$

The error in this approximation is given by

$$\bar{\ell}^{(i)}(\psi) = \hat{\bar{\ell}}^{(i)}(\psi) + O\left(\frac{1}{m}\right). \tag{A.3.29}$$

Since the observations between the strata are independent, we may write the likelihood and log-likelihood functions for the entire model as

$$L(\psi, \lambda) = \prod_{i=1}^{q} L^{(i)}(\psi, \lambda_i) \tag{A.3.30}$$

and

$$\ell(\psi, \lambda) = \sum_{i=1}^{q} \ell^{(i)}(\psi, \lambda_i), \tag{A.3.31}$$

respectively. If we define the weight function

$$G(\lambda;\psi) = \prod_{i=1}^{q} g(\lambda_i;\psi) \tag{A.3.32}$$

then the integrated likelihood function for ψ becomes separable. That is,

$$\begin{aligned}
\bar{L}(\psi) &= \int_{\Lambda^q} L(\psi,\lambda)G(\lambda;\psi)d\lambda \\
&= \int_{\Lambda}\cdots\int_{\Lambda}\left[\prod_{i=1}^{q} L^{(i)}(\psi,\lambda_i)\right]\left[\prod_{i=1}^{q} g(\lambda_i;\psi)\right]d\lambda_1\cdots d\lambda_q \\
&= \prod_{i=1}^{q}\int_{\Lambda} L^{(i)}(\psi,\lambda_i)g(\lambda_i;\psi)d\lambda_i \\
&= \prod_{i=1}^{q}\bar{L}^{(i)}(\psi).
\end{aligned} \tag{A.3.33}$$

Let $\bar{\ell}(\psi) = \log \bar{L}(\psi)$ denote the integrated log-likelihood function for ψ. Taking the logarithm of both sides in Equation A.3.33, we have

$$\bar{\ell}(\psi) = \sum_{i=1}^{q}\bar{\ell}^{(i)}(\psi). \tag{A.3.34}$$

Plugging in our approximation to $\bar{\ell}^{(i)}(\psi)$ and its error term in Equation A.3.28 and Equation A.3.29, respectively, yields

$$\begin{aligned}
\bar{\ell}(\psi) &= \sum_{i=1}^{q}\left[\ell_P^{(i)}(\psi) + \log g(\hat{\lambda}_{i\psi};\psi) - \log \mathcal{I}(\hat{\lambda}_{i\psi}) + O\!\left(\frac{1}{m}\right)\right] \\
&= \ell_P(\psi) + \sum_{i=1}^{q}\log g(\hat{\lambda}_{i\psi};\psi) - \frac{1}{2}\sum_{i=1}^{q}\log \mathcal{I}(\hat{\lambda}_{i\psi}) + O\!\left(\frac{q}{m}\right) \quad \text{as} \quad m\to\infty.
\end{aligned} \tag{A.3.35}$$

APPENDIX B

Chapter 3

B.1. Monte Carlo Methods

B.1.1. Simple Monte Carlo

B.1.2. Importance Sampling

To obtain an integrated likelihood for ψ alone, we can use the procedure described in the previous chapter to find an approximation to the integral in **?@eq-ZSE_IL11** where $\widetilde{L}(u;\psi)$ is the likelihood function reparameterized in terms of the ZSE parameter, and $\check{L}(u)$ and $\check{\pi}(u)$ are chosen such that $\check{\pi}$ is a conjugate prior for $\check{L}$.[1] In this case, a natural choice exists due to the fact that the Dirichlet distribution is a conjugate prior for the multinomial distribution. Since our original likelihood function L is based on a multinomial distribution, we can simply set $\check{L}(u) = L(u)$ and $\check{\pi}(u) \sim \text{Dir}(\boldsymbol{\alpha})$, where $\boldsymbol{\alpha} = (\alpha_1, ..., \alpha_d)$. Then the posterior distribution for u based on data $\mathbf{n} = (n_1, ..., n_d)$ is given by $L(u)\check{\pi}(u) \sim \text{Dir}(\mathbf{n} + \boldsymbol{\alpha})$. Consequently, we will take $\check{\pi}(u)$ to be the symmetric Dirichlet distribution on the probability simplex in $\mathbb{R}^d$ with $\alpha_j = 1$ for all j so that random variates of u can be sampled from a $\text{Dir}(\mathbf{n} + 1)$ distribution.

From **?@eq-ZSE_IL9**, finding $\widetilde{L}(u;\psi)$ is a matter of defining two functions, Q and T_ψ such that $\widetilde{L}(u;\psi) = L(T_\psi(Q(u))$. Q maps a random variate u sampled from the above posterior to an element ω in $\Omega_{\hat{\psi}}$, and T_ψ then maps ω to an element $\boldsymbol{\theta}$ in $\Theta(\psi)$. Since in this situation, these quantities u, ω, and $\boldsymbol{\theta}$ are all members of the probability simplex, the maps Q and T_ψ can be defined as returning the elements in their respective output spaces that are closest to the input element they have each received. That is, Q returns the element ω that minimizes the distance to an input u, subject to the constraints that the sum of the components of ω equal 1 and $\varphi(\omega) = \hat{\psi}$. Similarly, T_ψ returns the element $\boldsymbol{\theta}$ that minimizes

[1] This procedure is an adapted version of another algorithm developed by Severini (2022) for approximating the integrated likelihood for the entropy of a multinomial distribution.

the distance to an input ω, subject to the constraints that the sum of the components of $\boldsymbol{\theta}$ equal 1 and $\varphi(\boldsymbol{\theta}) = \psi$.

From here, we sample R random variates from the appropriate Dirichlet distribution, calculate $\widetilde{L}(u;\psi)$ and $L(u)$ for each one, and use **?@eq-ZSE__IL12** to approximate the integrated likelihood at a particular value of ψ. We then repeat this process over a finely spaced sequence of the possible values of ψ in order to get a shape of the overall integrated likelihood. See Appendix C for a graph comparing the plot of one such integrated likelihood to the profile likelihood, for observed data $(n_1, ..., n_6) = (1, 1, 4, 7, 10)$ and 250 samples of the appropriate Dirichlet distribution drawn for each value of ψ.[2]

[2]The samples were obtained using the 'LaplacesDemon' R package. The distance minimizations needed for Q and T_ψ were computed numerically using the 'nloptr' R package.

APPENDIX C

Chapter 4

C.1. Geometry of $\Omega_{\hat{\psi}}$ for Simpson's Index

Recall that for a model with full parameter $\boldsymbol{\theta} \in \Theta$ and parameter of interest $\psi = \varphi(\boldsymbol{\theta})$, the set

$$\Omega_{\hat{\psi}} = \left\{\boldsymbol{\omega} \in \Theta : \varphi(\boldsymbol{\omega}) = \hat{\psi}\right\} \tag{C.1.1}$$

is the level set of φ at the MLE $\hat{\psi}$. In the multinomial model with J categories, $\Theta = \Delta^{J-1}$, where

$$\Delta^{J-1} = \left\{\boldsymbol{\theta} \in \mathbb{R}^J : \theta_j > 0,\ \sum_{j=1}^{J} \theta_j = 1\right\}, \tag{C.1.2}$$

the open $(J-1)$-dimensional probability simplex. Taking Simpson's index

$$D(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|^2 = \sum_{j=1}^{J} \theta_j^2 \tag{C.1.3}$$

as the parameter of interest, i.e. $\varphi = D$, gives

$$\Omega_{\hat{\psi}} = \left\{\boldsymbol{\theta} \in \Delta^{J-1} : \|\boldsymbol{\theta}\|^2 = \hat{\psi}\right\}. \tag{C.1.4}$$

Since the level sets of D are intersections of the simplex with hyperspheres, they admit a clean geometric characterization. We will derive this characterization for arbitrary J first, then consider the special cases $J = 2$ and $J = 3$ in detail for some added intuition. While these low-dimensional cases do not produce nuisance parameters of sufficient dimension for the integrated likelihood to produce estimates that meaningfully outperform the profile likelihood, they nevertheless serve as concrete illustrations of the general result.

C.1.1. General J

C.1.1.1. Level Set Structure. Define the affine hyperplane

$$\Pi = \left\{ \mathbf{v} \in \mathbb{R}^J : \sum_{j=1}^{J} v_j = 1 \right\}, \tag{C.1.5}$$

the $(J-1)$-sphere of radius $\sqrt{\hat{\psi}}$ centered at the origin,

$$\mathcal{S}_{\hat{\psi}} = \left\{ \mathbf{v} \in \mathbb{R}^J : \|\mathbf{v}\|^2 = \hat{\psi} \right\}, \tag{C.1.6}$$

and their intersection

$$\mathcal{L}_{\hat{\psi}} = \Pi \cap \mathcal{S}_{\hat{\psi}}. \tag{C.1.7}$$

Note that $\Delta^{J-1} \subset \Pi$. Therefore

$$\begin{aligned}
\Omega_{\hat{\psi}} &= \left\{ \boldsymbol{\theta} \in \Delta^{J-1} : \|\boldsymbol{\theta}\|^2 = \hat{\psi} \right\} && \text{(C.1.4)} \\
&= \Delta^{J-1} \cap \mathcal{S}_{\hat{\psi}} && \text{(C.1.6)} \\
&= \Delta^{J-1} \cap \Pi \cap \mathcal{S}_{\hat{\psi}} && (\text{since } \Delta^{J-1} \subset \Pi) \\
&= \Delta^{J-1} \cap \mathcal{L}_{\hat{\psi}}, && \text{(C.1.7)}
\end{aligned}$$

so $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ if and only if $\mathcal{L}_{\hat{\psi}} \subset \Delta^{J-1}$.

To build intuition for the general case, consider $J = 3$. Here Π is the plane through $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, and Δ^2 is the triangle with those vertices. The set $\mathcal{S}_{\hat{\psi}}$ is a sphere of radius $\sqrt{\hat{\psi}}$ centered at the origin, and $\mathcal{L}_{\hat{\psi}}$ is the circle formed by its intersection with Π. As $\hat{\psi}$ grows the circle expands, remaining entirely inside Δ^2 for small values of $\hat{\psi}$ and eventually clipping through its edges. When clipping occurs, $\Omega_{\hat{\psi}}$ is no longer the full circle but the arcs of $\mathcal{L}_{\hat{\psi}}$ that remain inside Δ^2 (see Figure C.1 in Section C.1.3). Whether and when clipping occurs depends on $\hat{\psi}$ and J, as derived in Section C.1.1.2.

We may write any $\mathbf{v} \in \Pi$ as $\mathbf{v} = \mathbf{c} + \mathbf{d}$, where $\mathbf{c} = (1/J)\mathbf{1}$ is the vector extending from the origin to the centroid of Δ^{J-1}, and $\mathbf{d} = \mathbf{v} - \mathbf{c}$ is the displacement vector extending from the centroid to $\mathbf{v}$. Since

$$\mathbf{1} \cdot \mathbf{d} = \sum_{j=1}^{J} d_j = \sum_{j=1}^{J} (v_j - c_j) = \sum_{j=1}^{J} v_j - \sum_{j=1}^{J} c_j = 1 - 1 = 0,$$

$\mathbf{d}$ is orthogonal to $\mathbf{1}$ and so orthogonal to $\mathbf{c}$ as well. Now suppose $\mathbf{v} \in \mathcal{L}_{\hat{\psi}}$, so it further satisfies $\|\mathbf{v}\|^2 = \hat{\psi}$. Expanding, we have

$$\hat{\psi} = \|\mathbf{v}\|^2 = \|\mathbf{c} + \mathbf{d}\|^2 = \underbrace{\|\mathbf{c}\|^2}_{1/J} + \underbrace{2\mathbf{c} \cdot \mathbf{d}}_{0} + \|\mathbf{d}\|^2 = \frac{1}{J} + \|\mathbf{d}\|^2,$$

so $\|\mathbf{d}\|^2 = \hat{\psi} - 1/J$. Thus, every point in $\mathcal{L}_{\hat{\psi}}$ lies at a distance

$$r(\hat{\psi}, J) = \sqrt{\hat{\psi} - \frac{1}{J}} \tag{C.1.8}$$

from $\mathbf{c}$, so $\mathcal{L}_{\hat{\psi}}$ is a $(J-2)$-sphere of radius $r(\hat{\psi}, J)$ centered at $\mathbf{c}$ within Π, degenerating to the single point $\mathbf{c}$ only when $\hat{\psi} = 1/J$, i.e. when the observed counts are exactly equal across all categories.

C.1.1.2. The Clipping Threshold. The facets of Δ^{J-1} are its $(J-2)$-dimensional faces, each consisting of all points with one coordinate equal to zero. Each facet is uniquely identified by the vertex opposite to it, so there are exactly J facets in total. Since swapping two vertices also swaps the corresponding two facets while leaving all others unchanged, and any such swap can be achieved by a permutation of coordinates that leaves $\mathbf{c} = (1/J)\mathbf{1}$ unchanged, the distance from $\mathbf{c}$ to any one facet equals the distance to any other. This common distance is called the inradius of Δ^{J-1}, and the shape of $\Omega_{\hat{\psi}}$ depends on whether $r(\hat{\psi}, J)$ exceeds it.

To determine the inradius, we need to know the coordinates of the closest point on a facet to $\mathbf{c}$. Since Δ^{J-1} is a convex set and the distance function is strictly convex, this closest point is unique. Any permutation of the non-zero coordinates maps the facet to itself and fixes $\mathbf{c}$, so by uniqueness of the closest point it must fix that point too, meaning all its non-zero coordinates are equal. For the facet opposite $\mathbf{e}_1$, consisting of all points with first coordinate equal to zero, the simplex constraint $\sum_j \omega_j = 1$ then uniquely determines the closest point as the facet centroid $\mathbf{p} = (0, 1/(J-1), \dots, 1/(J-1))$.

The inradius is therefore

$$r_{\text{in}} = \|\mathbf{c} - \mathbf{p}\| = \left\| \left(\frac{1}{J}, \frac{-1}{J(J-1)}, \dots, \frac{-1}{J(J-1)} \right) \right\| = \sqrt{\frac{1}{J^2} + \frac{J-1}{J^2(J-1)^2}} = \sqrt{\frac{1}{J(J-1)}}.$$

Setting $r(\hat{\psi}, J) = r_{\text{in}}$ gives the clipping threshold $\hat{\psi} = 1/(J-1)$. When $\hat{\psi} < 1/(J-1)$, $\mathcal{L}_{\hat{\psi}}$ lies entirely within Δ^{J-1}, and $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ is a complete $(J-2)$-sphere. At $\hat{\psi} = 1/(J-1)$, $\mathcal{L}_{\hat{\psi}}$ is simultaneously tangent to all J facets, touching at $(1/(J-1), \dots, 1/(J-1), 0)$ and its permutations. When $\hat{\psi} > 1/(J-1)$, $\mathcal{L}_{\hat{\psi}}$ clips through all J facets, and $\Omega_{\hat{\psi}} \subset \mathcal{L}_{\hat{\psi}}$, breaking into J disconnected but identical spherical caps, one near each vertex. We will refer to the cases where $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ and $\Omega_{\hat{\psi}} \subset \mathcal{L}_{\hat{\psi}}$ as the connected and disconnected regimes, respectively.

C.1.1.3. Cap Geometry. Suppose we are in the disconnected regime. Let $\mathcal{C}_j$ denote the cap near vertex $\mathbf{e}_j$, so that

$$\Omega_{\hat{\psi}} = \bigcup_{j=1}^{J} \mathcal{C}_j. \tag{C.1.9}$$

For a point $\boldsymbol{\omega} = (\omega_1, \dots, \omega_J) \in \mathcal{C}_j$, the coordinate ω_j is strictly larger than all others, reflecting the proximity of the cap to $\mathbf{e}_j$. We will refer to ω_j as the dominant coordinate and ω_k for $k \neq j$ as the non-dominant coordinates. Since $\omega_j > \omega_k$ for all $k \neq j$ and $\sum_{k=1}^{J} \omega_k = 1$, it follows that $J\omega_j > 1$ and hence $\omega_j > 1/J$.

The cap axis of $\mathcal{C}_j$ is the line segment from $\mathbf{c}$ to $\mathbf{e}_j$, with unit direction

$$\mathbf{n}_j = \frac{\mathbf{e}_j - \mathbf{c}}{\|\mathbf{e}_j - \mathbf{c}\|}. \tag{C.1.10}$$

The cap center is the point where the cap axis intersects $\mathcal{C}_j$. A boundary point of $\mathcal{C}_j$ is any point in $\mathcal{C}_j$ that lies on a facet of Δ^{J-1}, i.e. where one of the non-dominant coordinates equals zero. The angular radius α of $\mathcal{C}_j$ is the angle with vertex at $\mathbf{c}$ between the cap axis and the direction to any boundary point $\mathbf{b}$ of $\mathcal{C}_j$.

Since $\mathbf{n}_j$ is a unit vector and $\mathbf{b} \in \mathcal{L}_{\hat{\psi}}$ lies at a distance $r(\hat{\psi}, J)$ from $\mathbf{c}$ (see Equation C.1.8), α satisfies

$$\mathbf{n}_j \cdot (\mathbf{b} - \mathbf{c}) = r \cos \alpha,$$

and thus

$$\alpha = \arccos\left(\frac{\mathbf{n}_j \cdot (\mathbf{b} - \mathbf{c})}{r}\right). \tag{C.1.11}$$

By the symmetry established above, α is the same for all J caps and all boundary points of each cap, so it suffices to compute α for any boundary point of $\mathcal{C}_1$. Since α completely characterizes the extent of each cap, its computation as a function of $\hat{\psi}$ and J is a prerequisite for the sampling procedure developed in Section C.1.1.6.

Of the J facets of Δ^{J-1}, $\mathcal{C}_1$ is bounded by the $J - 1$ on which one of the non-dominant coordinates $\omega_2, \dots, \omega_J$ vanishes. Each boundary point therefore has $\omega_1 > 1/J$, exactly one of $\omega_2, \dots, \omega_J$ equal to zero, and the remaining $J - 2$ strictly positive. Without loss of generality, take the boundary point where $\omega_2 = 0$. Since α is the same for all boundary points, we may seek a representative of symmetric form with $\omega_3 = \cdots = \omega_J = a$ for some $a > 0$, and verify its existence by solving the resulting system explicitly.

Set $\omega_2 = 0$ and $\omega_3 = \cdots = \omega_J = a$. Since $\boldsymbol{\omega} \in \mathcal{L}_{\hat{\psi}}$, the two constraints defining $\mathcal{L}_{\hat{\psi}}$ give

$$\omega_1 + (J-2)a = 1 \quad \text{and} \quad \omega_1^2 + (J-2)a^2 = \hat{\psi}.$$

Substituting $\omega_1 = 1-(J-2)a$ from the first equation into the second and rearranging yields the quadratic $(J-2)(J-1)a^2 - 2(J-2)a + (1-\hat{\psi}) = 0$, with roots

$$\frac{1 - \sqrt{\frac{(J-1)\hat{\psi}-1}{J-2}}}{J-1} \quad \text{and} \quad \frac{1 + \sqrt{\frac{(J-1)\hat{\psi}-1}{J-2}}}{J-1}.$$

Both roots are real for $\hat{\psi} \geq 1/(J-1)$, confirming the existence of the symmetric boundary point. Since ω_1 is the dominant coordinate, $a < \omega_1$. This implies $1 = \omega_1 + (J-2)a > a + (J-2)a = (J-1)a$, and so $a < 1/(J-1)$. Only the smaller root satisfies this upper bound for a, so we must have

$$a = \frac{1 - \sqrt{\frac{(J-1)\hat{\psi}-1}{J-2}}}{J-1}. \tag{C.1.12}$$

It remains to evaluate Equation C.1.11 at $\mathbf{b} = (\omega_1, 0, a, \dots, a)$. We first compute $\mathbf{n}_1$ from Equation C.1.10. The vector $\mathbf{e}_1 - \mathbf{c}$ has components

$$\begin{aligned}
\mathbf{e}_1 - \mathbf{c} &= (1, 0, \dots, 0) - \left(\frac{1}{J}, \dots, \frac{1}{J}\right) \\
&= \left(\frac{J-1}{J}, -\frac{1}{J}, \dots, -\frac{1}{J}\right) \\
&= \frac{1}{J}\Big(J-1, \underbrace{-1, \dots, -1}_{J-1}\Big),
\end{aligned}$$

with norm

$$\begin{aligned}
\|\mathbf{e}_1 - \mathbf{c}\| &= \frac{1}{J}\sqrt{(J-1)^2 + \sum_{j=1}^{J-1}(-1)^2} \\
&= \frac{1}{J}\sqrt{(J-1)^2 + (J-1)} \\
&= \frac{1}{J}\sqrt{J(J-1)} \\
&= \sqrt{\frac{J-1}{J}},
\end{aligned}$$

and hence

$$\mathbf{n}_1 = \frac{\mathbf{e}_1 - \mathbf{c}}{\|\mathbf{e}_1 - \mathbf{c}\|} = \frac{\frac{1}{J}(J-1, -1, \dots, -1)}{\sqrt{\frac{J-1}{J}}} = \frac{1}{\sqrt{J(J-1)}}(J-1, -1, \dots, -1).$$

The displacement $\mathbf{b} - \mathbf{c}$ has components

$$\mathbf{b} - \mathbf{c} = (\omega_1, 0, a, \dots, a) - \left(\frac{1}{J}, \dots, \frac{1}{J}\right) = \Big(\omega_1 - \frac{1}{J},\ -\frac{1}{J},\ \underbrace{a - \frac{1}{J}, \dots, a - \frac{1}{J}}_{J-2}\Big).$$

Taking the dot product,

$$\begin{aligned}
\mathbf{n}_1 \cdot (\mathbf{b} - \mathbf{c}) &= \frac{1}{\sqrt{J(J-1)}}(J-1, -1, \dots, -1) \cdot \left(\omega_1 - \frac{1}{J},\ -\frac{1}{J},\ a - \frac{1}{J},\ \dots,\ a - \frac{1}{J}\right) \\
&= \frac{(J-1)\left(\omega_1 - \frac{1}{J}\right) + \frac{1}{J} - (J-2)\left(a - \frac{1}{J}\right)}{\sqrt{J(J-1)}} \\
&= \frac{(J-1)\omega_1 - (J-2)a - \frac{J-2}{J} + \frac{J-2}{J}}{\sqrt{J(J-1)}} \\
&= \frac{(J-1)\omega_1 - (J-2)a}{\sqrt{J(J-1)}}.
\end{aligned}$$

Substituting $\omega_1 = 1-(J-2)a$ and a as in Equation C.1.12, the numerator simplifies to

$$\begin{aligned}
(J-1)\omega_1-(J-2)a &= (J-1)\left[1-(J-2)a\right]-(J-2)a \\
&= (J-1)-J(J-2)a \\
&= (J-1)-J(J-2)\left[\frac{1-\sqrt{\frac{(J-1)\hat{\psi}-1}{J-2}}}{J-1}\right] \\
&= \frac{(J-1)^2-J(J-2)}{J-1}+\frac{J\sqrt{(J-2)\left((J-1)\hat{\psi}-1\right)}}{J-1} \\
&= \frac{1+J\sqrt{(J-2)((J-1)\hat{\psi}-1)}}{J-1}.
\end{aligned}$$

Dividing by $r(\hat{\psi},J)$, we have

$$\frac{1}{r(\hat{\psi},J)}\mathbf{n}_1\cdot(\mathbf{b}-\mathbf{c}) = \frac{1}{\sqrt{\hat{\psi}-\frac{1}{J}}}\cdot\frac{\frac{1+J\sqrt{(J-2)\left((J-1)\hat{\psi}-1\right)}}{J-1}}{\sqrt{J(J-1)}} = \frac{1+J\sqrt{(J-2)((J-1)\hat{\psi}-1)}}{(J-1)\sqrt{(J-1)(J\hat{\psi}-1)}}.$$

Substituting into Equation C.1.11 gives the angular radius

$$\alpha(\hat{\psi},J) = \arccos\left(\frac{1+J\sqrt{(J-2)((J-1)\hat{\psi}-1)}}{(J-1)\sqrt{(J-1)(J\hat{\psi}-1)}}\right), \qquad \hat{\psi}\in\left[\frac{1}{J-1},1\right). \tag{C.1.13}$$

At $\hat{\psi}=1/(J-1)$, $\alpha=\arccos(1/(J-1))$; as $\hat{\psi}\to 1$, $\alpha\to 0$ and each cap collapses to its vertex.

C.1.1.4. Permutation Symmetry and Monte Carlo Sampling. The symmetric group S_J acts on $\Omega_{\hat{\psi}}$ by permuting category labels. The orbit of a point $\boldsymbol{\omega}\in\Omega_{\hat{\psi}}$ under this action is the set $\{\sigma(\boldsymbol{\omega}):\sigma\in S_J\}$ of all points reachable by permuting the coordinates of $\boldsymbol{\omega}$. In both regimes, S_J partitions $\Omega_{\hat{\psi}}$ into J geometrically identical components, one associated with each vertex of Δ^{J-1}: in the disconnected regime these are the J spherical caps, and in the connected regime they are J equal sectors of $\mathcal{L}_{\hat{\psi}}$. The $J!$ elements of S_J decompose into two types: those that map one component to another, and for each component, the $(J-1)!$ permutations that fix the dominant coordinate and map the component onto itself. This gives $J\times(J-1)! = J!$ total permutations, consistent with $|S_J|$.

The orbit size depends on how many within-component permutations fix $\boldsymbol{\omega}$. For a generic point with all non-dominant coordinates distinct, none do, and the orbit has $J!$ elements with $(J-1)!$ in each component. Partially degenerate points, where some but not all non-dominant coordinates coincide, have smaller orbits. Fully degenerate points with all non-dominant coordinates equal are fixed by all $(J-1)!$ within-component permutations, giving orbits of size J. These degenerate cases form a set of measure zero in $\Omega_{\hat{\psi}}$ and do not affect the Monte Carlo approximation in practice.

We can exploit the orbit structure of S_J to obtain equivalent coverage of $\Omega_{\hat{\psi}}$ at a fraction of the cost. Rather than drawing R elements independently from $\Omega_{\hat{\psi}}$, we draw base points one at a time from a single component, expand each to its full orbit of $J!$ by applying all elements of S_J, and screen the resulting candidates for acceptance. Accepted $\boldsymbol{\omega}$ candidates are accumulated across orbits until a total of R have been collected; the number of base $\boldsymbol{\omega}$ draws required is not fixed in advance but determined by how many candidates are accepted from each orbit. The within-orbit expansion is deterministic, and since with probability 1 each candidate in an orbit corresponds to a distinct permutation of the base $\boldsymbol{\omega}$ draw, the R accepted candidates contribute independently to the sum regardless of which orbits they came from. Since each orbit spans all J components by construction, partial acceptance of orbit elements still provides more balanced coverage than naive independent sampling, which offers no such guarantee. The precise procedures for sampling base points from a single component in each regime are developed in Sections C.1.1.5 and C.1.1.6. The resulting approximation is

$$\bar{L}(\psi) \approx \frac{1}{R}\sum_{k=1}^{R} L\Big(b\Big(\boldsymbol{\omega}^{(k)}, \psi;\, \hat{\psi}\Big)\Big)\,, \tag{C.1.14}$$

where $\boldsymbol{\omega}^{(1)}, \ldots, \boldsymbol{\omega}^{(R)}$ are the R accepted candidates, each of the form $\sigma(\boldsymbol{\omega})$ for some base draw $\boldsymbol{\omega}$ from a single component and some $\sigma \in S_J$, and $b(\boldsymbol{\omega}, \psi;\, \hat{\psi})$ denotes the maximizer of $\sum_j \omega_j \log \theta_j$ with respect to $\boldsymbol{\theta} \in \Delta^{J-1}$ subject to $D(\boldsymbol{\theta}) = \psi$.

C.1.1.5. Sampling in the Connected Regime. In the connected regime, $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ by definition, and sampling uniformly from $\Omega_{\hat{\psi}}$ reduces to sampling uniformly from the full $(J-2)$-sphere of radius $r(\hat{\psi}, J)$ (see Equation C.1.8) centered at $\mathbf{c}$ within Π. To do so, we may draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$ and project it onto $\mathbf{1}^\perp = \{\mathbf{v} \in \mathbb{R}^J : \mathbf{1} \cdot \mathbf{v} = 0\}$ by subtracting its mean,

$$\mathbf{z}' = \mathbf{z} - \bar{z}\mathbf{1}, \qquad \bar{z} = \frac{1}{J}\sum_{j=1}^{J} z_j,$$

so that $\mathbf{1} \cdot \mathbf{z}' = 0$. Since $\mathbf{z}'$ lies in $\mathbf{1}^\perp$ by construction, the relevant notion of symmetry is invariance under rotations that map $\mathbf{1}^\perp$ to itself. Such invariance implies that no direction within $\mathbf{1}^\perp$ is preferred over any other, so $\mathbf{z}'/\|\mathbf{z}'\|$ is uniform on the unit $(J-2)$-sphere within $\mathbf{1}^\perp$. This is precisely the property needed to ensure that a point $\mathbf{u} = \mathbf{c} + r \cdot \mathbf{z}'/\|\mathbf{z}'\|$ is uniform on $\mathcal{L}_{\hat{\psi}}$. It therefore suffices to show that the distribution of $\mathbf{z}'$ is invariant under all rotations that map $\mathbf{1}^\perp$ to itself.

Let $\mathbf{Q}$ be a rotation matrix — an orthogonal matrix with determinant $+1$ — that maps $\mathbf{1}^\perp$ to itself. The orthogonality of $\mathbf{Q}$ implies that it preserves inner products,

$$\langle \mathbf{Q}\mathbf{v}, \mathbf{Q}\mathbf{w} \rangle = \mathbf{v}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{w} = \mathbf{v}^\top \mathbf{w} = \langle \mathbf{v}, \mathbf{w} \rangle,$$

for any $\mathbf{v}, \mathbf{w} \in \mathbb{R}^J$. Therefore, if $\mathbf{1}$ is orthogonal to some $\mathbf{v} \in \mathbf{1}^\perp$, then

$$\langle \mathbf{1}, \mathbf{v} \rangle = 0 \implies \langle \mathbf{Q}\mathbf{1}, \mathbf{Q}\mathbf{v} \rangle = 0,$$

so $\mathbf{Q}\mathbf{1}$ is orthogonal to $\mathbf{Q}\mathbf{v}$. Since $\mathbf{Q}$ maps $\mathbf{1}^\perp$ to itself, $\mathbf{Q}\mathbf{v} \in \mathbf{1}^\perp$ for any $\mathbf{v} \in \mathbf{1}^\perp$. As $\mathbf{v}$ was arbitrary, $\mathbf{Q}\mathbf{1}$ must be orthogonal to every vector in $\mathbf{1}^\perp$, so $\mathbf{Q}\mathbf{1} \in \text{span}(\mathbf{1})$, i.e. $\mathbf{Q}\mathbf{1} = k\mathbf{1}$ for some scalar k. Rotation matrices preserve norms, so $\|\mathbf{Q}\mathbf{1}\| = \|\mathbf{1}\|$, implying $|k| = 1$. The determinant $+1$ condition then forces $k = 1$, so $\mathbf{Q}\mathbf{1} = \mathbf{1}$. Applying $\mathbf{Q}^{-1}$ to both sides gives $\mathbf{Q}^{-1}\mathbf{1} = \mathbf{1}$, and since $\mathbf{Q}^\top = \mathbf{Q}^{-1}$ by orthogonality, we have $\mathbf{Q}^\top\mathbf{1} = \mathbf{1}$. The sample mean of $\mathbf{Q}\mathbf{z}$ therefore satisfies $\overline{\mathbf{Q}\mathbf{z}} = \frac{1}{J}\mathbf{1}^\top\mathbf{Q}\mathbf{z} = \frac{1}{J}(\mathbf{Q}^\top\mathbf{1})^\top\mathbf{z} = \frac{1}{J}\mathbf{1}^\top\mathbf{z} = \bar{z}$, so

$$\text{proj}_{\mathbf{1}^\perp}(\mathbf{Q}\mathbf{z}) = \mathbf{Q}\mathbf{z} - \bar{z}\mathbf{1} = \mathbf{Q}\mathbf{z} - \bar{z}\mathbf{Q}\mathbf{1} = \mathbf{Q}(\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{Q}\mathbf{z}' = \mathbf{Q}\text{proj}_{\mathbf{1}^\perp}(\mathbf{z}).$$

That is, projection onto $\mathbf{1}^\perp$ and rotation within it commute. Note that $\mathbf{Q}\mathbf{z} \overset{d}{=} \mathbf{z}$ since $\mathbf{Q}\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}\mathbf{Q}^\top) = \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$. Therefore, we have

$$\mathbf{Q}\mathbf{z}' = \mathbf{Q}\text{proj}_{\mathbf{1}^\perp}(\mathbf{z}) = \text{proj}_{\mathbf{1}^\perp}(\mathbf{Q}\mathbf{z}) \overset{d}{=} \text{proj}_{\mathbf{1}^\perp}(\mathbf{z}) = \mathbf{z}',$$

so $\mathbf{Q}\mathbf{z}' \overset{d}{=} \mathbf{z}'$.

Thus, the distribution of $\mathbf{z}'$ is invariant under any rotation that maps $\mathbf{1}^\perp$ to itself. A random vector with this property must assign equal probability to all directions within $\mathbf{1}^\perp$, since any two directions can be mapped to one another by such a rotation. It follows that $\mathbf{z}'/\|\mathbf{z}'\|$ is distributed uniformly on the unit $(J-2)$-sphere within $\mathbf{1}^\perp$. Rescaling by r gives a vector $\mathbf{d} = r \cdot \mathbf{z}'/\|\mathbf{z}'\|$ that is uniform on the $(J-2)$-sphere of radius r within $\mathbf{1}^\perp$, meaning all points at distance r from the origin in $\mathbf{1}^\perp$ are equally likely. Translating by $\mathbf{c}$ shifts the center of this $(J-2)$-sphere from the origin to $\mathbf{c}$, giving a vector $\mathbf{v} = \mathbf{c} + \mathbf{d}$ that is distributed uniformly on the $(J-2)$-sphere of radius r centered at $\mathbf{c}$ within the affine subspace $\mathbf{c} + \mathbf{1}^\perp = \Pi$. This is precisely $\mathcal{L}_{\hat{\psi}}$ (see Equation C.1.7 and Equation C.1.8). Hence, to sample a point uniformly at random from $\Omega_{\hat{\psi}}$ in the connected regime, we need merely to draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$, project onto $\mathbf{1}^\perp$ by setting $\mathbf{z}' = \mathbf{z} - \bar{z}\mathbf{1}$, and calculate $\mathbf{u} = \mathbf{c} + r \cdot \mathbf{z}'/\|\mathbf{z}'\|$.

C.1.1.6. Sampling in the Disconnected Regime. For $J \geq 3$, the disconnected regime arises when $\hat{\psi} \geq 1/(J-1)$, and $\Omega_{\hat{\psi}}$ consists of J disjoint spherical caps, one near each vertex of Δ^{J-1}. We need not consider $J = 2$ here since, as shown in Section C.1.2, it is impossible for the disconnected regime to arise in such a case. By the permutation symmetry of $\Omega_{\hat{\psi}}$ under S_J, all J caps are geometrically identical, so without loss of generality we may describe how to sample only from $\mathcal{C}_1$, the cap near $\mathbf{e}_1$.

Any vector $\mathbf{v} \in \mathcal{C}_1$ lies in Π and can be written as $\mathbf{v} = \mathbf{c} + \mathbf{d}$, where $\mathbf{d}$ is the displacement from the centroid to $\mathbf{v}$. Since $\Pi = \mathbf{c} + \mathbf{1}^\perp$ is an affine subspace with $\sum_j p_j = 1$ for all $\mathbf{v} \in \Pi$, the difference between any two vectors in Π sums to zero and therefore lies in $\mathbf{1}^\perp$. So $\mathbf{d} = \mathbf{v} - \mathbf{c}$ lies in $\mathbf{1}^\perp$ and has length $r = r(\hat{\psi}, J)$ (see Equation C.1.8).

Any vector in $\mathbf{1}^\perp$ can be decomposed into the sum of two components, one pointing along a chosen reference direction and one that is orthogonal to it. The natural reference direction here is $\mathbf{n}_1$, the unit vector from $\mathbf{c}$ toward $\mathbf{e}_1$. Like $\mathbf{d}$, this vector lies in $\mathbf{1}^\perp$, since it is proportional to $\mathbf{e}_1 - \mathbf{c}$, the difference between two points in Π. Writing γ for the angle between $\mathbf{d}$ and $\mathbf{n}_1$ and $h = \cos\gamma$, the component of $\mathbf{d}$ pointing along $\mathbf{n}_1$ has length rh, and the component orthogonal to $\mathbf{n}_1$ has length $r\sqrt{1-h^2}$. When $J = 3$ the subspace of $\mathbf{1}^\perp$ orthogonal to $\mathbf{n}_1$ is zero-dimensional, so the orthogonal component has a fixed direction and the displacement $\mathbf{d}$ is fully determined by h alone. For $J \geq 4$ this subspace has dimension $J-3 \geq 1$; letting $\mathbf{u}$ denote a unit vector within it, the decomposition is

$$\mathbf{d} = rh\,\mathbf{n}_1 + r\sqrt{1-h^2}\,\mathbf{u} = r\Big(h\,\mathbf{n}_1 + \sqrt{1-h^2}\,\mathbf{u}\Big)\,. \tag{C.1.15}$$

The scalar h controls how far along the cap axis the vector $\mathbf{v}$ lies. When $h = 1$ the displacement points directly along the cap axis, placing $\mathbf{v}$ at the tip of the cap, the point nearest to $\mathbf{e}_1$. When $h = \cos\alpha$ the displacement makes the maximum angle α with the axis, placing $\mathbf{v}$ at one of the boundaries of the cap. For $J \geq 4$, a point on $\mathcal{C}_1$ is fully specified by the pair $(h, \mathbf{u})$, which separately control how far the point is from the tip of the cap and in which direction around the cap axis it lies.

To sample $\mathbf{v}$ uniformly from $\mathcal{C}_1$ when $J \geq 4$, it suffices to draw h and $\mathbf{u}$ independently from their respective marginal distributions. The decomposition in Equation C.1.15 separates $\mathbf{d}$ into a component along $\mathbf{n}_1$ and a component orthogonal to it, and under a uniform distribution on the sphere these two components are independent, so the joint distribution of $(h, \mathbf{u})$ factors into the product of the marginals. The component $\mathbf{u}$ is sampled by the same Gaussian projection used in the connected regime, extended to project out both the $\mathbf{1}$ direction and the $\mathbf{n}_1$ direction before normalizing: draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$, subtract its mean to project onto $\mathbf{1}^\perp$, then subtract the component along $\mathbf{n}_1$, and normalize. By the same rotational symmetry argument as in the connected regime, the result is uniform on the unit $(J-3)$-sphere within the orthogonal subspace. When $J = 3$, the orthogonal subspace is one-dimensional, so the unit $(J-3)$-sphere within it is zero-dimensional, consisting of just two points $\pm\mathbf{u}_0$ for a fixed unit vector $\mathbf{u}_0$, and all randomness in the draw comes from h.

Sampling h requires the marginal density of γ. Consider the set of all points on $\mathcal{C}_1$ at angle γ relative to $\mathbf{n}_1$. Call this set the latitude ring at angle γ from the cap axis. From the decomposition Equation C.1.15, fixing γ fixes the component of $\mathbf{d}$ along $\mathbf{n}_1$ to $r\cos\gamma$, while the orthogonal component is free to point in any direction $\mathbf{u}$ within the $(J-3)$-dimensional subspace orthogonal to $\mathbf{n}_1$, at a fixed distance $r\sin\gamma$ from the axis. It follows that the latitude ring is a $(J-3)$-sphere of radius $r\sin\gamma$. Since the surface area of a k-sphere of radius ρ scales as ρ^k, the surface area of this ring is proportional to $(r\sin\gamma)^{J-3}$. The surface element of $\mathcal{L}_{\hat{\psi}}$ at angle γ then decomposes as the product of the area of the latitude ring and the arc length width of an infinitesimal band around it,

$$dA \propto (r \sin \gamma)^{J-3} \cdot r d\gamma.$$

Under a uniform distribution on $\mathcal{C}_1$, the probability of landing in an infinitesimal band between γ and $\gamma + d\gamma$ is proportional to the band's surface area, which is the surface area of the latitude ring times the arc length width $r d\gamma$. Thus, the probability of landing in the band is

$$f(\gamma)\, d\gamma \propto (r \sin \gamma)^{J-3} \cdot r\, d\gamma.$$

Dividing through by $d\gamma$ and absorbing the constant factors of r into the proportionality gives the marginal density of γ,

$$f(\gamma) \propto \sin(\gamma)^{J-3}, \qquad \gamma \in [0, \alpha]. \tag{C.1.16}$$

For $J = 3$ the exponent vanishes so $f(\gamma) \propto 1$, meaning γ is uniform on $[0, \alpha]$ and $h = \cos(U\alpha)$ for $U \sim \text{Uniform}(0, 1)$ suffices. For $J \geq 4$ the density Equation C.1.16 has no closed-form CDF in general, but the substitution $s = \sin^2 \gamma$ transforms it into a recognizable family. Differentiating gives $ds = 2 \sin \gamma \cos \gamma\, d\gamma$, so $d\gamma = ds/(2\sqrt{s}\sqrt{1-s})$, and substituting into Equation C.1.16 yields the density of s,

$$\begin{aligned}
f_s(s) &= f(\gamma) \cdot \left| \frac{d\gamma}{ds} \right| \\
&\propto \sin(\gamma)^{J-3} \cdot \frac{1}{2\sqrt{s}\sqrt{1-s}} \\
&= s^{(J-3)/2} \cdot \frac{1}{2\sqrt{s}\sqrt{1-s}} \\
&= \frac{1}{2} s^{(J-3)/2 - 1/2} (1-s)^{-1/2} \\
&= \frac{1}{2} s^{(J-2)/2 - 1} (1-s)^{1/2 - 1}, \qquad s \in [0, \sin^2 \alpha].
\end{aligned}$$

We recognize this as the kernel of a Beta($(J-2)/2$, $1/2$) distribution truncated to $[0, \sin^2 \alpha]$. An exact draw is obtained by inverting the truncated Beta CDF. Draw $U \sim \text{Uniform}(0, 1)$, compute

$$q = U \cdot I_{\sin^2 \alpha}\left(\tfrac{J-2}{2}, \tfrac{1}{2}\right),$$

where $I_x(a,b)$ denotes the regularized incomplete beta function, and set

$$s = I_q^{-1}\big(\tfrac{J-2}{2},\ \tfrac{1}{2}\big)\,. \tag{C.1.17}$$

Scaling U by $I_{\sin^2\alpha}((J-2)/2, 1/2)$ before inversion restricts the draw to $[0, \sin^2\alpha]$, enforcing the truncation. The height coordinate is then recovered as $h = \sqrt{1-s}$. This approach is exact and numerically stable for any cap width, including the narrow caps that arise when $\hat{\psi}$ is large and α is small.

Hence, to draw a vector uniformly at random from $\mathcal{C}_1$, draw $U \sim \text{Uniform}(0,1)$ and sample h as follows: for $J = 3$ set $h = \cos(U\alpha)$, and for $J \geq 4$ compute $q = U \cdot I_{\sin^2\alpha}((J-2)/2, 1/2)$, set $s = I_q^{-1}((J-2)/2, 1/2)$, and recover $h = \sqrt{1-s}$. In both cases, sample $\mathbf{u}$ by drawing $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_J)$, subtracting its mean, subtracting the component along $\mathbf{n}_1$, and normalizing. The desired vector is then $\mathbf{v} = \mathbf{c} + r(h\,\mathbf{n}_1 + \sqrt{1-h^2}\,\mathbf{u})$.

C.1.2. The Case $J = 2$

When $J = 2$, the simplex Δ^1 is a line segment and $D(\boldsymbol{\theta}) = \theta_1^2 + \theta_2^2 < 1$ for all interior points, so the clipping threshold $1/(J-1) = 1/(2-1) = 1$ is never reached. Consequently, the disconnected regime never arises, and $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ degenerates to a 0-sphere, consisting of a single pair of points for any $\hat{\psi} > 1/2$.

The parameter space is $\Delta^1 = \{(\theta_1, \theta_2) : \theta_1 + \theta_2 = 1,\ \theta_1, \theta_2 \in (0,1)\}$, and Simpson's index reduces to

$$D(\boldsymbol{\theta}) = \theta_1^2 + (1-\theta_1)^2 = 2\theta_1^2 - 2\theta_1 + 1. \tag{C.1.18}$$

Setting $D(\boldsymbol{\theta}) = \hat{\psi}$ and solving gives $\theta_1 = (1 \pm \sqrt{2\hat{\psi} - 1})/2$, so the two points are

$$\boldsymbol{\omega}^{(1)} = \left(\frac{1+\sqrt{2\hat{\psi}-1}}{2},\ \frac{1-\sqrt{2\hat{\psi}-1}}{2}\right), \qquad \boldsymbol{\omega}^{(2)} = \left(\frac{1-\sqrt{2\hat{\psi}-1}}{2},\ \frac{1+\sqrt{2\hat{\psi}-1}}{2}\right).$$

These are the $J = 2$ instances of the J degenerate caps identified in the general case, each collapsing to a single point near one of the two vertices. They are related by the coordinate swap $(\theta_1, \theta_2) \mapsto (\theta_2, \theta_1)$, which is the $J = 2$ instance of the permutation symmetry established above, and migrate toward the corners $(1,0)$ and $(0,1)$ as $\hat{\psi} \to 1$. The integrated likelihood at ψ is

$$\bar{L}(\psi) = L(b(\boldsymbol{\omega}^{(1)}, \psi;\, \hat{\psi})) + L(b(\boldsymbol{\omega}^{(2)}, \psi;\, \hat{\psi})), \tag{C.1.19}$$

up to a multiplicative constant. Since $\boldsymbol{\omega}^{(1)} = \mathbf{1} - \boldsymbol{\omega}^{(2)}$ and the constraint $D(\boldsymbol{\theta}) = \psi$ is symmetric in θ_1 and θ_2, the maximizer $b(\boldsymbol{\omega}^{(2)}, \psi;\, \hat{\psi})$ is simply $\mathbf{1} - b(\boldsymbol{\omega}^{(1)}, \psi;\, \hat{\psi})$, so only one constrained optimization is needed. The likelihood must still be evaluated separately at each point, however, since $\hat{\psi} > 1/2$ implies $N_1 \neq N_2$, and consequently $L(\boldsymbol{\theta}) = \theta_1^{N_1}\theta_2^{N_2}$ will not be invariant under permutation of $\boldsymbol{\theta}$.

When $\hat{\psi} = 1/2$, the two points coincide at the centroid $\mathbf{c} = (1/2, 1/2)$, which is also the uniform distribution $\hat{\boldsymbol{\theta}} = (1/2, 1/2)$, and $\Omega_{\hat{\psi}} = \{\hat{\boldsymbol{\theta}}\}$. The integrated likelihood then reduces to the single evaluation $L(b(\hat{\boldsymbol{\theta}}, \psi;\, \hat{\psi}))$, which is precisely the profile likelihood maximizer, so the two are identical in this case. More generally, since $\Omega_{\hat{\psi}}$ contains only two points for any $\hat{\psi}$, there is little room for the integrated likelihood to differ from the profile likelihood through averaging over meaningfully distinct nuisance parameters, and the two will tend to agree closely in practice. This is consistent with the well-established result that the integrated likelihood only begins to outperform the profile likelihood when the nuisance parameter is high-dimensional (Berger et al., 1999; De Bin et al., 2015).

C.1.3. The Case $J = 3$

When $J = 3$, $\mathcal{L}_{\hat{\psi}}$ is a circle of radius $r(\hat{\psi}, 3) = \sqrt{\hat{\psi} - 1/3}$ centered at $\mathbf{c}$ within Π (see Equation C.1.8). The parameter space Δ^2 is an equilateral triangle, with the facets reducing to three one-dimensional edges. Each edge has length $\sqrt{2}$ and the inradius is $r_{\text{in}} = 1/\sqrt{6}$. The clipping threshold is $1/(J-1) = 1/2$, and $\Omega_{\hat{\psi}}$ is a complete circle when $\hat{\psi} < 1/2$ and three disjoint arcs when $\hat{\psi} \geq 1/2$, corresponding to the connected and disconnected regimes of the general case. These are illustrated in Figure C.1.

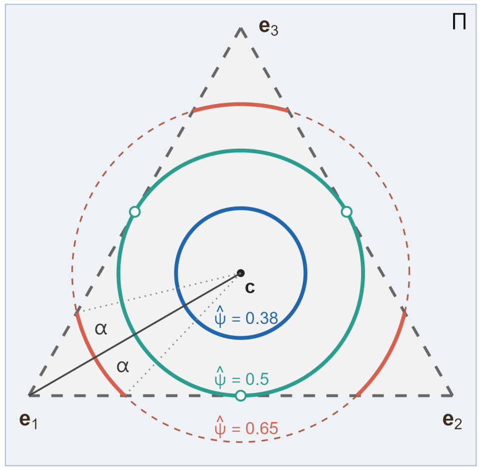

Figure C.1. Level Sets of Simpson's Index on the Probability Simplex Δ^2. The circle $\mathcal{L}_{\hat{\psi}}$ of radius $r(\hat{\psi}, 3) = \sqrt{\hat{\psi} - 1/3}$ centered at the centroid $\mathbf{c}$ and embedded in the affine plane Π is shown for three values of $\hat{\psi}$. When $\hat{\psi} < 1/2$, $\mathcal{L}_{\hat{\psi}}$ lies entirely within Δ^2 and $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ is a complete circle. At $\hat{\psi} = 1/2$, $\mathcal{L}_{\hat{\psi}}$ is tangent to all three edges of Δ^2 simultaneously, touching at the open circles marking the uniform distributions over two categories. When $\hat{\psi} > 1/2$, $\mathcal{L}_{\hat{\psi}}$ extends beyond Δ^2 and $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}} \cap \Delta^2$ consists of three disjoint arcs, one near each vertex. Each arc subtends an angle 2α at $\mathbf{c}$, with the centroid-to-vertex line bisecting the arc.

Setting $J = 3$ in Equation C.1.13 gives the angular radius of each arc:

$$\alpha(\hat{\psi}) = \arccos\left(\frac{1 + 3\sqrt{\hat{\psi} - \frac{1}{2}}}{2\sqrt{2\hat{\psi} - 1}}\right), \qquad \hat{\psi} \in \left[\tfrac{1}{2}, 1\right). \tag{C.1.20}$$

At $\hat{\psi} = 1/2$, $\alpha = \pi/3$, so the three arcs tile the full circle, consistent with $\Omega_{\hat{\psi}} = \mathcal{L}_{\hat{\psi}}$ at the threshold. As $\hat{\psi} \to 1$, $\alpha \to 0$ and each arc shrinks to a point at its vertex, consistent with Equation C.1.13.

Evaluating Equation C.1.14 at $J = 3$ gives

$$\bar{L}(\psi) \approx \frac{1}{R}\sum_{k=1}^{R} L\Big(b\Big(\boldsymbol{\omega}^{(k)}, \psi;\, \hat{\psi}\Big)\Big), \tag{C.1.21}$$

where $\boldsymbol{\omega}^{(1)}, \ldots, \boldsymbol{\omega}^{(R)}$ are the R accepted candidates, each of the form $\sigma(\boldsymbol{\omega})$ for some base draw $\boldsymbol{\omega}$ from a single arc and some $\sigma \in S_3$. The six elements of S_3 decompose into three cyclic permutations mapping each arc to another, and three reflections mapping each arc onto itself across the axis from $\mathbf{c}$ to its vertex. The transition point $\hat{\psi} = 1/2$ marks the spot where the sector boundaries coincide exactly with the simplex edges.

APPENDIX D

Chapter 6

Proposition: $\psi^*_\omega = \hat{\psi}$ for all $\omega \in \Omega_{\hat{\psi}}$.

Proof: It suffices to show that an arbitrary branch curve $b_\omega(\psi)$ has a stationary point at $\psi = \hat{\psi}$, i.e. $b'_\omega(\hat{\psi}) = 0$. Since $\ell(\theta; Y)$ is concave in θ and the mapping $\psi \mapsto \tilde{\theta}(\psi; \omega)$ is smooth, such a point will be the unique maximizer of $b_\omega(\psi)$. Differentiating the branch yields

$$b'_\omega(\psi) = \nabla_\theta \ell(\tilde{\theta}(\psi; \omega); Y)^\top \frac{\partial \tilde{\theta}(\psi; \omega)}{\partial \psi}. \tag{D.0.1}$$

Thus, we need to be able to evaluate both the ZSE optimizer $\tilde{\theta}(\psi; \omega)$ and its partial derivative with respect to ψ at $\psi = \hat{\psi}$. Recall that to evaluate $\tilde{\theta}(\psi; \omega)$ at a point ψ for a given ω is to solve the constrained maximization problem

$$\arg\max_\theta \sum_{g=1}^{G} \left(t_g \omega_g \log \theta_g - t_g \theta_g\right) \text{ s.t. } \alpha^\top \theta = \psi.$$

We proceed by the method of Lagrange multipliers. For some multiplier λ (which may depend on ψ and ω), define the Lagrangian function

$$\mathcal{L}(\theta, \lambda) = \sum_{g=1}^{G} \left(t_g \omega_g \log \theta_g - t_g \theta_g\right) + \lambda \left[\psi - \sum_{g=1}^{G} \alpha_g \theta_g\right].$$

Per the Lagrange multiplier theorem, the solution to the maximization problem will satisfy

$$\left.\frac{\partial \mathcal{L}(\theta, \lambda)}{\theta_g}\right|_{\theta_g = \tilde{\theta}_g} = 0, \text{ for } g = 1, ..., G \tag{D.0.2}$$

and

$$\frac{\partial \mathcal{L}(\theta, \lambda)}{\lambda} = 0. \tag{D.0.3}$$

From Equation D.0.2, we have

$$\frac{t_g \omega_g}{\tilde{\theta}_g} - t_g - \lambda \alpha_g = 0.$$

This implies that the optimal solution $\tilde{\theta}$, viewed here as a function of λ, satisfies

$$\tilde{\theta}_g(\lambda) = \frac{t_g \omega_g}{t_g + \lambda \alpha_g}. \tag{D.0.4}$$

Paired with the constraint condition enforced by Equation D.0.3, this in turn implies that the optimal value of λ will satisfy

$$\sum_{g=1}^{G} \alpha_g \tilde{\theta}_g(\lambda) = \psi. \tag{D.0.5}$$

Note that $\lambda = 0 \iff \tilde{\theta} = \omega \iff \psi = \hat{\psi}$. It follows that $\tilde{\theta}(\hat{\psi}; \omega) = \omega$.

Turning our attention now to the derivative of the ZSE optimizer with respect to ψ, we have that

$$\frac{\partial \tilde{\theta}_g}{\partial \psi} = \frac{\partial \tilde{\theta}_g}{\partial \lambda} \frac{d\lambda}{d\psi}.$$

From Equation D.0.4, we can compute

$$\frac{\partial \tilde{\theta}_g}{\partial \lambda} = -\frac{t_g \omega_g \alpha_g}{(t_g + \lambda \alpha_g)^2}.$$

At $\lambda = 0$, this evaluates to

$$\left. \frac{\partial \tilde{\theta}_g}{\partial \lambda} \right|_{\lambda=0} = -\frac{\omega_g \alpha_g}{t_g}.$$

Similarly, differentiating both sides of Equation D.0.5 with respect to ψ gives

$$\sum_{g=1}^{G} \alpha_g \frac{\partial \tilde{\theta}_g}{\partial \lambda} \frac{d\lambda}{d\psi} = 1 \implies \frac{d\lambda}{d\psi} = \frac{1}{\sum_{g=1}^{G} \alpha_g \frac{\partial \tilde{\theta}_g}{\partial \lambda}}.$$

At $\psi = \hat{\psi}$, this evaluates to

$$\left. \frac{d\lambda}{d\psi} \right|_{\psi=\hat{\psi}} = \frac{1}{\sum_{g=1}^{G} \alpha_g \left. \frac{\partial \tilde{\theta}_g}{\partial \lambda} \right|_{\lambda=0}} = -\frac{1}{\sum_{g=1}^{G} \alpha_g^2 \omega_g / t_g}.$$

Thus

$$\begin{aligned}
\left.\frac{\partial\tilde{\theta}_g}{\partial\psi}\right|_{\psi=\hat{\psi}} &= \left.\frac{\partial\tilde{\theta}_g}{\partial\lambda}\right|_{\lambda=0} \cdot \left.\frac{d\lambda}{d\psi}\right|_{\psi=\hat{\psi}} \\
&= \left(-\frac{\omega_g\alpha_g}{t_g}\right)\left(-\frac{1}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}\right) \\
&= \frac{\alpha_g\omega_g/t_g}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h},
\end{aligned}$$

where we have changed the index of summation from g to h in the denominator of the final two lines to avoid confusion with the particular index g being considered in the numerator.

Returning to Equation D.0.1, we can now compute

$$\begin{aligned}
b'_\omega(\hat{\psi}) &= \left[\nabla_\theta \ell(\tilde{\theta}(\psi;\omega);Y)\Big|_{\psi=\hat{\psi}}\right]^\top \left[\left.\frac{\partial\tilde{\theta}(\psi;\omega)}{\partial\psi}\right|_{\psi=\hat{\psi}}\right] \\
&= \sum_{g=1}^{G}\left(\left.\frac{\partial\ell}{\partial\theta_g}\right|_{\theta_g=\tilde{\theta}_g(\hat{\psi};\omega)}\right)\left(\left.\frac{\partial\tilde{\theta}_g}{\partial\psi}\right|_{\psi=\hat{\psi}}\right) \\
&= \sum_{g=1}^{G}\left(\frac{Y_g}{\tilde{\theta}_g(\hat{\psi};\omega)} - t_g\right)\left(\frac{\alpha_g\omega_g/t_g}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}\right) \\
&= \sum_{g=1}^{G}\left(\frac{Y_g}{\omega_g} - t_g\right)\left(\frac{\alpha_g\omega_g/t_g}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}\right) \\
&= \frac{1}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}\sum_{g=1}^{G}\alpha_g\left(\frac{Y_g}{t_g} - \omega_g\right) \\
&= \frac{1}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}\left(\sum_{g=1}^{G}\alpha_g\hat{\theta}_g - \sum_{g=1}^{G}\alpha_g\omega_g\right) \\
&= \frac{1}{\sum_{h=1}^{H}\alpha_h^2\omega_h/t_h}(\hat{\psi} - \hat{\psi}) \\
&= 0.
\end{aligned}$$

$\therefore\ b'_\omega(\hat{\psi}) = 0.$

Since the choice of branch was arbitrary, this holds true for all $\omega \in \Omega_{\hat{\psi}}$.

$\therefore \psi^*_\omega = \hat{\psi}$ for all $\omega \in \Omega_{\hat{\psi}}$. ■